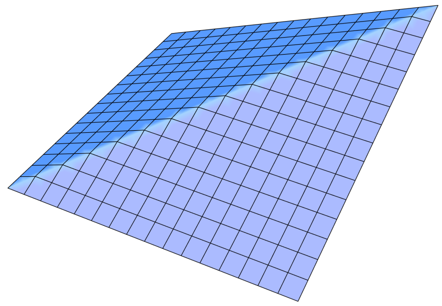

# ACTSC 445: Quantitative Risk Management


Prof. David Saunders • Fall 2015 • University of Waterloo

TR 1:00–2:20PM in MC 2065

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Transcribed, LaTeXed, and edited by Rob Zimmerman


**Note:** These are personal lecture notes, not official course materials. They may contain errors (which by default you should attribute to me, Rob Zimmerman, rather than the course instructor). The permanent home of these — and all of my other lecture notes — is `https://rob-zimmerman.github.io/coursenotes`.

## Contents

**Editorial Note** At the time of this offering, much of ACTSC 445's course content was based on *Quantitative Risk Management: Concepts, Techniques and Tools (Revised Edition)* by Alexander J. McNeil, Rüdiger Frey, and Paul Embrechts (Princeton University Press, 2015), which is unquestionably the greatest textbook written on the subject.

**Lecture 1 — Tuesday, September 15, 2015**

# 1. Enterprise Risk Management Foundations

## Risk Management and Enterprise Risk Management

All firms manage risk. In financial services, risk management is a core part of the business, so firms need processes and frameworks for identifying, measuring, and managing risk. These processes should be **top-down**, meaning that risk management should be integral to corporate governance rather than confined to an isolated technical function.

**Enterprise risk management** takes this top-down idea seriously. It implies a holistic approach in which risk management is incorporated into the firm's values, strategy, operations, and financial reporting.

Historically, risk management practice developed through several related areas: insurable risk management, financial risk management, actuarial risk management, asset-liability management, quantitative risk management, project risk management, and operational risk management. Enterprise risk management attempts to integrate these areas into a single enterprise-wide framework.

## Insurable Risks and Financial Risk Management

The traditional management of insurable risks is studied in risk management and insurance. These are **pure risks**: they involve downside exposure rather than upside opportunity, and they are risks that can often be insured. The classical risk management strategies are to transfer the risk, retain it, avoid it, or reduce it.

**Financial risk management** begins with portfolio theory, especially the Markowitz-Sharpe mean-variance framework. Variance is used as a measure of risk, although it is a two-sided measure because it penalizes upside and downside deviations symmetrically. Mean-variance portfolio analysis balances risk and return, and it eliminates portfolios that are not on the efficient frontier.

The capital asset pricing model (CAPM) gives

$$\mathbb{E}\left[R_k\right] = r_f + \beta_k\big(\mathbb{E}\left[R_M\right] - r_f\big).$$

Here $R_k$ is the return on asset $k$, $r_f$ is the risk-free rate, $R_M$ is the return on the market portfolio, and $\beta_k$ is the asset's beta.

If we subtract the risk-free rate from both sides, we obtain

$$\mathbb{E}\left[R_k\right] - r_f = \beta_k\big(\mathbb{E}\left[R_M\right] - r_f\big),$$

so the expected excess return on the asset is its beta times the **market risk premium**. The coefficient $\beta_k$ is the **systematic risk** of the asset, meaning the part of the asset's risk that is attributable to movements in the overall market and cannot be diversified away in the CAPM framework.

A European call option pays

$$\max(S_T - K, 0)$$

at time $T$, while a European put option pays

$$\max(K - S_T, 0).$$

Here $K$ is the strike price and $S_T$ is the underlying asset value at maturity. Writing these payoffs out by cases makes the economics clearer: the call pays $S_T - K$ when $S_T > K$ and otherwise pays 0, while the put pays $K - S_T$ when $S_T < K$ and otherwise pays 0. Black-Scholes-Merton analysis showed how to price and hedge such contingent claims through replication, and this became a powerful framework for financial risk management.

These classical tools also have important limitations. Standard deviation is not always a good measure of portfolio risk because it treats upside and downside deviations symmetrically, even though firms are usually more concerned with losses than gains. It may also fail to capture rare but high-impact events. In practice, means, standard deviations, and correlations are not known parameters; they must be estimated, often with substantial uncertainty, and different investors may have different views about them. From an enterprise perspective, beta is also too narrow because it studies assets or investments in isolation, ignores the interaction among different parts of the enterprise, and ignores mismatches between assets and liabilities. Likewise, Black-Scholes-Merton analysis depends on strong assumptions such as Gaussian log-returns, no transaction costs or taxes, continuous trading, and constant volatility.

There is therefore both a contrast and a connection between insurable risk management and financial risk management. Insurable risk management is traditionally concerned with pure downside exposures and with the choice among transfer, retention, avoidance, and reduction. Financial risk management is more explicitly concerned with the joint determination of risk and return, with market prices, hedging, and valuation. Even so, both areas require measurement, modelling, and deliberate treatment of uncertainty, and both ultimately feed into enterprise-wide decision making.

## Actuarial Risk Management, Asset-Liability Management, Quantitative Risk Management, and Project Risk Management

**Actuarial risk management** combines insurable and financial risks. Business risk has both upside and downside, and actuaries often manage solvency and, to some extent, profitability for insurers and pension plans through product design, pricing, valuation, and investment strategy. Dynamic financial analysis is one of the main actuarial frameworks here, and it makes use of scenario tests, stress tests, and stochastic simulation.

**Asset-liability management** is concerned with mismatching between assets and liabilities. Interest rate risk is a central issue, and duration and immunization analysis are standard tools. The same general framework also extends to exchange-rate risk, credit spreads, and other sources of mismatch.

**Quantitative risk management** provides the technical toolkit used across these areas. Important examples include extreme value theory, dependency modelling, risk measures such as value-at-risk and conditional tail expectation, and econometric models of financial processes.

**Project risk management** is aimed at business projects with a definite beginning and end. It is multidisciplinary and must deal with complex budgeting and scheduling issues. In practice, this means that risk management methods must be adapted not only to financial and insurance exposures, but also to operational and organizational contexts.

**Enterprise risk management** is a process, effected by an entity's board of directors, management, and other personnel, applied in strategy setting and across the enterprise, designed to identify potential events that may affect the entity and to manage risk so that it remains within the entity's risk appetite, thereby providing reasonable assurance regarding the achievement of the entity's objectives.

The Committee of Sponsoring Organizations of the Treadway Commission (COSO) is a widely cited source for enterprise risk management. It describes enterprise risk management as "a process, effected by an entity's board of directors, management, and other personnel, applied in strategy setting and across the enterprise, designed to identify potential events that may affect the entity and to manage risk so that it remains within the entity's risk appetite, thereby providing reasonable assurance regarding the achievement of the entity's objectives." This definition emphasizes the process nature of enterprise risk management, its connection to strategy, its scope across the enterprise, and its focus on risk appetite and the achievement of objectives.

Meanwhile, the Risk Management Society describes enterprise risk management as "a strategic business discipline that supports the achievement of an organization's objectives by addressing the full spectrum of its risks and by managing the combined impact of those risks as an interrelated risk portfolio."

The Risk Management Society description can be unpacked in several ways. First, enterprise risk management encompasses all major sources of organizational exposure, including financial, operational, reporting, compliance, governance, strategic, and reputational risks. Second, it prioritizes and manages these exposures as an interrelated portfolio rather than as separate silos. Third, it evaluates that portfolio in the context of internal and external environments, systems, circumstances, and stakeholders. Fourth, it recognizes that risks interact, so the combined exposure can differ materially from the simple sum of the individual risks. Fifth, it provides a structured process that applies to both quantitative and qualitative risks. Sixth, it treats effective risk management as a possible competitive advantage. Seventh, it seeks to embed risk management into all critical organizational decisions.

Under ISO 31000, enterprise risk management should create and protect value. It should be integral to all processes and decisions, should explicitly address uncertainty, and should proceed through a systematic, structured, and timely process using the best available information. It should be tailored to the organization, should account for human and cultural factors, should be transparent and inclusive, should be dynamic and iterative, and should contribute to continuous improvement.

The ISO 31000 risk management process consists of establishing the context, identifying risks, analyzing risks, evaluating risks, and treating risks. Communication and consultation occur throughout the process, and monitoring and review also continue throughout rather than appearing only at the end. Risk management is therefore a cycle embedded in governance rather than a one-time calculation.

## Governance, Strategy, and Strategic Plans

Enterprise risk management begins with corporate governance. The board of directors sets the firm's mission, vision, and values, and these should incorporate the firm's risk appetite. The strategic plan should therefore be framed at the board level and should interpret the mission, vision, and values in operational terms. Stakeholders must be identified explicitly, and the resulting objectives should be linked to performance metrics, often through a balanced scorecard.

At the financial level, the ultimate objective is to maximize shareholder value. This is supported by customer-level objectives such as exceeding customer expectations and inspiring loyalty. Those customer outcomes, in turn, depend on internal business process objectives such as creating quality partnerships, maximizing operational effectiveness, and creating high-quality products. At the base are learning-and-growth objectives, namely recruiting quality staff and training employees. Strategy is read from the bottom upward: capabilities support processes, processes support customer outcomes, and customer outcomes support financial performance.

The Society of Actuaries (SOA) strategic plan provides a concrete example. Its mission is to advance actuarial knowledge and to enhance the ability of actuaries to provide expert advice and relevant solutions for financial, business, and societal problems involving uncertain future events. Its vision is that actuaries should be the leading professionals in the measurement and management of risk.

The stakeholder portion of the SOA plan identifies several constituencies and the value proposition for each. Candidates want the skill set and credentials needed to advance. Members want opportunities to continue evolving and to use their skill sets. Employers and clients want highly qualified experts who are relevant to their needs. The public wants trust to be preserved and the common good to be served.

The strategic themes in the SOA plan are also explicit. The first theme is to develop knowledge by producing and supporting research that expands the boundaries of actuarial science, by promoting the development of intellectual capital, and by identifying opportunities for its application. The second theme is to transfer knowledge by offering a prestigious credential, maintaining its value through leading-edge educational systems and programs, and leveraging a variety of delivery channels to disseminate and facilitate knowledge transfer. The third theme is to cultivate opportunities by establishing and expanding the actuarial brand and marketing message, by creating and promoting new areas of practice, and by ensuring a robust stream of well-qualified new candidates and credentialed actuaries.

The same plan also emphasizes the development and promotion of professional networks. In particular, sections are to function as knowledge communities and networking facilitators, long-term partnerships are to be built with actuarial and other professional organizations around the world, and relationships are to be expanded across academia, think tanks, and policy influencers. Workforce and staff objectives include attracting, developing, and retaining talented people, focusing on outcomes and rewarding execution, identifying and developing leaders at all levels, and promoting a culture of commitment, innovation, service, and excellence. Financially, the plan seeks to generate margins in order to fund strategic activities, align the use of SOA financial resources with the strategic plan, and deliver cost-efficient support and services. The professional foundation of the plan is specialized knowledge, integrity, and public interest.

A strategic plan should not be separated from risk management. A **balanced scorecard** requires explicit performance metrics for each strategic element, and those metrics should include risk measures, triggers, and thresholds. The plan should identify whether the organization is operating within its risk appetite and should specify the risk management process that will be used when limits are approached or breached.

### Risk Management Process

The risk management process has five steps. The first step is to determine the firm's risk appetite and risk tolerance. The second step is to identify the risks. The third step is to analyze the risks. The fourth step is to evaluate the risks. The fifth step is to manage or treat the risks.

The firm's **risk appetite** is its high-level statement about the amount and type of risk it is willing to seek or accept in pursuit of its objectives. **Risk tolerance** is a more detailed and usually more quantitative articulation of that appetite. **Risk limits** are then set at the business-unit or operational level so that the broad appetite is translated into day-to-day constraints.

One common interpretation is that the risk appetite specifies the target level of risk within the firm's risk corridor, while risk tolerance describes the acceptable range of deviation around that target. On that interpretation, risk limits are the concrete operational constraints that keep the firm from drifting too far away from its chosen risk appetite.

The board of directors is responsible for determining risk appetite, risk tolerance, and risk limits. These can be quantitative or qualitative. They may take into account external signals such as analyst or rating agency views, and they should be specific about the kinds of risks the firm is prepared to seek, avoid, or accept. They must also be stated with an understanding of the economic capital that will be required to support those choices.

**Economic capital** is the capital required by a bank or insurer to limit the risk of insolvency to a specified level over a given time horizon. It is distinct from regulatory capital, since it is based on the firm's own internal standards and models rather than solely on external rules.

A good economic capital model promotes firm-wide consistency. It supports the assessment of risk, the pricing of risk, and the evaluation of return on risk capital. This is closely related to the Own Risk and Solvency Assessment (ORSA), which is an internal process through which the organization evaluates the adequacy of its capital relative to its risk profile.

**Example 1.1** A firm might state that it wants to limit the probability of a 20% drop in earnings over the next two years to no more than 5%. A bank might say that it has a very low appetite for credit risk. A university might state that preserving its reputation is critical and therefore that it has a low appetite for activities that could place that reputation in jeopardy or lead to undue adverse publicity. A balance-sheet constraint such as ensuring that long-term borrowings never exceed 20% of net assets is another example of a risk limit expressed in operational terms.

**Risk identification** must be rigorous and regular. It should involve front-line staff in a bottom-up process as well as senior officers in a top-down process. The organization should identify the risks themselves, their types, changes and trends in previously identified risks, and new risks as they emerge. It is also necessary to keep in mind the **unknown unknowns**, that is, exposures that have not yet been clearly conceptualized or measured. This is closely related to **Knightian uncertainty**, where uncertainty is present but cannot be fully described by known probabilities.

**Risk analysis** then studies the potential financial consequences of adverse outcomes. This may begin with qualitative assessment and high-level tools such as heat maps. It may proceed through deterministic analysis, scenario testing, and stress testing, and it may culminate in stochastic analysis. The analysis should also account for the effect of mitigation strategies and for the continuing maintenance of those strategies.

Risk evaluation compares the results of the analysis with the firm's targets and risk appetite. At this stage, aggregation matters. Risks cannot be evaluated only one by one, because interactions among them may increase or decrease the total exposure of the enterprise.

A **top-down assessment** identifies and evaluates risk potential at the enterprise-wide level and supports strategic and risk-finance decisions. It often relies on stress testing, scenario analysis, and risk mapping, together with surveys or interviews of senior managers and the use of internal or external data. Its weakness is that it may fail to identify operating issues and may not translate into action at the front line.

By contrast, a **bottom-up assessment** identifies and evaluates risk potential at the transaction level or business-unit level. It relies more on front-line staff, surveys, interviews, what-if analysis, self-assessment, new-product reviews, and internal data analysis. It is particularly useful for process risks. Its weaknesses are that it may lack sufficient perspective or credibility, may not be given enough priority, and may not be timely. Both approaches are needed.

## Risk Treatment

For insurable risks, the classical responses are to avoid the risk, reduce or mitigate it, transfer it, or retain it without mitigation. In the broader ISO 31000 enterprise framework, the menu of treatments is expanded. The firm may avoid the risk, pursue it when it is consistent with strategic objectives, remove the risk source, change the probability of the event, change the severity of the consequences, share or transfer the risk, or retain it by an informed decision. Treatment is not always about eliminating risk. In many cases the firm deliberately takes risk, but it does so consciously, with governance, limits, and capital support in place.

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### Lecture 2 — Tuesday, September 22, 2015

**Example 2.1** The Deepwater Horizon case involved an explosion on a drilling rig in the Gulf of Mexico. Eleven people were killed, sixteen were injured, and the resulting oil spill caused severe environmental damage.

Several causes were identified. The cement used in the construction phase was defective, the crew was not well trained in shutdown procedures, equipment maintenance was poor, alarms and automatic shutdown processes were bypassed, and the overall safety culture was weak. Cheap materials and methods were used, and modelling software outputs indicating risk were ignored. The resulting losses included cleanup costs, legal liabilities, and severe reputational damage.

**Example 2.2** In 2007, Manulife had roughly \$75 billion of exposure through segregated funds, which are mutual-fund-type products with repayment guarantees. The firm did not hedge the equity exposure and also appears to have gamed the capital requirements. When the 2008 equity crash arrived, the firm was forced into an emergency stock issue, a \$3 billion bank loan, a credit rating downgrade, class action lawsuits, and about a 75% drop in its share price.

The alleged causes included a strategic decision not to hedge equity exposure, an overly dominant chief executive officer, an underqualified board, thin-tailed models, the use of regulatory capital as a target rather than a baseline, poor understanding of finance by senior actuarial leadership, poor contract design, and aggressive marketing. The resulting losses included capital strain, liquidity risk, reputational damage, and legal liabilities.

A **risk taxonomy** is a classification system for organizing the major risks faced by an enterprise. The goal is not to force risks into rigidly separate categories. Many risks fit naturally into more than one class, and one adverse event often generates further risks elsewhere in the organization.

Nevertheless, a broad classification is useful for structuring analysis and governance.

One common distinction is between external and internal risks. External risks include financial market risk, political risk, regulatory risk, macroeconomic risk, and environmental risk. Internal risks include operational risk, strategic risk, and reputational risk. Even this distinction is imperfect, since some risks have both internal and external components.

## Financial Risks

**Stock market risk** is the risk arising from movements in equity market values. This may affect the asset side of the balance sheet, the liability side through embedded options, or the relationship between assets and liabilities. In some situations a firm can even be exposed to upward market movements, such as when liabilities increase as equity markets rise.

**Interest rate risk** is the risk arising from movements in borrowing and lending rates. Interest rate movements affect borrowing costs and the value of loans and bonds held by the firm. Recall that for a fixed-coupon bond with cash flows $C_t$ and yield $y$, the price (as a function of $y$) is

$$P(y) = \sum_{t=1}^{n} \frac{C_t}{(1+y)^t}.$$

Differentiating with respect to $y$ gives

$$\frac{\mathrm{d}P}{\mathrm{d}y} = -\sum_{t=1}^{n} \frac{tC_t}{(1+y)^{t+1}} < 0,$$

so bond prices fall when interest rates rise. Interest rate risk also includes **reinvestment risk**, since falling rates may prevent the firm from earning the return it had expected on reinvested cash flows. **Spread risk** is the risk arising from changes in the difference between short- and long-term rates or, more broadly, between different borrowing rates. **Asset-liability management risk** is a further aspect of interest rate risk and arises when the duration of assets is mismatched with the duration of liabilities.

**Exchange rate risk** (or **foreign exchange risk** or **currency risk**) is the risk arising from unexpected movements in relative currency values. **Transaction risk** arises from contracts denominated in different currencies. **Economic risk** refers to the broader exposure created by imports, exports, foreign investments, foreign subsidiaries, foreign competition, and supplier or customer relationships. **Translation risk** arises in financial reporting when foreign currency items must be converted into the reporting currency (it is not related to language translation).

**Market risk** is the risk of bond defaults or credit downgrades of bond issuers. This includes **bond default risk**, **downgrade risk**, **credit spread risk**, and **sovereign risk**. Credit ratings such as AAA, BB, C, and D summarize perceived credit quality, with higher ratings generally corresponding to higher bond values because investors demand lower compensation for default risk. A downgrade usually reduces the market value of the bond because the required yield increases.

**Systemic risk** is the risk arising from instability or failure in the financial system. This is often treated as a subset of market risk. It is particularly associated with domino or contagion effects in which the failure of one financial institution contributes to failures elsewhere in the system.

**Liquidity risk** is the risk that assets cannot be realized quickly enough to meet liabilities, without a severe adverse impact on price, even though the firm is technically solvent. It may be caused by illiquid assets, unanticipated cash flows, a reputational shock at the company level, or stressed market conditions. Note that liquidity is not the same as marketability! An asset may be marketable in normal times and still be difficult to sell quickly in the volume required. **Liquidation value risk** arises when unexpected timing or amounts of required cash force the firm to sell assets at depressed prices. **Affiliated company risk** arises when capital is tied up in a subsidiary and cannot be realized when needed. **Capital market risk** arises when the firm cannot raise money through the markets, for example by issuing equity or debt.

### External Risks

**Business cycle risk** is the risk arising from economy-wide fluctuations in the economic environment. These fluctuations are difficult to predict in timing and duration. They are closely connected with financial market risks, and economic troughs are generally harmful to firms, although some firms may benefit from them. For example, a recession may reduce demand for luxury goods but increase demand for discount retailers.

**Inflation risk** is the reduction in real returns caused by unexpected inflation. Long-term fixed cash flows are especially vulnerable. The risk is fundamentally a cash-flow mismatch problem. A firm that receives fixed nominal income while making inflation-linked payments is exposed to losses in real terms.

**Political risk** arises from adverse political changes in a firm's home country or in a foreign country where it has exposure. For example, a change in government or political instability can affect the firm's operations and profitability. Political instability can interrupt business operations, and a change in the political climate can directly harm a particular enterprise. Political risk is often connected with economic risk and exchange rate risk.

**Regulatory risk** is the risk that changes in law adversely affect the business. Examples include labour laws, currency transaction laws, financial reporting laws, licensing laws, regulatory capital laws, and tax laws.

**Environmental risk** consists of risks arising from environmental events and trends. Examples include changing patterns of natural disasters, soil degradation, drought, and the effects of climate change on markets, suppliers, and transportation costs. The term is also sometimes used for liability arising from environmental events, such as inadequate waste disposal or the failure of an oil tanker or drilling platform. In that second sense it overlaps with internal risk categories.

### Non-Financial and Operational Risks

The Basel Committee defines **operational risk** as the risk of loss arising from inadequate or failed internal processes, people, and systems, or from external events. Operational risk is often broken down into people risk, information technology risk, process risk, project risk, legal risk, pricing risk, and related categories.

**People risk** arises when employees or agents fail to follow the firm's rules, processes, or procedures. The failures may be deliberate, as in fraud, theft, sabotage, or bypassing systems to save time or increase remuneration. They may also be non-deliberate, as in error, carelessness, inadequate training, inadequate competence, poor resource allocation, or weak management controls. Human resource and management processes can also fail. This includes **key person risk** (where the loss of a key individual can significantly impact the firm's operations, revenue, or reputation), failure to recruit suitably qualified candidates, and inadequate processes for controlling fraud or expenses.

**IT risk** includes accidental loss or corruption of data, viruses, malware, denial-of-service attacks, unidentified program bugs, theft of data or intellectual property through security breaches, capacity failures, service outages, and failures by suppliers or consultants.

**Project risk** arises in project development and implementation. Common examples are scope risk, defect risk, schedule risk, and resource risk. **Scope risk** arises when the project scope is not well defined or controlled, leading to scope creep and cost overruns. **Defect risk** arises when the project delivers a product that does not meet quality standards or user requirements. **Schedule risk** arises when the project timeline is unrealistic or not adhered to, leading to delays. **Resource risk** arises when there are insufficient resources (such as personnel, budget, or equipment) to complete the project successfully.

**Legal risk** covers lawsuits against the enterprise, the risks of pursuing litigation, and legal defects in transactions. It includes liability arising from breaches of laws or regulations, negligence leading to civil liability, defective transactions, the costs of defending or opposing patents and copyright claims, and adverse changes in law.

**Pricing risk** is the risk of underpricing or undervaluing products. It is especially serious when there is a delay between contract formation and delivery of goods or services, or when currency risk is present. Model risk and adverse selection are important contributors to pricing risk.

**Process risk** is specific to the firm's business processes. Examples include health and safety failures caused by defective equipment, flawed procedures or human error, manufacturing and engineering failures caused by defective components, **model risk** (where a model is applied in a setting that violates its assumptions), **basis risk** (where a model is applied in a context where the assumptions are invalid), and failures in maintenance processes.

**Strategic risk** is the risk arising from strategic directives and strategic decision making. All strategic decisions involve risk, including the decision not to act. Some strategic decisions fail and cause investments to be lost. Strategic risk is especially significant when decisions are made using flawed processes, weak planning, poor data, or insufficient expertise.

**Reputational risk** is the risk that the enterprise is harmed through damage to its reputation. Reputational risk often follows another risk event. The BP Deepwater Horizon disaster from Example 2.1 is a standard example. The **Enron scandal** is another standard example: Enron used accounting fraud and off-balance-sheet entities to conceal losses and overstate performance, and when the truth emerged the firm collapsed and the reputation of Arthur Andersen, their accounting firm and auditor, was effectively destroyed along with it.

The **Tylenol case** is a contrasting example. After several people died from cyanide-laced Tylenol capsules in 1982, Johnson & Johnson rapidly recalled the product, cooperated with authorities, communicated openly with the public, and later reintroduced Tylenol with tamper-resistant packaging. The case is therefore widely viewed as an example in which good crisis management strengthened reputation rather than destroying it. As Warren Buffett observed, it can take twenty years to build a reputation and five minutes to destroy it.

**Example 2.3**

The Deepwater Horizon case from Example 2.1 involves environmental risk, operational risk, legal risk, reputational risk, and strategic risk. The operational component includes defective equipment, poor maintenance, inadequate training, and failures in process and safety culture. The environmental losses then generated legal liabilities and severe reputational damage.

The Manulife segregated fund case from Example 2.2 involves stock market risk, liquidity risk, strategic risk, pricing and model risk, regulatory risk, and reputational risk. The unhedged guarantee exposure made the firm highly vulnerable to falling equity markets, while weak governance and poor model assumptions amplified the losses. Capital strain, emergency financing, and market reaction then created liquidity and reputational consequences.

**Lecture 3 — Tuesday, September 29, 2015**

# 2. Risk Measures and Market Risk

## Risk Measures

For quantifiable risks, potential losses can often be modelled with a probability distribution. In simple settings one may use standard parametric families such as the normal, lognormal, Pareto, or compound Poisson distributions. In more complicated settings, such as operational risk losses, Monte Carlo methods are often used instead.

**Definition 3.1** Fix a time horizon and let $L$ be the loss random variable over that horizon. A **risk measure** is a functional $\rho$ that maps the loss $L$ to a real number $\rho(L)$ intended to quantify the risk associated with $L$.

Risk measures are used in premium calculations, benchmarking, regulatory capital, and economic capital. They are sometimes described as estimates of the maximum possible loss, but that interpretation is too strong, since most risk measures summarize only part of the loss distribution.

There are several standard principles for calculating insurance premiums. The simplest is the **net premium principle**, which sets the premium equal to the expected loss, that is,

$$\rho(X) = \mathbb{E}\left[X\right].$$

Three other standard premium principles are

$$\rho(X) = (1+\alpha)\mathbb{E}\left[X\right], \qquad \alpha \geq 0,$$

which is the **expected value principle**,

$$\rho(X) = \mathbb{E}[X] + \alpha\sqrt{\operatorname{Var}(X)}, \qquad \alpha \geq 0,$$

which is the **standard deviation principle**, and

$$\rho(X) = \mathbb{E}[X] + \alpha \operatorname{Var}(X), \qquad \alpha \geq 0,$$

which is the **variance principle**.

## Value-at-Risk and Conditional Tail Expectation

**Definition 3.2** For $\alpha \in (0,1)$, the $\alpha$**-Value-at-Risk** (or $\alpha$**-VaR**) of the loss random variable $L$ is

$$\operatorname{VaR}_\alpha(L) = Q_\alpha := \inf\{q : \mathbb{P}(L \leq q) \geq \alpha\}.$$

This is the loss level exceeded with probability $1-\alpha$, which is simply the generalized inverse (or quantile function) of the distribution function of $L$, which we all know and love. If $L$ is a continuous random variable, then

$$\mathbb{P}(L \leq Q_\alpha) = \alpha \qquad \text{and} \qquad Q_\alpha = F_L^{-1}(\alpha).$$

Usually $\alpha \geq 0.95$ is used (typically $\alpha \in \{0.95, 0.99, 0.999\}$), so that the VaR is a high quantile of the loss distribution. For example, $\operatorname{VaR}_{0.99}(L)$ is the loss level exceeded with probability 1%. The VaR is often interpreted as a worst-case loss, but that interpretation is too strong, since it only describes the tail of the distribution and ignores the magnitude of losses beyond that tail.

**Example 3.3** Suppose

$$\mathbb{P}(L = 100) = 0.005, \qquad \mathbb{P}(L = 50) = 0.045, \qquad \mathbb{P}(L = 10) = 0.10, \qquad \mathbb{P}(L = 0) = 0.85.$$

The distribution function therefore satisfies

$$F_L(0) = 0.85, \qquad F_L(10) = 0.95, \qquad F_L(50) = 0.995, \qquad F_L(100) = 1.$$

It follows that

$$\operatorname{VaR}_{0.99}(L) = 50, \qquad \operatorname{VaR}_{0.95}(L) = 10, \qquad \operatorname{VaR}_{0.80}(L) = 0.$$

**Definition 3.4** For $\alpha \in (0,1)$, the **conditional tail expectation** (**CTE**) (or **expected shortfall** or **Tail-VaR** or **CVaR**) of the loss random variable $L$ is defined as

$$\mathrm{CTE}_\alpha(L) := \frac{1}{1-\alpha}\int_\alpha^1 Q_u \,\mathrm{d}u.$$

In words, this is the average loss in the worst $1-\alpha$ fraction of the distribution.

**Proposition 3.5** If $L$ is continuous at $Q_\alpha$, then

$$\mathrm{CTE}_\alpha(L) = \mathbb{E}\left[L \mid L > Q_\alpha\right].$$

**Proof.** We prove the result under the slightly stronger assumption that $L$ is continuous. The weaker assumption of continuity at $Q_\alpha$ is sufficient to ensure that $\mathbb{P}(L > Q_\alpha) = 1-\alpha$, so the result still holds under that weaker assumption, but the proof uses Stieltjes integrals and is more complicated.

Let $U = F_L(L)$. Since $F_L$ is continuous, the probability integral transform implies that $U \sim \mathrm{Unif}(0,1)$. Also, continuity gives $F_L(Q_\alpha) = \alpha$ almost surely, so

$$L > Q_\alpha \qquad \Longleftrightarrow \qquad F_L(L) > \alpha \qquad \Longleftrightarrow \qquad U > \alpha.$$

Therefore,

$$\mathbb{E}\left[L \mid L > Q_\alpha\right] = \mathbb{E}\left[Q_U \mid U > \alpha\right] = \frac{1}{1-\alpha}\int_\alpha^1 Q_u \,\mathrm{d}u = \mathrm{CTE}_\alpha(L),$$

because $U$ is uniformly distributed on $(0,1)$. □

If there is mass at the quantile (i.e., there exists some $\varepsilon > 0$ such that $Q_\alpha = Q_{\alpha+\varepsilon}$), then the conditional expectation formula uses too little of the worst $1-\alpha$ tail and needs a correction:

**Proposition 3.6** Let

$$\beta' := \sup\{\beta : Q_\beta = Q_\alpha\}.$$

If $\beta' < 1$, then

$$\mathrm{CTE}_\alpha(L) = \frac{(\beta'-\alpha)Q_\alpha + (1-\beta')\mathbb{E}\left[L \mid L > Q_\alpha\right]}{1-\alpha},$$

**Proof.** Let $U \sim \text{Unif}(0,1)$ and let $X = Q_U$. Then $X$ has the same distribution as $L$ by the probability integral transform. Since the quantile function is nondecreasing and $\beta'$ is the largest level at which the quantile is still equal to $Q_\alpha$, we have

$$Q_u = Q_\alpha \quad \text{for } \alpha \le u \le \beta', \qquad \text{and} \qquad Q_u > Q_\alpha \quad \text{for } u > \beta'.$$

Therefore,

$$\begin{aligned}
\mathrm{CTE}_\alpha(L) &= \frac{1}{1-\alpha}\int_\alpha^1 Q_u \,\mathrm{d}u \\
&= \frac{1}{1-\alpha}\left(\int_\alpha^{\beta'} Q_u \,\mathrm{d}u + \int_{\beta'}^1 Q_u \,\mathrm{d}u\right) \\
&= \frac{1}{1-\alpha}\left((\beta'-\alpha)Q_\alpha + \int_{\beta'}^1 Q_u \,\mathrm{d}u\right).
\end{aligned}$$

Also, by the characterization above,

$$X > Q_\alpha \qquad \Longleftrightarrow \qquad U > \beta'.$$

Since $X$ and $L$ have the same distribution,

$$\mathbb{E}\left[L \mid L > Q_\alpha\right] = \mathbb{E}\left[X \mid X > Q_\alpha\right] = \mathbb{E}\left[Q_U \mid U > \beta'\right] = \frac{1}{1-\beta'}\int_{\beta'}^1 Q_u \,\mathrm{d}u,$$

because $U \sim \text{Unif}(0,1)$. Substituting this into the previous expression gives

$$\mathrm{CTE}_\alpha(L) = \frac{(\beta'-\alpha)Q_\alpha + (1-\beta')\mathbb{E}\left[L \mid L > Q_\alpha\right]}{1-\alpha}.$$

□

**Example 3.7** For the loss distribution in Example 3.3,

$$\mathrm{CTE}_{0.99}(L) = \frac{0.005 \cdot 50 + 0.005 \cdot 100}{0.01} = 75.$$

Also,

$$\mathrm{CTE}_{0.95}(L) = \mathbb{E}\left[L \mid L > 10\right] = \frac{100(0.005) + 50(0.045)}{0.05} = 55.$$

Finally,

$$\mathrm{CTE}_{0.80}(L) = \frac{0.05 \cdot 0 + 0.15 \cdot \mathbb{E}\left[L \mid L > 0\right]}{0.20} = \frac{100(0.005) + 50(0.045) + 10(0.10)}{0.20} = 18.75.$$

Several basic facts are immediate. One always has $\mathrm{CTE}_\alpha(L) \geq \mathrm{VaR}_\alpha(L)$, with equality only when the quantile is already the maximum possible loss. Also,

$$\mathrm{CTE}_0(L) = \mathbb{E}\left[L\right], \qquad \lim_{\alpha \downarrow 0} \mathrm{VaR}_\alpha(L) = \min L, \qquad \mathrm{VaR}_{0.5}(L) = \mathrm{median}(L).$$

VaR and CTE are available in closed form for a number of distributions.

**Example 3.8** If $L \sim \mathcal{N}(\mu, \sigma^2)$, then for $q \in (0,1)$,

$$\mathrm{VaR}_q(L) = \mu + \sigma\Phi^{-1}(q) \quad \text{and} \quad \mathrm{CTE}_q(L) = \mu + \sigma\frac{\varphi(\Phi^{-1}(q))}{1-q}.$$

**Proof.** Let $Z = (L-\mu)/\sigma$, so that $Z \sim N(0,1)$. If $z_q = \Phi^{-1}(q)$, then

$$\mathbb{P}(L \leq \mu + \sigma z_q) = \mathbb{P}(Z \leq z_q) = \Phi(z_q) = q,$$

so

$$\mathrm{VaR}_q(L) = \mu + \sigma\Phi^{-1}(q).$$

Since the normal distribution is continuous, Proposition 3.5 gives

$$\mathrm{CTE}_q(L) = \mathbb{E}\left[L \mid L > \mathrm{VaR}_q(L)\right] = \mu + \sigma\,\mathbb{E}\left[Z \mid Z > z_q\right].$$

Now

$$\mathbb{E}\left[Z \mid Z > z_q\right] = \frac{1}{1-q}\int_{z_q}^{\infty} z\varphi(z)\ \mathrm{d}z = \frac{1}{1-q}\int_{z_q}^{\infty} -\varphi'(z)\ \mathrm{d}z = \frac{\varphi(z_q)}{1-q},$$

because $\varphi'(z) = -z\varphi(z)$ and $\varphi(z) \to 0$ as $z \to \infty$. Therefore,

$$\mathrm{CTE}_q(L) = \mu + \sigma\frac{\varphi(\Phi^{-1}(q))}{1-q}.$$

□

**Example 3.9** If $L \sim \text{Pareto}(\gamma, \theta)$ with $\gamma > 1$, then for $q \in (0,1)$,

$$\text{VaR}_q(L) = \theta\left((1-q)^{-1/\gamma} - 1\right) \quad \text{and} \quad \text{CTE}_q(L) = \frac{\gamma\theta}{\gamma-1}(1-q)^{-1/\gamma} - \theta.$$

**Proof.** From the distribution function

$$F_L(x) = 1 - \left(\frac{\theta}{x+\theta}\right)^{\gamma},$$

the $q$-quantile is obtained by solving

$$q = 1 - \left(\frac{\theta}{x+\theta}\right)^{\gamma}.$$

This gives

$$\frac{\theta}{x+\theta} = (1-q)^{1/\gamma} \quad \text{and} \quad x+\theta = \theta(1-q)^{-1/\gamma},$$

and hence

$$\text{VaR}_q(L) = \theta\left((1-q)^{-1/\gamma} - 1\right).$$

Let $d = \text{VaR}_q(L)$. Since the Pareto distribution is continuous, Proposition 3.5 gives

$$\text{CTE}_q(L) = \mathbb{E}\left[L \mid L > d\right].$$

For $y \geq 0$,

$$\mathbb{P}(L-d > y \mid L > d) = \frac{\mathbb{P}(L > d+y)}{\mathbb{P}(L > d)} = \frac{\left(\frac{\theta}{d+y+\theta}\right)^{\gamma}}{\left(\frac{\theta}{d+\theta}\right)^{\gamma}} = \left(\frac{d+\theta}{d+y+\theta}\right)^{\gamma}.$$

Therefore $L-d \mid L > d \sim \text{Pareto}(\gamma, d+\theta)$ in the same parameterization. Its mean is $(d+\theta)/(\gamma-1)$, so

$$\text{CTE}_q(L) = d + \frac{d+\theta}{\gamma-1}.$$

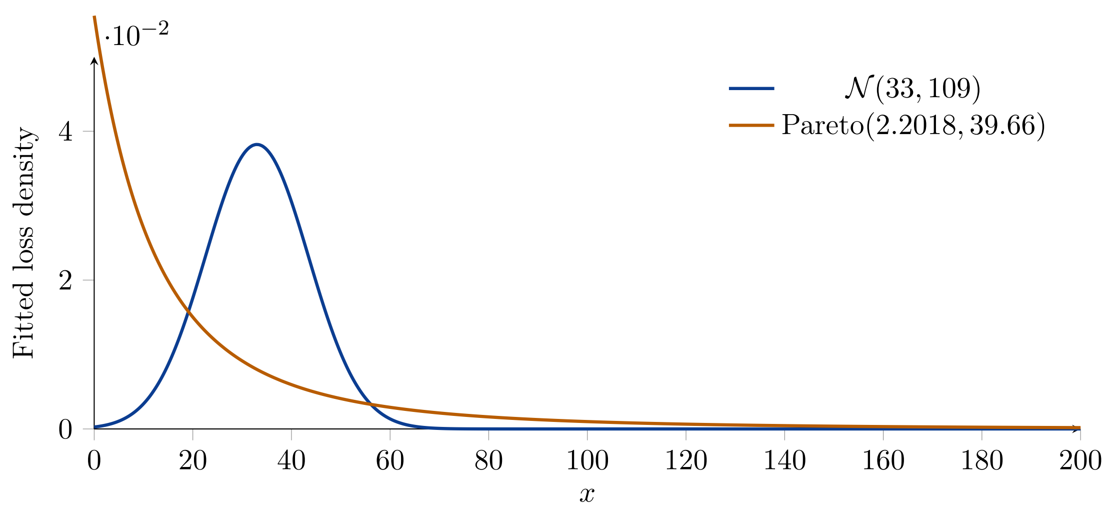
Since $d+\theta = \theta(1-q)^{-1/\gamma}$, this becomes

$$\text{CTE}_q(L) = \theta\left((1-q)^{-1/\gamma} - 1\right) + \frac{\theta(1-q)^{-1/\gamma}}{\gamma-1} = \frac{\gamma\theta}{\gamma-1}(1-q)^{-1/\gamma} - \theta.$$

□

The Pareto distribution is much heavier-tailed than the normal distribution, so its VaR and CTE are much larger at high quantiles. For example, suppose that we use empirical losses to estimate parameters by moment matching. If the sample mean is 33 and the sample variance is 109, then the fitted normal distribution has $\mu = 33$ and $\sigma^2 = 109$, while the fitted Pareto distribution has $\theta = 39.66$ and $\gamma = 2.2018$.

Matching only low-order moments does not determine tail risk. If $L \sim \mathcal{N}(33, 109)$, then $\mathrm{VaR}_{0.999}(L) \approx 370$ and $\mathrm{CTE}_{0.999}(L) \approx 400$, while if $L \sim \mathrm{Pareto}(2.2018, 39.66)$, then $\mathrm{VaR}_{0.999}(L) \approx 874$ and $\mathrm{CTE}_{0.999}(L) \approx 1635$. The fitted normal and Pareto distributions have similar central behaviour but very different extreme losses, so tail modelling matters far more than moment matching when high-quantile capital is the objective.




VaR is widely used in banking and in Solvency II, while CTE is especially common in insurance practice in Canada and the United States. Basel regulation has also moved away from VaR towards CTE. Which risk measure is better?

## Coherence

**Definition 3.10** A risk measure $\rho$ is called **coherent** if it satisfies the following four properties:

1. **Translation invariance**: For any constant $c$, $\rho(L + c) = \rho(L) + c$.
2. **Positive homogeneity**: For any $\lambda > 0$, $\rho(\lambda L) = \lambda\rho(L)$.
3. **Monotonicity**: If $\mathbb{P}(L_1 \leq L_2) = 1$, then $\rho(L_1) \leq \rho(L_2)$.
4. **Subadditivity**: For any $L_1$ and $L_2$, $\rho(L_1 + L_2) \leq \rho(L_1) + \rho(L_2)$.

Each property has a natural interpretation. Translation invariance means that adding a certain amount to the loss increases the risk by the same amount. Positive homogeneity means that scaling the loss by a positive factor scales the risk by the same factor. Monotonicity means that if one loss is always less than or equal to another, then its risk should be less than or equal to the other's risk. Subadditivity means that diversification should not increase risk: the risk of a combined loss should be no greater than the sum of the risks of the individual losses.

**Proposition 3.11** VaR satisfies translation invariance, positive homogeneity, and monotonicity.

**Proof.** Let
$$\mathrm{VaR}_\alpha(L) = \inf\{x : \mathbb{P}(L \leq x) \geq \alpha\}.$$
For translation invariance, let $c$ be a constant. Then

$$\begin{aligned}
\mathrm{VaR}_\alpha(L + c) &= \inf\{x : \mathbb{P}(L + c \leq x) \geq \alpha\} \\
&= \inf\{x : \mathbb{P}(L \leq x - c) \geq \alpha\} \\
&= c + \inf\{y : \mathbb{P}(L \leq y) \geq \alpha\} \\
&= c + \mathrm{VaR}_\alpha(L).
\end{aligned}$$

For positive homogeneity, let $\lambda > 0$. Then

$$\begin{aligned}
\mathrm{VaR}_\alpha(\lambda L) &= \inf\{x : \mathbb{P}(\lambda L \leq x) \geq \alpha\} \\
&= \inf\{x : \mathbb{P}(L \leq x/\lambda) \geq \alpha\}
\end{aligned}$$

$$= \lambda \inf\{y : \mathbb{P}(L \le y) \ge \alpha\}$$
$$= \lambda \operatorname{VaR}_\alpha(L).$$

For monotonicity, suppose $L_1 \le L_2$ almost surely. Then, for every $x$,

$$\{L_2 \le x\} \subseteq \{L_1 \le x\},$$

so

$$\mathbb{P}(L_1 \le x) \ge \mathbb{P}(L_2 \le x).$$

If $x$ belongs to the set

$$\{x : \mathbb{P}(L_2 \le x) \ge \alpha\},$$

then it also belongs to the set

$$\{x : \mathbb{P}(L_1 \le x) \ge \alpha\}.$$

Therefore the second set lies weakly to the left of the first, and hence

$$\operatorname{VaR}_\alpha(L_1) \le \operatorname{VaR}_\alpha(L_2).$$

$\square$

VaR is not subadditive in general, so it is not coherent in the sense of Definition 3.10. To see this, let $X$ and $Y$ be independent and identically distributed with

$$\mathbb{P}(X = 0) = 0.96, \qquad \mathbb{P}(X = 1) = 0.04,$$

and take $\alpha = 0.95$. Then

$$\operatorname{VaR}_{0.95}(X) = \operatorname{VaR}_{0.95}(Y) = 0,$$

because

$$\mathbb{P}(X \le 0) = 0.96 > 0.95.$$

However,

$$\mathbb{P}(X + Y = 0) = 0.96^2 = 0.9216,$$

so

$$\operatorname{VaR}_{0.95}(X + Y) = 1.$$

Thus

$$\operatorname{VaR}_{0.95}(X + Y) = 1 > 0 + 0 = \operatorname{VaR}_{0.95}(X) + \operatorname{VaR}_{0.95}(Y),$$

which violates subadditivity.

**Proposition 3.12** CTE is coherent in the sense of Definition 3.10.

**Proof.** For translation invariance, let $c$ be a constant. Since

$$\mathbb{P}(L + c \le x) = \mathbb{P}(L \le x - c),$$

the $u$-quantile of $L + c$ is $Q_u(L) + c$. Therefore

$$\mathrm{CTE}_\alpha(L + c) = \frac{1}{1-\alpha}\int_\alpha^1 Q_u(L + c)\ \mathrm{d}u = \frac{1}{1-\alpha}\int_\alpha^1 \big(Q_u(L) + c\big)\ \mathrm{d}u = \mathrm{CTE}_\alpha(L) + c.$$

For positive homogeneity, let $\lambda > 0$. Then the $u$-quantile of $\lambda L$ is $\lambda Q_u(L)$, so

$$\mathrm{CTE}_\alpha(\lambda L) = \frac{1}{1-\alpha}\int_\alpha^1 Q_u(\lambda L)\ \mathrm{d}u = \frac{1}{1-\alpha}\int_\alpha^1 \lambda Q_u(L)\ \mathrm{d}u = \lambda\,\mathrm{CTE}_\alpha(L).$$

For monotonicity, suppose $L_1 \le L_2$ almost surely. Then for every $x$,

$$\mathbb{P}(L_1 \le x) \ge \mathbb{P}(L_2 \le x),$$

so the quantiles satisfy $Q_u(L_1) \le Q_u(L_2)$ for every $u \in (0,1)$. Integrating gives

$$\mathrm{CTE}_\alpha(L_1) = \frac{1}{1-\alpha}\int_\alpha^1 Q_u(L_1)\ \mathrm{d}u \le \frac{1}{1-\alpha}\int_\alpha^1 Q_u(L_2)\ \mathrm{d}u = \mathrm{CTE}_\alpha(L_2).$$

It remains to prove subadditivity. Let $X$ and $Y$ be two losses, and write $Z = X + Y$. We use the an order statistic representation of CTE: if $X_1, \ldots, X_n$ is an independent sample from the distribution of $X$, with order statistics

$$X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)},$$

then

$$\mathrm{CTE}_\alpha(X) = \lim_{n\to\infty} \frac{1}{n(1-\alpha)} \sum_{j=n\alpha+1}^{n} X_{(j)},$$

which is essentially a strong law of large numbers for the tail of the distribution (however, the proof takes some care because the tail is shrinking as $n$ grows, and is beyond the scope of the course).

For simplicity, assume that $n\alpha$ is an integer; otherwise, we may replace $n\alpha$ by $\lfloor n\alpha \rfloor$, which does not affect the limit. The same representation holds for $Y$ and for $Z$. Now take an independent sample

$$(X_1, Y_1), \ldots, (X_n, Y_n)$$

from the joint distribution of $(X, Y)$, and define

$$Z_j = X_j + Y_j, \qquad j = 1, \ldots, n.$$

Let

$$Z_{(1)} \leq Z_{(2)} \leq \cdots \leq Z_{(n)}$$

be the order statistics of the sample $Z_1, \ldots, Z_n$. Let $A_n$ be the set of indices corresponding to the largest $n(1-\alpha)$ observations among the $Z_j$'s. Then

$$\sum_{j=n\alpha+1}^{n} Z_{(j)} = \sum_{i \in A_n} Z_i = \sum_{i \in A_n} X_i + \sum_{i \in A_n} Y_i.$$

But the sum of any $n(1-\alpha)$ observations from the $X_i$'s is at most the sum of the $n(1-\alpha)$ largest order statistics of the $X_i$'s. Hence

$$\sum_{i \in A_n} X_i \leq \sum_{j=n\alpha+1}^{n} X_{(j)}.$$

Similarly,

$$\sum_{i \in A_n} Y_i \leq \sum_{j=n\alpha+1}^{n} Y_{(j)}.$$

Therefore,

$$\sum_{j=n\alpha+1}^{n} Z_{(j)} \leq \sum_{j=n\alpha+1}^{n} X_{(j)} + \sum_{j=n\alpha+1}^{n} Y_{(j)}.$$

Dividing by $n(1-\alpha)$ and taking limits gives

$$\mathrm{CTE}_\alpha(Z) \leq \mathrm{CTE}_\alpha(X) + \mathrm{CTE}_\alpha(Y).$$

Since $Z = X + Y$, this is precisely

$$\mathrm{CTE}_\alpha(X+Y) \leq \mathrm{CTE}_\alpha(X) + \mathrm{CTE}_\alpha(Y).$$

Thus CTE is subadditive. Since all four properties hold, CTE is coherent. $\square$

But does subadditivity actually matter in practice, or is this just mathematical claptrap?

It can matter a great deal! In banking risk management, a VaR limit can sometimes be **gamed**: because VaR looks only at a single quantile and ignores the severity of losses beyond that quantile, a trader may be able to satisfy a VaR constraint while nevertheless shifting the portfolio toward positions with more severe tail losses. In that sense, a VaR restriction may actually incentivize riskier trading rather than safer trading. This concern is developed formally by Basak and Shapiro (2001), who show that risk constraints based on VaR can distort portfolio choice toward undesirable tail-risk exposures.

There is also a broader market-wide concern. If many institutions are managed using similar VaR limits, then those institutions may respond to changing market conditions in similar ways, for example by cutting positions simultaneously when measured risk rises. Such common reactions can amplify market stress, worsen liquidity conditions, and increase **systemic risk**. Danielsson et al. (2001) argue that this is one reason subadditivity is not merely an abstract mathematical nicety: the choice of risk measure can affect incentives, trading behaviour, and financial stability.

## Estimating VaR and CTE

Risk measures may be estimated empirically from historical data, by Monte Carlo simulation, or by fitting a parametric model to the loss distribution. Each approach introduces estimation uncertainty.

If a Monte Carlo sample of size $N$ is generated and arranged as

$$L_{(1)} \leq L_{(2)} \leq \cdots \leq L_{(N)},$$

then the empirical $\alpha$-VaR is estimated by one of the nearby order statistics, often a smoothed version of the empirical quantile. In the unsmoothed version one typically takes an order statistic near rank $N\alpha$, for example $L_{(N\alpha)}$ or $L_{(N\alpha+1)}$ when those indices are integers. The practical issue is that different conventions give slightly different estimates in finite samples, although they are asymptotically equivalent. For CTE, a natural estimator is

$$\widehat{\mathrm{CTE}}_\alpha = \frac{1}{N(1-\alpha)} \sum_{j=N\alpha+1}^{N} L_{(j)}.$$

If $\widehat{Q}_\alpha$ denotes the estimated VaR and $f$ is the density at the true quantile, then asymptotically

$$\operatorname{se}(\widehat{Q}_\alpha) = \frac{\sqrt{\alpha(1-\alpha)}}{\sqrt{N}\, f(Q_\alpha)}.$$

An approximate 95% confidence interval is therefore

$$\widehat{Q}_\alpha \pm 1.96\,\operatorname{se}(\widehat{Q}_\alpha).$$

This approximation can be unreliable in small samples or deep in the tail, and density estimation near the tail is often unstable.

A non-parametric alternative uses order statistics directly and avoids estimating the density at the quantile. Define indicators

$$I_j = \begin{cases} 1, & L_j \le Q_\alpha, \\ 0, & L_j > Q_\alpha, \end{cases} \qquad j = 1, \dots, N,$$

and let

$$M = \sum_{j=1}^{N} I_j.$$

Then $M \sim \operatorname{Bin}(N, \alpha)$. If integers $s < r$ are chosen so that

$$\mathbb{P}(s \le M < r) = \sum_{j=s}^{r-1} \binom{N}{j} \alpha^j (1-\alpha)^{N-j} = q,$$

then this event is equivalent to

$$L_{(s)} \le Q_\alpha < L_{(r)},$$

so

$$\mathbb{P}\big(L_{(s)} \le Q_\alpha < L_{(r)}\big) = q.$$

This gives an exact non-parametric confidence interval for the true VaR in terms of order statistics. A convenient symmetric approximation chooses

$$s = N\alpha - k \quad \text{and} \quad r = N\alpha + k.$$

Since

$$\mathbb{E}[M] = N\alpha \quad \text{and} \quad \operatorname{Var}(M) = N\alpha(1-\alpha),$$

the normal approximation to the binomial gives

$$k \approx z_{(1+q)/2}\sqrt{N\alpha(1-\alpha)}.$$

Hence an approximate symmetric $q$-confidence interval is

$$\big(L_{(N\alpha-k)},\, L_{(N\alpha+k)}\big), \text{where} k = z_{(1+q)/2}\sqrt{N\alpha(1-\alpha)},$$

provided that $N\alpha$ is large enough for the normal approximation to be reliable.

Repeated simulation is another practical approach. One generates several independent Monte Carlo samples, computes the VaR estimate from each, and then uses the sample standard deviation of those estimates as an estimated standard error.

For CTE, one can also study estimation uncertainty more explicitly. A useful starting point is the variance decomposition

$$\mathrm{Var}\big(\widehat{\mathrm{CTE}}_\alpha\big) = \mathbb{E}\left[\mathrm{Var}\big(\widehat{\mathrm{CTE}}_\alpha \mid \widehat{Q}_\alpha\big)\right] + \mathrm{Var}\left(\mathbb{E}\big[\widehat{\mathrm{CTE}}_\alpha \mid \widehat{Q}_\alpha\big]\right).$$

Carrying this calculation through yields the large-sample approximation

$$\mathrm{Var}\Big(\widehat{\mathrm{CTE}}_\alpha\Big) \approx \frac{\mathrm{Var}(L \mid L > Q_\alpha) + \alpha(\mathrm{CTE}_\alpha - Q_\alpha)^2}{N(1-\alpha)}.$$

The first term reflects the variability of losses inside the tail beyond the quantile. The second term reflects the additional uncertainty created by estimating the tail cutoff itself.

In practice, the unknown quantities are replaced by their empirical analogues. If

$$s^2 = \frac{1}{N(1-\alpha)-1}\sum_{j=N\alpha+1}^{N}\big(L_{(j)} - \widehat{\mathrm{CTE}}_\alpha\big)^2$$

is the sample variance of the losses lying in the estimated tail, then a practical estimator of the variance of $\widehat{\mathrm{CTE}}_\alpha$ is

$$\widehat{\mathrm{Var}}\big(\widehat{\mathrm{CTE}}_\alpha\big) = \frac{s^2 + \alpha\big(\widehat{\mathrm{CTE}}_\alpha - \widehat{Q}_\alpha\big)^2}{N(1-\alpha)}.$$

This makes it possible to report an estimated standard error for the CTE estimator in the same way as for VaR.

VaR and CTE are not the only risk measures used in practice. Variance and **semivariance** are also used as measures of risk. Semivariance focuses only on downside deviations below a benchmark such as the mean, so it is often more appealing than full variance when upside volatility is not viewed as harmful. These measures are useful in portfolio theory and in describing variability, especially downside variability in the case of semivariance. However, they do not directly provide capital requirements in the same way as tail-based measures such as VaR and CTE, and they do not target the far tail as directly as extreme-quantile measures do.

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### Lecture 4 — Tuesday, October 06, 2015

In banking, market risk is often measured over very short horizons such as one day or ten days. If a portfolio has value $V_t$ at time $t$ and the VaR horizon is $h$, the loss random variable is

$$L_t = V_t - V_{t+h}.$$

One may also work with the centered loss

$$L_t^{\text{mean}} = \mathbb{E}_t\left[V_{t+h}\right] - V_{t+h}.$$

For small horizons it is often assumed that $\mathbb{E}_t\left[V_{t+h}\right] \approx V_t$, so the two definitions lead to essentially the same VaR.

If the portfolio return over $(t, t+h)$ is

$$R_V = \frac{V_{t+h}}{V_t} - 1,$$

then

$$V_{t+h} = V_t(1 + R_V),$$

and hence

$$L_t = V_t - V_t(1 + R_V) = -R_V V_t.$$

Therefore,

$$Q_\alpha(L_t) = Q_\alpha(-R_V)V_t = -Q_{1-\alpha}(R_V)V_t.$$

## VaR in Banking

Assume a single stock price $S(t)$ has return

$$R = \frac{S(t+h) - S(t)}{S(t)},$$

and that

$$R \sim N(0, h\sigma^2).$$

The loss on one share over the horizon is

$$L = -RS(t).$$

Since $R$ is normal,

$$Q_{1-\alpha}(R) = z_{1-\alpha}\sigma\sqrt{h},$$

so

$$\mathrm{VaR}_\alpha(L) = -Q_{1-\alpha}(R)S(t) = -z_{1-\alpha}\sigma\sqrt{h}\,S(t) = z_\alpha\sigma\sqrt{h}\,S(t),$$

where $z_\alpha = \Phi^{-1}(\alpha)$.

Next, consider a portfolio with positions $a_j$ in stocks $j = 1, \ldots, n$. Let

$$R_j = \frac{S_j(t+h) - S_j(t)}{S_j(t)}$$

denote the return on stock $j$ over the VaR horizon. Assume the return vector $\boldsymbol{R} = (R_1, \ldots, R_n)$ is multivariate normal with

$$R_j \sim N(0, \sigma_j^2 h), \qquad \mathrm{Cov}(R_j, R_k) = \rho_{jk}\sigma_j\sigma_k h.$$

The portfolio value at time $t$ is

$$V(t) = \sum_{j=1}^{n} a_j S_j(t),$$

and the loss is

$$L_V = V(t) - V(t+h) = -\sum_{j=1}^{n} a_j\big(S_j(t+h) - S_j(t)\big) = -\sum_{j=1}^{n} a_j S_j(t) R_j.$$

Because $L_V$ is a linear combination of jointly normal random variables, it is normal. Its mean is zero, and its variance is

$$\mathrm{Var}(L_V) = h \sum_{j=1}^{n} \sum_{k=1}^{n} a_j S_j(t) a_k S_k(t) \rho_{jk} \sigma_j \sigma_k.$$

If

$$\sigma_L^2 = \sum_{j=1}^{n} \sum_{k=1}^{n} a_j S_j(t) a_k S_k(t) \rho_{jk} \sigma_j \sigma_k,$$

then

$$\mathrm{VaR}_\alpha(L_V) = z_\alpha \sigma_L \sqrt{h}.$$

The previous approach requires linear dependence on the underlying assets, which is not always appropriate, especially when the portfolio contains derivatives. The assumption of normality is also often unrealistic, and the estimation of many correlations can be difficult in any case. Instead, one can use a first-order Taylor expansion to approximate the loss, which leads to the **delta-VaR** approach.

## Delta-VaR

Suppose the portfolio value depends on time and on risk factors $X_1, \ldots, X_n$. A first-order Taylor expansion gives

$$V(t+h, X(t+h)) \approx V(t, X(t)) \; + \; \frac{\partial V}{\partial t} h \; + \; \sum_{j=1}^{n} \frac{\partial V}{\partial x_j} \big( X_j(t+h) - X_j(t) \big).$$

Hence the loss is approximately

$$L_V = V(t) - V(t+h) \approx -\frac{\partial V}{\partial t} h - \sum_{j=1}^{n} \frac{\partial V}{\partial x_j} \big( X_j(t+h) - X_j(t) \big).$$

If some factors are modelled in relative-return form and others in absolute-change form, this becomes

$$L_V \approx -\frac{\partial V}{\partial t} h - \sum_{j=1}^{m} X_j \frac{\partial V}{\partial x_j} R_j - \sum_{j=m+1}^{n} \frac{\partial V}{\partial x_j} \Delta X_j.$$

Under multivariate normality, the approximate loss is again normal and VaR can be computed in closed form.

## Historical Simulation and Monte Carlo Methods

**Historical simulation** takes observed past changes in the risk factors,

$$\Delta X_{t-k} = X(t-k+1) - X(t-k), \qquad k = 1, \dots, T,$$

and treats them as an empirical distribution with probabilities $1/T$. Revaluing the portfolio under each scenario gives losses

$$L_k = V(t, X_0) - V(t+h, X_0 + \Delta X_k), \qquad k = 1, \dots, T.$$

These losses are then treated in the same way as simulated losses from a Monte Carlo simulation.

Historical simulation is intuitive, simple to implement, and non-parametric, so it can reflect skewness and heavy tails. Its weaknesses are that it must be modified to account for seasonal trends, it is sensitive to **window effects** (e.g., when a large historical loss drops out of the time window used for the simulation), it takes a long time to account for structural changes in the market, and it is not clear how to incorporate new information that has not yet been observed in the historical data.

**Monte Carlo simulation** is more flexible. It can incorporate arbitrary portfolios, large numbers of scenarios, heavy tails, skewness, volatility clustering, and autocorrelation. Its drawbacks are that it is slower, harder to implement, and capable of hiding strong modelling assumptions inside a complicated simulation engine.

## Backtesting

To backtest a VaR model, define the **violation indicators**

$$I_t = \begin{cases} 1, & L_t > \mathrm{VaR}_{\alpha,t}, \\ 0, & L_t \le \mathrm{VaR}_{\alpha,t}\,. \end{cases}$$

Under a correctly calibrated model, the indicators should behave like independent Bernoulli random variables with success probability

$$p = 1 - \alpha.$$

If $T$ is the sample size and

$$T_1 = \sum_{t=1}^{T} I_t$$

is the number of violations, then the usual MLE

$$\widehat{p} = \frac{T_1}{T}$$

estimates $p$, with

$$\mathbb{E}\left[\widehat{p}\right] = p, \qquad \operatorname{Var}(\widehat{p}) = \frac{p(1-p)}{T}.$$

An asymptotic mean test is based on

$$M_T = \frac{\sqrt{T}(\widehat{p} - p)}{\sqrt{p(1-p)}} \approx N(0,1).$$

One can also use a likelihood ratio test. The Bernoulli likelihood is

$$L(\pi) = (1-\pi)^{T_0}\pi^{T_1},$$

where $T_0 = T - T_1$. The unrestricted maximizer is $\widehat{p} = T_1/T$, and the likelihood ratio statistic is

$$-2\log\left(\frac{L(p)}{L(\widehat{p})}\right) \approx \chi^2_1.$$

For small samples, simulation-based finite-sample critical values may be preferable.

The violations should also show no time pattern. For example, if a model ignores volatility clustering, then VaR violations will often occur in clusters. This can be checked using the autocorrelation function

$$\gamma_k = \operatorname{Corr}(I_t, I_{t-k}).$$

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**Lecture 5 — Tuesday, October 13, 2015**

## 3. Frequency-Severity Analysis

Many informal descriptions of quantitative risk say that

$$\text{Risk} = \text{Probability} \times \text{Severity}.$$

Under restrictive assumptions this leads to an expected loss, but expected value alone is only a limited summary of risk. A fuller analysis models both the number of loss events and their severities.

The basic assumption is that the number of losses and the severity of each loss can be analysed separately. This is most appropriate for idiosyncratic, diversifiable, insurable losses such as many operational and environmental losses. It is also usually assumed that frequency is independent of individual loss severity. This last assumption strikes us as being incredibly dumb, but we press onward.

## Aggregate Loss Models and Insurance

Let $S$ be the aggregate loss in a given period, let $N$ be the number of loss events, and let $Y_j$ be the severity of the $j$th event. Then

$$S = \begin{cases} \sum_{j=1}^{N} Y_j, & N \geq 1, \\ 0, & N = 0. \end{cases}$$

The standard assumptions are that $N$ is independent of the sequence $\{Y_j\}$ and that the severities are independent and identically distributed.

Two useful identities are

$$\mathbb{E}\left[S\right] = \mathbb{E}\left[\mathbb{E}\left[S \mid N\right]\right]$$

and

$$\mathrm{Var}(S) = \mathbb{E}\left[\mathrm{Var}(S \mid N)\right] + \mathrm{Var}(\mathbb{E}\left[S \mid N\right]).$$

Let $X$ be a random variable. Recall that if $M_X(z) = \mathbb{E}\left[e^{zX}\right]$ is the moment generating function, $P_X(z) = \mathbb{E}\left[z^X\right]$ is the probability generating function, and $C_X(z) = \log M_X(z)$ is the cumulant generating function, then

$$M_S(z) = \mathbb{E}\left[e^{zS}\right] = P_N(M_Y(z)).$$

## Frequency Models

The frequency random variable $N$ takes values in $\{0, 1, 2, \dots\}$, so common models are counting distributions such as the binomial, Poisson, and negative binomial. Let us recall several facts about these distributions.

If $N \sim \mathrm{Bin}(m, q)$, then

$$\mathbb{P}(N = k) = \binom{m}{k} q^k (1-q)^{m-k}$$

$$\begin{aligned}
\mathbb{E}[N] &= mq \\
\mathrm{Var}(N) &= mq(1-q) \\
\mathbb{E}[(N-\mathbb{E}[N])^3] &= mq(1-q)(1-2q) \\
M_N(z) &= \mathbb{E}[e^{zN}] = (1+q(e^z-1))^m \\
P_N(t) &= \mathbb{E}[t^N] = (1+q(t-1))^m.
\end{aligned}$$

If $N \sim \text{Poisson}(\lambda)$, then

$$\begin{aligned}
\mathbb{P}(N=k) &= e^{-\lambda}\frac{\lambda^k}{k!} \\
\mathbb{E}[N] &= \lambda \\
\mathrm{Var}(N) &= \lambda \\
\mathbb{E}[(N-\mathbb{E}[N])^3] &= \lambda \\
M_N(z) &= \mathbb{E}[e^{zN}] = \exp(\lambda(e^z-1)) \\
P_N(t) &= \mathbb{E}[t^N] = e^{\lambda(t-1)}.
\end{aligned}$$

The Poisson model is very convenient analytically, particularly when aggregating or disaggregating risks. If independent Poisson random variables satisfy

$$N_j \sim \text{Poi}(\lambda_j), \qquad j = 1, \ldots, r,$$

then

$$\sum_{j=1}^{r} N_j \sim \text{Poisson}\left(\sum_{j=1}^{r} \lambda_j\right).$$

This result is often used to simulate or aggregate risks from different sources. Conversely, if $N \sim \text{Poisson}(\lambda)$ and each event is independently assigned to category $j$ with probability $q_j$, then the number of events in category $j$ is

$$N_j \sim \text{Poi}(\lambda q_j),$$

which is useful for *disaggregating* risks.

If $N \sim \text{NB}(r, \beta)$, then

$$\begin{aligned}
\mathbb{P}(N=k) &= \frac{\Gamma(k+r)}{k!\Gamma(r)}\left(\frac{\beta}{1+\beta}\right)^k\left(\frac{1}{1+\beta}\right)^r \\
\mathbb{E}[N] &= r\beta \\
\mathrm{Var}(N) &= r\beta(1+\beta)
\end{aligned}$$

$$\mathbb{E}[(N-\mathbb{E}[N])^3] = r\beta(1+\beta)(1+2\beta)$$
$$M_N(z) = \mathbb{E}[e^{zN}] = \left(\frac{1}{1-\beta(e^z-1)}\right)^r$$
$$P_N(t) = \mathbb{E}[t^N] = \left(\frac{1}{1-\beta(t-1)}\right)^r.$$

The negative binomial distribution is flexible. For example, it allows overdispersion, meaning that the variance exceeds the mean.

## The $(a, b, 0)$ and $(a, b, 1)$ Distributions

**Definition 5.1** If $N$ is a discrete random variable whose probabilities $p_k = \mathbb{P}(N = k)$ satisfy the **Panjer recursion formula**

$$\frac{p_{k+1}}{p_k} = a + \frac{b}{k+1}, \qquad k = 0, 1, 2, \ldots,$$

then $N$ has an $(a, b, 0)$ **distribution**.

**Proposition 5.2** The Poisson, binomial, and negative binomial distributions are the only $(a, b, 0)$ distributions.

**Proof.** To begin with, observe that we must have $a + b = p_1/p_0 \geq 0$. If $a + b = 0$, then we get $p_n = \mathbb{1}_{n>0}$, which is not a valid probability mass function. Thus, we have $a + b > 0$. We analyze three cases:

*Case* $a = 0$. We have

$$p_1 = p_0 b, \qquad p_2 = p_1\frac{b}{2} = \frac{p_0 b^2}{2!}, \qquad p_3 = p_2\frac{b}{3} = p_0\frac{b^3}{3!},$$

and it follows by induction that $p_k = p_0 b^k/k!$, and for a valid probability distribution we must have $p_0 = e^{-b}$. Replacing $b$ with $\lambda$ gives the Poisson$(\lambda)$ distribution.

*Case* $a > 0$. Let $\beta = (a + b)/a$. Then

$$\frac{p_{k+1}}{p_k} = a\left(1 + \frac{\beta - 1}{k+1}\right) = a\frac{k+\beta}{k+1}.$$

Iterating the recursion gives

$$p_k = p_0 a^k \prod_{j=0}^{k-1} \frac{j+\beta}{j+1} = p_0 a^k \frac{\Gamma(k+\beta)}{\Gamma(\beta)\, k!}, \qquad k \geq 0.$$

Since $p_k \geq 0$ for all $k$, we must have $a \in (0,1)$; otherwise the probabilities do not sum to a finite total. Using the generalized binomial expansion,

$$\sum_{k=0}^{\infty} \frac{\Gamma(k+\beta)}{\Gamma(\beta)\, k!} a^k = (1-a)^{-\beta},$$

so normalization requires $p_0 = (1-a)^{\beta}$. Therefore

$$p_k = (1-a)^{\beta} \frac{\Gamma(k+\beta)}{\Gamma(\beta)\, k!} a^k.$$

This is the negative binomial distribution with parameters

$$r = \beta, \qquad a = \frac{\beta}{1+\beta}, \qquad b = a(\beta-1) = \frac{(r-1)\beta}{1+\beta}.$$

*Case* $a < 0$. Since $a + b > 0$, the factors $a + b/(k+1)$ are initially positive. However, because they decrease to $a < 0$ as $k \to \infty$, there is a first integer $m \geq 0$ for which

$$a + \frac{b}{m+1} = 0.$$

Then $p_{m+1} = 0$, and the recursion implies $p_k = 0$ for all $k > m+1$. Solving for $b$ gives $b = -a(m+1)$, so for $0 \leq k \leq m$ we have

$$\frac{p_{k+1}}{p_k} = a\left(1 - \frac{m}{k+1}\right) = (-a)\frac{m-k}{k+1}.$$

Let $q = -a/(1-a)$, so that $0 < q < 1$ and $-a = q/(1-q)$. Iterating the recursion gives

$$p_k = p_0 \binom{m}{k} \left(\frac{q}{1-q}\right)^k, \qquad 0 \leq k \leq m.$$

Summing from $k = 0$ to $m$ yields

$$1 = \sum_{k=0}^{m} p_k = p_0 \sum_{k=0}^{m} \binom{m}{k} \left(\frac{q}{1-q}\right)^k = p_0 \left(1 + \frac{q}{1-q}\right)^m = p_0 (1-q)^{-m},$$

so $p_0 = (1-q)^m$ and hence

$$p_k = \binom{m}{k} q^k (1-q)^{m-k}, \qquad 0 \le k \le m.$$

This is the $\mathrm{Bin}(m, q)$ distribution. $\square$

The proof above provides $a$ and $b$ for each of these three distributions:

$$\text{Poisson: } a = 0,\ b = \lambda,$$

$$\text{Binomial: } a = -\frac{q}{1-q},\ b = \frac{(m+1)q}{1-q},$$

$$\text{Negative binomial: } a = \frac{\beta}{1+\beta},\ b = \frac{(r-1)\beta}{1+\beta}.$$

**Definition 5.3** If $N$ is a discrete random variable whose probabilities $p_k = \mathbb{P}(N = k)$ satisfy

$$\frac{p_{k+1}}{p_k} = a + \frac{b}{k+1}, \qquad k = 1, 2, \ldots,$$

then $N$ has an $(a, b, 1)$ **distribution**.

The difference is that the recursion is imposed only for $k \ge 1$. This allows the zero-loss probability to be modified, including zero-truncated cases with $p_0 = 0$.

## Severity Models

Severity is the loss amount conditional on a loss event having occurred. It is usually modelled by a continuous non-negative distribution and is often positively skewed and heavy-tailed.

If $X \sim \mathrm{Exp}(\theta)$ with $\theta > 0$, then

$$\begin{aligned}
f(x) &= \frac{1}{\theta} e^{-x/\theta}, \qquad x > 0,\\
F(x) &= 1 - e^{-x/\theta},\\
\mathbb{E}[X] &= \theta,\\
\mathrm{Var}(X) &= \theta^2,
\end{aligned}$$

$$M_X(z) = \mathbb{E}[e^{zX}] = \frac{1}{1-\theta z}, \qquad z < 1/\theta.$$

The exponential distribution is very convienent and tractable. However, it has a constant hazard rate and relatively light tails, so it is often not a good fit for loss severity.

If $Y \sim \text{Exp}(\theta)$, then $X = Y^{1/\tau} \sim \text{Weibull}(\theta, \tau)$ with $\theta > 0$ and $\tau > 0$, with

$$\begin{aligned} f(x) &= \frac{\tau}{\theta}\left(\frac{x}{\theta}\right)^{\tau-1} e^{-(x/\theta)^\tau}, \qquad x > 0, \\ F(x) &= 1 - e^{-(x/\theta)^\tau}, \\ \mathbb{E}[X] &= \theta\Gamma\left(1 + \frac{1}{\tau}\right). \end{aligned}$$

It has heavier tails than the exponential distributionwhen $\tau < 1$ and lighter tails when $\tau > 1$.

If $X \sim \Gamma(\alpha, \beta)$ with $\alpha > 0$ and $\beta > 0$, then

$$\begin{aligned} f(x) &= \frac{x^{\alpha-1}e^{-x/\beta}}{\beta^\alpha\Gamma(\alpha)}, \qquad x > 0, \\ F(x) &= \frac{1}{\Gamma(\alpha)}\gamma\left(\alpha, \frac{x}{\beta}\right), \\ \mathbb{E}[X] &= \alpha\beta, \\ \text{Var}(X) &= \alpha\beta^2. \end{aligned}$$

The gamma distribution is flexible and tractable, but it has relatively light tails.

If $X \sim \text{Pareto}(\alpha, \theta)$ with $\alpha > 0$ and $\theta > 0$, then

$$\begin{aligned} f(x) &= \frac{\alpha\theta^\alpha}{(x+\theta)^{\alpha+1}}, \qquad x > 0, \\ F(x) &= 1 - \left(\frac{\theta}{x+\theta}\right)^\alpha, \\ \mathbb{E}[X] &= \frac{\theta}{\alpha-1}, \qquad \alpha > 1, \\ \text{Var}(X) &= \frac{\alpha\theta^2}{(\alpha-1)^2(\alpha-2)}, \qquad \alpha > 2. \end{aligned}$$

The Pareto distribution is a standard model for heavy-tailed losses. It has a power-law tail, so the tail probability decays like $x^{-\alpha}$ as $x \to \infty$. The mean exists only for $\alpha > 1$, and the variance exists only for $\alpha > 2$. The smaller $\alpha$ is, the heavier the tail.

If $X \sim \text{Lognormal}(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$ and $\sigma > 0$, then

$$\begin{aligned} f(x) &= \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log x - \mu)^2}{2\sigma^2}\right), \qquad x > 0, \\ F(x) &= \Phi\left(\frac{\log x - \mu}{\sigma}\right), \\ \mathbb{E}\left[X^k\right] &= \exp\left(k\mu + \frac{k^2\sigma^2}{2}\right), \\ \operatorname{Var}(X) &= \exp\bigl(2\mu + \sigma^2\bigr)\left(e^{\sigma^2} - 1\right). \end{aligned}$$

Lognormal distributions are positively skewed and heavy-tailed, though not as heavy-tailed as Pareto distributions.

If $Y_1, \ldots, Y_N$ are independent and identically distributed with distribution $F_Y$, then the aggregate loss $S$ has a **compound distribution** with frequency distribution $F_N$ and severity distribution $F_Y$. The moment generating function of $S$ is

$$M_S(z) = \mathbb{E}\left[e^{zS}\right] = P_N(M_Y(z)).$$

If $\mathbb{E}[Y] = \mu_Y$ and $\operatorname{Var}(Y) = \sigma_Y^2$, then

$$\mathbb{E}[S \mid N] = N\mu_Y \qquad \text{and} \qquad \operatorname{Var}(S \mid N) = N\sigma_Y^2.$$

Applying iterated expectation and variance yields

$$\mathbb{E}[S] = \mu_Y \mathbb{E}[N]$$

and

$$\operatorname{Var}(S) = \sigma_Y^2 \mathbb{E}[N] + \mu_Y^2 \operatorname{Var}(N).$$

In particular, if $N$ is Poisson with parameter $\lambda$, then $S$ has a **compound Poisson distribution** with rate $\lambda$, often written $S \sim \text{CoPoi}(\lambda, F_Y)$. In that case

$$\mathbb{E}[S] = \lambda\mathbb{E}[Y], \qquad \operatorname{Var}(S) = \lambda\mathbb{E}\left[Y^2\right], \qquad \mathbb{E}\left[(S - \mathbb{E}[S])^3\right] = \lambda\mathbb{E}\left[Y^3\right].$$

For large $\lambda$, the central limit theorem suggests the approximation

$$S \approx N(\lambda\mu_Y, \lambda\mathbb{E}\left[Y^2\right]).$$

## Estimating the Aggregate Loss Distribution

We consider three main approaches: approximation by another distribution, **Panjer recursion**, and the **discrete Fourier transform**.

We use the following running example:

**Example 5.4** Suppose that the frequency and severity distributions are

$$S \sim \text{CoPoi}(\lambda = 50, F) \quad \text{and} \quad Y \sim \text{Lognormal}(5, 1).$$

The severity moments are

$$\mathbb{E}[Y] = e^{5+1/2} = e^{11/2} \approx 244.692,$$

$$\mathbb{E}\left[Y^2\right] = e^{2\cdot 5+2} = e^{12} \approx 162754.791, \qquad \mathbb{E}\left[Y^3\right] = e^{3\cdot 5+9/2} = e^{39/2} \approx 294267566.042.$$

Hence

$$\mathbb{E}[S] = 50\mathbb{E}[Y] = 50e^{11/2} \approx 12234.6,$$

$$\text{Var}(S) = 50\mathbb{E}\left[Y^2\right] = 50e^{12} \approx 8137739.57, \qquad \text{sd}(S) = \sqrt{8137739.57} \approx 2852.67,$$

and

$$\mathbb{E}\left[(S - \mathbb{E}[S])^3\right] = 50\mathbb{E}\left[Y^3\right] = 50e^{39/2} \approx 1.47134 \times 10^{10},$$

so the distribution is positively skewed.

Approximating $S$ by a normal distribution with the same mean and variance works better when $\mathbb{E}[N]$ is large and we are interested in the center of the distribution, rather than the tails. The normal distribution has zero skewness, so it cannot capture the positive skewness of $S$. The normal approximation is therefore expected to underestimate tail probabilities. In this example, the normal approximation gives

$$S^* \sim \mathcal{N}(\mathbb{E}[S], \text{Var}(S)),$$

so

$$\mathbb{P}(S \le s) \approx \Phi\left(\frac{s - \mathbb{E}[S]}{\sqrt{\text{Var}(S)}}\right).$$

For $s = 20000$,

$$z = \frac{20000 - 12234.6}{2852.67} \approx 2.722,$$

and therefore

$$\mathbb{P}(S > 20000) \approx 1 - \Phi(2.722) \approx 0.00324.$$

For $s = 25000$,
$$z = \frac{25000 - 12234.6}{2852.67} \approx 4.475,$$
and therefore
$$\mathbb{P}(S > 25000) \approx 1 - \Phi(4.475) \approx 0.00000382.$$
These values are much too small in the tail because the normal approximation ignores positive skewness.

A **translated gamma approximation** writes
$$S \approx W + k, \qquad W \sim \Gamma(\alpha, \beta),$$
and matches the first three central moments:
$$\alpha\beta + k = \mathbb{E}\left[S\right], \qquad \alpha\beta^2 = \mathrm{Var}(S), \qquad 2\alpha\beta^3 = \mathbb{E}\left[(S - \mathbb{E}\left[S\right])^3\right].$$
From the last two equations,
$$\beta = \frac{\mathbb{E}\left[(S - \mathbb{E}\left[S\right])^3\right]}{2\,\mathrm{Var}(S)} \approx \frac{1.47134 \times 10^{10}}{2(8137739.57)} \approx 904.021,$$
and then
$$\alpha = \frac{\mathrm{Var}(S)}{\beta^2} \approx \frac{8137739.57}{(904.021)^2} \approx 9.957.$$
Finally,
$$k = \mathbb{E}\left[S\right] - \alpha\beta \approx 12234.6 - (9.957)(904.021) \approx 3232.88.$$
Therefore
$$\mathbb{P}(S > s) \approx \mathbb{P}(W > s - k).$$
For $s = 20000$,
$$\mathbb{P}(S > 20000) \approx \mathbb{P}(W > 20000 - 3232.88) = \mathbb{P}(W > 16767.12) \approx 0.0110.$$
For $s = 25000$,
$$\mathbb{P}(S > 25000) \approx \mathbb{P}(W > 25000 - 3232.88) = \mathbb{P}(W > 21767.12) \approx 0.00039.$$
These values are much closer to the more accurate numerical methods.

The normal approximation is justified by the central limit theorem, but its tails may be too thin for Poisson and negative binomial frequency models (unless $\mathbb{E}[N]$ is large). The translated gamma approximation is more ad hoc; it is not justified by a limit theorem, but it can capture skewness and heavy tails. It often works quite well for the compound Poisson model, but it can perform poorly for other frequency models, such as the negative binomial. It has a positive skewness of $2/\sqrt{\alpha}$, so it cannot capture very heavy tails. However, left tail of the translated gamma approximation is also often too thin, which can lead to underestimation of the zero-loss probability.

### Panjer Recursion

Suppose the severity distribution is discrete with $f_k = \mathbb{P}(Y = k)$ and the aggregate loss probabilities are $g_s = \mathbb{P}(S = s)$. If $N$ is an $(a, b, 0)$ frequency distribution, then

$$g_0 = P_N(f_0)$$

and, for $s \geq 1$,

$$g_s = \frac{1}{1 - af_0} \sum_{j=1}^{s} \left( a + \frac{bj}{s} \right) f_j g_{s-j}.$$

For a compound Poisson model this reduces to

$$g_0 = \exp\big(-\lambda(1 - f_0)\big)$$

and

$$g_s = \frac{\lambda}{s} \sum_{j=1}^{s} j f_j g_{s-j}, \qquad s = 1, 2, \ldots.$$

For $(a, b, 1)$ distributions, the analogous recursion is

$$g_s = \frac{1}{1 - af_0} \left[ \big(p_1 - (a + b)p_0\big) f_s + \sum_{j=1}^{s} \left( a + \frac{bj}{s} \right) f_j g_{s-j} \right].$$

The recursive nature of these formulas makes them computationally efficient. In practice, we would stop the recursion at the maximum claim size (if there is one). Computing $g_s$ is then $O(s^2)$, which is much faster than the $O(2^s)$ complexity of a naive convolution approach. Panjer recursion is exact for discrete severity distributions, but it can be used with discretized continuous severity distributions.

In the Poisson-lognormal running example, take $f_0 = 0$ and $g_0 = e^{-\lambda} = e^{-50}$. If we discretize the severity distribution in units from $f_1$ to $f_{2^{16}}$, then Panjer recursion gives

$$\mathbb{P}(S > 20000) \approx 0.0111, \qquad \mathbb{P}(S > 25000) \approx 0.00070.$$

The computation takes mere seconds in R.

## Discrete Fourier Transforms

The characteristic function approach uses

$$\phi_S(t) = \mathbb{E}\left[e^{itS}\right] = P_N(\phi_Y(t)).$$

After discretizing the severity distribution onto a grid $\{0, 1, \ldots, M-1\}$ with probabilities $f_0, f_1, \ldots, f_{M-1}$, define the discrete Fourier transform by

$$\widehat{f}(s) = \sum_{k=0}^{M-1} f_k e^{2\pi iks/M}, \qquad s = 0, 1, \ldots, M-1.$$

The aggregate loss transform is then obtained pointwise from the probability generating function of the frequency distribution:

$$\widehat{g}(s) = P_N\big(\widehat{f}(s)\big), \qquad s = 0, 1, \ldots, M-1,$$

where $\widehat{g}$ is the discrete transform of the aggregate loss probabilities $g_0, g_1, \ldots, g_{M-1}$. Finally, the aggregate loss probabilities are recovered from the inverse transform

$$g_s = \frac{1}{M} \sum_{k=0}^{M-1} \widehat{g}(k) e^{-2\pi isk/M}, \qquad s = 0, 1, \ldots, M-1.$$

Thus the Fourier transform method reduces the compound distribution problem to three steps: transform the severity distribution, apply the frequency probability generating function pointwise, and invert. The **fast Fourier transform** makes these computations highly efficient.

**Algorithm 1: FFT Method for Compound Distributions**

**Data:** Discretized severity probabilities $f_0, f_1, \ldots, f_{M-1}$ on the grid $\{0, 1, \ldots, M-1\}$, where one typically chooses $M = 2^m$.

**Result:** Aggregate loss probabilities $g_0, g_1, \ldots, g_{M-1}$ and cumulative probabilities $G(s) = \sum_{j=0}^{s} g_j$.

1. Form the vector $\boldsymbol{f} = (f_0, f_1, \ldots, f_{M-1})$ of discretized severity probabilities.
2. Compute the discrete Fourier transform $\widehat{f}(t)$ for $t = 0, 1, \ldots, M-1$ using the FFT.
3. Compute $\widehat{g}(t) = P_N\big(\widehat{f}(t)\big)$ pointwise for $t = 0, 1, \ldots, M-1$.
4. Apply the inverse FFT to $\widehat{g}$ to recover the aggregate loss probabilities $g_s$ for $s = 0, 1, \ldots, M-1$.
5. Compute the cumulative sums $G(s) = \sum_{j=0}^{s} g_j$.

For a compound Poisson model, the corresponding R implementation is concise:

```r
M <- length(f)
fhat <- fft(f, inverse = FALSE)
ghat <- exp(lambda * (fhat - 1))
g <- fft(ghat, inverse = TRUE) / M
g <- Re(g)
G <- cumsum(g)
```

The main numerical appeal is speed. Direct convolution is slow, whereas the FFT computes these transforms in order $M \log M$ operations. In practice one chooses $M$ large enough that the support of the discretized aggregate distribution is well captured. Otherwise, wrap-around effects can contaminate the tail probabilities. A common improvement is **exponential tilting** before inversion, which stabilizes the far tail; this involves multiplying the severity probabilities by $e^{\theta k}$ for some $\theta > 0$ before applying the FFT, and then adjusting the aggregate loss probabilities by $e^{-\theta s}$ after inversion.

For Example 5.4, the FFT gives

$$\mathbb{P}(S > 20000) \approx 0.0108, \qquad \mathbb{P}(S > 25000) \approx 0.00068.$$

FFT is faster than Panjer recursion but may produce poor far-tail behaviour unless improvements such as exponential tilting are used.

## Aggregate Loss Risk Measures

Under the normal approximation,

$$\mathrm{VaR}_\alpha(S) \approx \mathbb{E}\left[S\right] + z_\alpha \sqrt{\mathrm{Var}(S)}$$

and

$$\mathrm{CTE}_\alpha(S) \approx \mathbb{E}\left[S\right] \; + \; \sqrt{\mathrm{Var}(S)}\frac{\phi(z_\alpha)}{1-\alpha}.$$

For Example 5.4, this gives

$$\mathrm{VaR}_{0.95}(S) \approx 16927, \qquad \mathrm{CTE}_{0.95}(S) \approx 18117.$$

Under the translated gamma approximation,

$$\mathrm{VaR}_q(S) \approx F_W^{-1}(q) + k$$

and

$$\mathrm{CTE}_q(S) \approx \mathrm{CTE}_q(W) + k,$$

where

$$\mathrm{CTE}_q(W) = \left(\frac{1 - F^*(Q_q)}{1-q}\right)\mathbb{E}[W]$$

and $F^*$ is the distribution function of the size-biased distribution of $W$, which is $\Gamma(\alpha+1, \beta)$ in this case. The quantile function of the gamma distribution can be computed using numerical root-finding methods, and the CTE can be computed from the gamma distribution function and mean. For Example 5.4,

$$\mathrm{VaR}_{0.95}(S) \approx 17382, \qquad \mathrm{CTE}_{0.95}(S) \approx 19096.$$

For Panjer recursion and the FFT, VaR is approximately the largest $s$ such that $G(s) \le \alpha$, and CTE is approximated as

$$\mathrm{CTE}_\alpha(S) \approx \frac{1}{1-\alpha} \sum_{s=Q_\alpha}^{M} s\, g_s$$

for large $M$. For the example in (5.4), we have

$$\mathrm{VaR}_{0.95}(S) \approx 17342 \text{ and } 17286,$$

$$\mathrm{VaR}_{0.99}(S) \approx 20192 \text{ and } 20133,$$

$$\mathrm{CTE}_{0.95}(S) \approx 19101 \text{ and } 19079,$$

$$\mathrm{CTE}_{0.99}(S) \approx 21881 \text{ and } 21958.$$

The normal approximation understates the tail risk in this example because the aggregate distribution is positively skewed. The translated gamma approximation is materially better, and the numerical methods based on Panjer recursion and the FFT agree very closely. In practice, when Panjer and the FFT agree to this extent, one gains confidence that the discretization error is small.

## Partial Insurance

Frequency-severity analysis is especially natural for insurable risks. A risk is typically insurable when the losses are reasonably quantifiable, random from the policyholder's perspective, and sufficiently diversifiable for the insurer. Under **full insurance**, the insurer indemnifies the policyholder for the entire loss. Under **partial insurance**, the policyholder retains part of the risk. This reduces the moral hazard risk for the insurer and can make insurance more affordable for the policyholder; the expenses of small claims are also reduced for the insurer.

Partial insurance can take various forms, including **proportional insurance** (or **coinsurance**) and **insurance with a deductible** (or **excess loss insurance** or **stop loss insurance**). In proportional insurance, with retention rate $a \in (0,1)$, the policyholder retains $aS$ and the insurer pays $(1-a)S$. Under a **deductible** $D$, the policyholder pays the first $D$ dollars of each loss and the insurer pays the excess, possibly subject to a maximum claim amount.

**Example 5.5** A firm is interested in insuring losses from employee liability. Losses, measured in thousands of dollars, follow a compound Poisson model with frequency parameter $\lambda = 10$ and severity distribution

$$Y_j \sim \mathrm{Pareto}(\alpha = 11, \theta = 1000).$$

The insurer uses the standard deviation premium principle,

$$\mathrm{Premium} = \mathbb{E}[\mathrm{Claim}] + \xi \ \mathrm{sd}(\mathrm{Claim}).$$

Full insurance is offered with loading $\xi = 0.5$. A 90% proportional insurance contract is also offered, under which the insurer covers 90% of each loss and the firm retains the remaining 10%; for this contract, $\xi = 0.4$. Finally, the insurer offers a deductible contract with deductible $D = 10$ and no upper limit on each loss, again with $\xi = 0.4$.

If the firm does not fully insure its losses, the regulator requires it to hold capital equal to the 95% CTE of the uninsured losses. This capital is treated as a loan from the shareholders, with an annual service charge equal to 10% of that CTE. Ignoring investment returns, the firm's expected annual outgo is therefore

$$\text{Premium} + \mathbb{E}[U] + 0.10\ \text{CTE}_{0.95}(U),$$

where $U$ denotes uninsured losses.

Compute the expected annual outgo under each of the following four options:

1. full insurance,
2. no insurance,
3. 90% proportional insurance,
4. deductible insurance with $D = 10$.

Use the FFT for the aggregate loss distribution where required.

**Solution.** We first compute the moments of the claim severity distribution. For a Pareto$(\alpha, \theta)$ severity with $\alpha = 11$ and $\theta = 1000$,

$$\mathbb{E}[Y_j] = \frac{\theta}{\alpha - 1} = \frac{1000}{10} = 100,$$

and

$$\mathbb{E}[Y_j^2] = \frac{2\theta^2}{(\alpha-1)(\alpha-2)} = \frac{2 \cdot 1000^2}{10 \cdot 9} = 22222.2.$$

Since $S$ is compound Poisson with parameter $\lambda = 10$,

$$\mathbb{E}[S] = \lambda \mathbb{E}[Y_j] = 10 \cdot 100 = 1000,$$

and

$$\text{Var}(S) = \lambda \mathbb{E}[Y_j^2] = 10 \cdot 22222.2 = 222222,$$

so

$$\text{sd}(S) = \sqrt{222222} \approx 471.4.$$

We now consider the four insurance choices.

*(i) Full insurance.* In this case the insurer covers the entire loss, so the premium is

$$\mathbb{E}[S] + 0.5\,\mathrm{sd}(S) = 1000 + 0.5(471.4) = 1235.7.$$

There is no uninsured loss, so $U = 0$ and the expected annual outgo is simply

$$1235.7.$$

*(ii) No insurance.* Here there is no premium, and the uninsured loss is $U = S$. Thus the expected uninsured loss is

$$\mathbb{E}[U] = \mathbb{E}[S] = 1000.$$

Using the FFT approach described above, the aggregate loss distribution gives

$$\mathrm{VaR}_{0.95}(S) = 1858, \qquad \mathrm{CTE}_{0.95}(S) = 2167.$$

Therefore the expected annual outgo is

$$1000 + 0.10(2167) = 1216.7.$$

*(iii)* 90% *proportional insurance.* The uninsured loss is $U = 0.1S$ and the insured loss is $S^I = 0.9S$. The premium is therefore

$$\mathbb{E}[S^I] + 0.4\,\mathrm{sd}(S^I) = 0.9\mathbb{E}[S] + 0.4(0.9\,\mathrm{sd}(S)) = 900 + 0.36(471.4) = 1069.7.$$

The expected uninsured loss is

$$\mathbb{E}[U] = 0.1\mathbb{E}[S] = 100.$$

Since CTE is positively homogeneous,

$$\mathrm{CTE}_{0.95}(U) = 0.1\,\mathrm{CTE}_{0.95}(S) = 0.1(2167) = 216.7.$$

Hence the capital service charge is

$$0.10(216.7) = 21.7,$$

and the expected annual outgo is

$$1069.7 + 100 + 21.7 = 1191.4.$$

*(iv) Deductible insurance with $D = 10$.* Let $Y_j^I$ and $Y_j^U$ denote the insured and uninsured portions

of a single loss. Then

$$Y_j^U = \min\{Y_j, D\}, \qquad Y_j^I = (Y_j - D)_+.$$

To compute the premium, we first analyze the insured severity. Let

$$Z = Y_j - D \mid Y_j > D.$$

Then, for $z \geq 0$,

$$\mathbb{P}(Z > z) = \mathbb{P}(Y_j - D > z \mid Y_j > D) = \frac{\mathbb{P}(Y_j > z + D)}{\mathbb{P}(Y_j > D)} = \left(\frac{\theta + D}{\theta + D + z}\right)^{\alpha},$$

which is the survival function of a Pareto distribution with parameters $\alpha = 11$ and $\theta + D = 1010$. Therefore

$$\mathbb{E}[Y_j^I] = \mathbb{E}[Y_j - D \mid Y_j > D]\mathbb{P}(Y_j > D) = \frac{\theta + D}{\alpha - 1}\left(\frac{\theta}{\theta + D}\right)^{\alpha} \approx 90.529,$$

and similarly

$$\mathbb{E}\big[(Y_j^I)^2\big] = \mathbb{E}\big[(Y_j - D)^2 \mid Y_j > D\big]\mathbb{P}(Y_j > D) = \frac{2(\theta + D)^2}{(\alpha - 1)(\alpha - 2)}\left(\frac{\theta}{\theta + D}\right)^{\alpha} \approx 20318.60.$$

Hence, for the aggregate insured loss $S^I$,

$$\mathbb{E}[S^I] = \lambda\mathbb{E}[Y_j^I] = 905.29, \qquad \mathrm{sd}(S^I) = \sqrt{\lambda\mathbb{E}[(Y_j^I)^2]} = \sqrt{203186} \approx 450.76.$$

So the premium is

$$905.29 + 0.4(450.76) = 1085.59.$$

The expected uninsured loss is

$$\mathbb{E}[U] = \mathbb{E}[S] - \mathbb{E}[S^I] = 1000 - 905.29 = 94.71.$$

Using the FFT approach for the uninsured aggregate loss gives

$$\mathrm{CTE}_{0.95}(U) = 149.7.$$

Therefore the capital service charge is

$$0.10(149.7) = 15.0,$$

and the expected annual outgo is

$$1085.6 + 94.7 + 15.0 = 1195.3.$$

Thus the expected annual outgo under the four options is

$$\text{Full insurance } = 1235.7, \qquad \text{No insurance } = 1216.7,$$

$$90\% \text{ proportional insurance } = 1191.4, \qquad \text{Deductible insurance } = 1195.3.$$

The deductible and proportional contracts have very similar expected costs, but the deductible leaves the firm with less tail risk because small losses are eliminated rather than merely scaled down. □

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### Lecture 6 — Tuesday, October 20, 2015

**Extreme value theory** studies rare, extreme outcomes. The focus here is on large (positive) losses. Two main approaches are used: modelling maxima and modelling exceedances over a high threshold.

# 4. Extremes, Dependence, and Financial Return Models

## Maximum Losses

Let $X_1, \ldots, X_n$ be independent and identically distributed with distribution function $F$, and let

$$M_n = \max\{X_1, \ldots, X_n\}.$$

Then

$$\mathbb{P}(M_n \leq m) = \mathbb{P}(X_1 \leq m, \ldots, X_n \leq m) = [F(m)]^n.$$

As $n \to \infty$, this distribution either degenerates to 0 or 1, so some kind of normalization is required, which is analogous to the central limit theorem for sums or averages. In that setting one first centers and rescales before taking a limit, for example

$$\mathbb{P}\left(\frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}} \leq x\right) \to \Phi(x).$$

Maxima also need centering and scaling. Without such normalization, the limit is typically trivial.

If the number of losses is random, say $N$, then conditioning on $N$ gives

$$\mathbb{P}(M_N \le x) = \sum_{n=0}^{\infty} \mathbb{P}(M_N \le x \mid N = n)\mathbb{P}(N = n) = \sum_{n=0}^{\infty} [F(x)]^n \mathbb{P}(N = n) = P_N(F(x)),$$

where $P_N$ is the pgf of $N$.

**Definition 6.1** Let $M_n$ be the maximum of an independent and identically distributed sample from $F$. If there exist deterministic $c_n > 0$ and $d_n$ such that

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{M_n - d_n}{c_n} \le x\right) = H(x),$$

then $F$ is said to belong to the **maximum domain of attraction** of $H$, written

$$F \in \mathrm{MDA}(H).$$

## GEV Distributions

There are three possible limit types for normalized maxima: Gumbel, Fréchet, and Weibull. In a location-scale parameterization, they can be written as follows.

**Definition 6.2** The **Gumbel distribution** has distribution function

$$F(x) = \exp\left(-\exp\left(-\frac{x-\mu}{\sigma}\right)\right), \qquad \sigma > 0.$$

The **Fréchet distribution** has distribution function

$$F(x) = \exp\left(-\left(\frac{x-\mu}{\sigma}\right)^{-\alpha}\right), \qquad \alpha > 0, \qquad x > \mu.$$

The **(negative) Weibull distribution** has distribution function

$$F(x) = \exp\left(-\left(-\frac{x-\mu}{\sigma}\right)^{\alpha}\right), \qquad \alpha > 0, \qquad x < \mu.$$

These three families are combined into the **generalized extreme value (GEV)** family

$$H_\gamma(x) = \begin{cases} \exp\left(-(1+\gamma x)^{-1/\gamma}\right), & \gamma \neq 0, \\ \exp(-e^{-x}), & \gamma = 0, \end{cases}$$

defined where $1 + \gamma x > 0$.

With location parameter $\mu$ and scale parameter $\sigma > 0$, the distribution becomes

$$H_{\gamma,\mu,\sigma}(x) = H_\gamma\left(\frac{x-\mu}{\sigma}\right).$$

If $\mu = 0$ and $\sigma = 1$, this reduces to the standard form $H_\gamma$. The parameter $\gamma$ is the shape parameter. Fréchet corresponds to $\gamma > 0$, Gumbel to $\gamma = 0$, and Weibull to $\gamma < 0$. When $\gamma > 0$, the **tail index** is

$$\alpha = \frac{1}{\gamma}.$$

**Theorem 6.3 (Fisher-Tippett-Gnedenko theorem)** If $F \in \mathrm{MDA}(H)$ for some non-degenerate distribution $H$, then $H$ is a GEV distribution; that is, $H = H_\gamma$ for some $\gamma \in \mathbb{R}$.

The Fisher-Tippett-Gnedenko theorem states that any non-degenerate limit law for normalized maxima must be a member of the GEV family. It is often convenient to express this convergence in terms of the **survival function** $S(x) = 1 - F(x)$ for $x$ in the support of the limit law.

**Theorem 6.4** $F \in \mathrm{MDA}(H_\gamma)$ if and only if there exist normalizing constants $c_n > 0$ and $d_n$ such that

$$\lim_{n\to\infty} nS(c_n x + d_n) = -\log H_\gamma(x) = \begin{cases} (1+\gamma x)^{-1/\gamma}, & \gamma \neq 0, \\ e^{-x}, & \gamma = 0, \end{cases}.$$

This characterization is often easier to verify than the convergence of the full distribution of the normalized maxima.

**Example 6.5** Determine the limiting distribution for a maximum of Pareto$(\alpha, \lambda)$ random variables, where $\alpha > 0$ and $\lambda > 0$, and find the normalizing constants $c_n$ and $d_n$.

**Solution.** For the Pareto distribution,

$$S(x) = \mathbb{P}(X > x) = \left(\frac{\lambda}{\lambda + x}\right)^{\alpha}, \qquad x > 0.$$

Hence

$$nS(c_n x + d_n) = n\left(\frac{\lambda}{\lambda + d_n + c_n x}\right)^{\alpha} = n\left(1 + \frac{d_n}{\lambda} + \frac{c_n}{\lambda}x\right)^{-\alpha}.$$

Rewriting this gives

$$nS(c_n x + d_n) = \left(n^{-1/\alpha} + \frac{n^{-1/\alpha}d_n}{\lambda} + \frac{n^{-1/\alpha}c_n}{\lambda}x\right)^{-\alpha}.$$

This has the Fréchet form $(k_1 + k_2 x)^{-\alpha}$, so the limit is in the Fréchet family with shape parameter

$$\gamma = \frac{1}{\alpha} > 0.$$

To match the standard Fréchet form

$$-\log H_\gamma(x) = (1 + \gamma x)^{-1/\gamma} = (1 + \gamma x)^{-\alpha},$$

we choose the constants so that, asymptotically,

$$n^{-1/\alpha} + \frac{n^{-1/\alpha}d_n}{\lambda} = 1, \qquad \frac{n^{-1/\alpha}c_n}{\lambda} = \gamma = \frac{1}{\alpha}.$$

Therefore

$$d_n = (n^{1/\alpha} - 1)\lambda, \qquad c_n = \frac{\lambda n^{1/\alpha}}{\alpha}.$$

In summary, the Pareto distribution belongs to the maximum domain of attraction of the Fréchet distribution with shape parameter $\gamma = 1/\alpha$, and one suitable choice of normalizing constants is

$$c_n = \frac{\lambda n^{1/\alpha}}{\alpha}, \qquad d_n = (n^{1/\alpha} - 1)\lambda.$$

□

Let us discuss the main tail classes.

Fréchet limits arise from heavy-tailed distributions such as Pareto, inverse gamma, and Student's

$t$. More generally, **Pareto tails**, which are of the form

$$cx^{-\alpha}$$

or, more generally,

$$L(x)x^{-\alpha},$$

where $L$ is slowly varying at infinity, fall into the Fréchet domain. This includes most distributions with a finite number of moments, since the tail index $\alpha$ is the smallest moment order that fails to exist. For example, a Pareto distribution with $\alpha = 3$ has finite mean and variance but infinite third moment, and it belongs to the Fréchet domain with $\gamma = 1/3$.

**Definition 6.6** A function $L : (0, \infty) \to (0, \infty)$ is **slowly varying at infinity** if, for every $t > 0$,

$$\lim_{x\to\infty} \frac{L(tx)}{L(x)} = 1.$$

A function $h : (0, \infty) \to (0, \infty)$ is **regularly varying at infinity** with index $\rho$ if, for every $t > 0$,

$$\lim_{x\to\infty} \frac{h(tx)}{h(x)} = t^{\rho}.$$

The moments of the Gumbel distribution are all finite, so it arises from distributions with lighter tails. For example, the normal, lognormal, exponential, gamma, and inverse Gaussian families all belong to the Gumbel domain. The Weibull domain arises when the underlying distribution is bounded above, as it does for distributions with tails of the form $ke^{-x/\theta}$ or, more generally, $L(x)e^{-x/\theta}$, where $L$ is slowly varying at infinity. One can show that $F$ is in the Gumbel MDA if and only if the survival function$S = 1 - F$ satisfies

$$\lim_{n\to\infty} nS(c_n x + d_n) = e^{-x}$$

for some normalizing constants $c_n > 0$ and $d_n$.

The Weibull distribution is bounded above, so it arises from distributions with finite upper endpoints. For example, the Unif$[0, 1]$ distribution belongs to the Weibull domain with $\gamma = -1$. More generally, if $F$ has a finite upper endpoint $x_F$, then $F$ is in the Weibull MDA if and only if the survival function $S$ satisfies

$$\lim_{n\to\infty} nS(x_F - c_n x) = e^{-x}$$

for some normalizing constants $c_n > 0$. Beacuse the tail is bounded, the Weibull domain is not relevant for loss modelling, but it can be relevant for financial return modelling, where there may be a natural upper bound on losses.

Although many common loss models belong to one of these three maximum domains of attraction, not every distribution does. Classical counterexamples can be built from extremely slowly varying tails. For example,

$$F(x) = 1 - \frac{1}{\log x}, \quad x > e$$

and

$$F(x) = 1 - \frac{1}{\log \log x}, \quad x > e^e$$

have extremely heavy tails that do not belong to any of the three maximum domains of attraction. However, such examples are not common in practice.

## Fitting GEV Distributions by Block Maxima

The **block maxima method** divides the data into blocks of size $n$, records the maximum in each block, and then treats those maxima as a sample from a three-parameter GEV distribution. If $m_1, \ldots, m_k$ are the block maxima, then the GEV density is

$$h(x) = \frac{1}{\sigma}\left(1 + \gamma\frac{x-\mu}{\sigma}\right)^{-1-1/\gamma} \exp\left[-\left(1 + \gamma\frac{x-\mu}{\sigma}\right)^{-1/\gamma}\right],$$

for $1 + \gamma(x-\mu)/\sigma > 0$, and the log-likelihood is

$$\ell(\gamma, \mu, \sigma) = \sum_{j=1}^{k} \log h(m_j).$$

For $\gamma \neq 0$, this can be written more explicitly as

$$\ell(\gamma, \mu, \sigma) = -k\log(\sigma) - \left(1 + \frac{1}{\gamma}\right)\sum_{j=1}^{k} \log\left(1 + \gamma\frac{m_j - \mu}{\sigma}\right) - \sum_{j=1}^{k}\left(1 + \gamma\frac{m_j - \mu}{\sigma}\right)^{-1/\gamma},$$

with the Gumbel case obtained by continuity as $\gamma \to 0$. Maximum likelihood estimation is then used for $\gamma$, $\mu$, and $\sigma$. When the main question is whether the tail is genuinely heavy, one often focuses on testing $\gamma > 0$ against a lighter-tailed alternative using a likelihood ratio test; under the null hypothesis,

$$-2\big(\ell(0, \widehat{\mu}_0, \widehat{\sigma}_0) - \ell(\widehat{\gamma}, \widehat{\mu}, \widehat{\sigma})\big) \xrightarrow{d} \chi^2_1,$$

where $(\widehat{\gamma}, \widehat{\mu}, \widehat{\sigma})$ and $(0, \widehat{\mu}_0, \widehat{\sigma}_0)$ are the unconstrained and constrained maximum likelihood estimates, respectively.

There is a tradeoff in block choice. Longer blocks give us more confidence that the maxima lie in the tail of the distribution, but they also reduce the number of block maxima available for estimation. Shorter blocks give more data but less confidence that the limiting approximation is accurate. The standard error of $\gamma$ is often especially large, which matters because $\gamma$ determines the tail type.

### Threshold Exceedances and the GPD

The block maxima method discards most of the data, so an alternative is to study exceedances above a threshold $d$. Define the exceedance random variable

$$Y_d = X - d \mid X > d.$$

Its distribution function is

$$F_d(x) = \mathbb{P}(X - d \leq x \mid X > d),$$

and its survival function is obtained from the parent tail. If $S_X(x) = 1 - F_X(x)$ is the survival function of $X$, then the survival function of the exceedance is

$$S_d(x) = \mathbb{P}(X - d > x \mid X > d) = \frac{S_X(x+d)}{S_X(d)}.$$

The **mean excess loss** is

$$e(d) = \mathbb{E}\left[X - d \mid X > d\right].$$

**Definition 6.7** For $\beta > 0$, the **generalized Pareto distribution** is

$$G_{\gamma,\beta}(x) = \begin{cases} 1 - (1 + \gamma x/\beta)^{-1/\gamma}, & \gamma \neq 0, \\ 1 - e^{-x/\beta}, & \gamma = 0, \end{cases}$$

with $x \geq 0$ when $\gamma \geq 0$ and $0 \leq x \leq -\beta/\gamma$ when $\gamma < 0$.

When $\gamma > 0$, the GPD is Pareto; when $\gamma = 0$, it is exponential; and when $\gamma < 0$, it is a generalized beta distribution, which is bounded above.

If $Y \sim \mathrm{GPD}(\gamma, \beta)$ in the sense of Definition 6.7 and $Z = Y - d \mid Y > d$, then

$$Z \sim \mathrm{GPD}(\gamma, \beta + \gamma d).$$

Therefore the mean excess function is linear:

$$e(d) = \frac{\beta + \gamma d}{1-\gamma}, \qquad \gamma < 1.$$

**Theorem 6.8 (Pickands-Balkema-de Haan theorem)** $F \in \mathrm{MDA}(H_{\gamma,\mu,\theta})$ if and only if

$$\lim_{d\to\infty} \sup_{x\geq 0} \left| F_d(x) - G_{\gamma,\beta(d)}(x) \right| = 0$$

for some function $\beta(d)$ such that $\beta(d) \to \infty$ as $d \to \infty$ when $\gamma \geq 0$ and $\beta(d) \to -\infty$ as $d \to \infty$ when $\gamma < 0$.

One direction of the Pickands-Balkema-de Haan theorem states that if $F_X \in \mathrm{MDA}(H_\gamma)$ in the sense of Definition 6.1, then for high thresholds $d$ the exceedance distribution is approximately generalized Pareto with the same shape parameter $\gamma$; see Definition 6.7. This makes threshold modelling extremely useful even when the full underlying distribution is unknown.

We now derive risk measures under a GPD tail approximation. Suppose the tail beyond threshold $d$ is approximated by a GPD with parameters $\gamma$ and $\beta$. Then, for $x > d$,

$$\mathbb{P}(X > x) = \mathbb{P}(X > d)\mathbb{P}(X > x \mid X > d) = S_X(d) \left(1 + \frac{\gamma(x-d)}{\beta}\right)^{-1/\gamma}.$$

Setting $x = Q_\alpha$ and solving for the quantile gives

$$1 - \alpha = S_X(d) \left(1 + \frac{\gamma(Q_\alpha - d)}{\beta}\right)^{-1/\gamma},$$

so

$$Q_\alpha = d + \frac{\beta}{\gamma}\left[\left(\frac{1-\alpha}{S_X(d)}\right)^{-\gamma} - 1\right].$$

In practice $S_X(d)$ is estimated empirically. If $d = x_{(n-j)}$ is the $(n-j)$th order statistic from a sample of size $n$, then

$$\widehat{S}_X(d) = \frac{j}{n}.$$

For the CTE, observe that

$$\mathrm{CTE}_\alpha = \mathbb{E}\left[X \mid X > Q_\alpha\right] = Q_\alpha + e(Q_\alpha).$$

Assuming that in the tail beyond $d < Q_\alpha$ the GPD approximation holds, we have that $X - Q_\alpha \mid X > Q_\alpha$ is approximately $\text{GPD}(\gamma, \beta + \gamma(Q_\alpha - d))$. If $\gamma < 1$, then the mean of this distribution exists and is given by

$$e(Q_\alpha) = \frac{\beta + \gamma(Q_\alpha - d)}{1-\gamma}.$$

Therefore

$$\text{CTE}_\alpha = Q_\alpha + \frac{\beta + \gamma(Q_\alpha - d)}{1-\gamma} = \frac{Q_\alpha + \beta - \gamma d}{1-\gamma}.$$

The parameters $\gamma$ and $\beta$ may be estimated from exceedances above a threshold $D$ by maximum likelihood. Confidence intervals follow from the asymptotic covariance matrix, and one may test $\gamma > 0$ against $\gamma = 0$ by a likelihood ratio test.

A standard diagnostic is the **empirical mean excess plot**. If the ordered data are

$$x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)},$$

then the empirical mean excess function at threshold $x_{(j)}$ is

$$\widehat{e}(x_{(j)}) = \frac{1}{n-j} \sum_{i=j+1}^{n} \big(x_{(i)} - x_{(j)}\big).$$

One chooses a threshold $D$ above which this plot is approximately linear, balancing bias against variance. If the mean excess plot slopes upward, a Pareto-type tail is indicated. If it is roughly flat, an exponential tail is more plausible.

Once a threshold $D$ has been selected, one fits the GPD to the exceedances

$$x_i - D, \qquad x_i > D,$$

and then studies the stability of the fitted parameters and their confidence intervals as the threshold varies. In a good threshold region, neighboring thresholds should lead to similar fitted tail distributions and similar extreme-quantile estimates.

Estimation of tail quantities is often difficult because the most extreme part of the sample is volatile, the number of exceedances is small, and threshold choice can materially affect the fitted model. The **Hill estimator** provides another way to estimate the tail index and is taken up next.

### Lecture 7 — Tuesday, October 27, 2015

## The Hill Method

For distributions in the maximum domain of attraction of the Fréchet law, as in Definition 6.1, the survival function has a regularly varying form

$$S(x) = L(x)x^{-1/\gamma} = L(x)x^{-\alpha},$$

where $L$ is slowly varying at infinity and $\alpha = 1/\gamma$; see Definition 6.6. This representation suggests using the tail behaviour of the logarithms of the losses to estimate the tail index.

Assume $X > 0$. The mean excess function of $\log X$ above threshold $\log u$ is

$$e_{\log X}(\log u) = \mathbb{E}\left[\log X - \log u \mid X > u\right].$$

Using the survival function, this can be written as

$$e_{\log X}(\log u) = \frac{1}{S_X(u)} \int_u^\infty (\log x - \log u)\, \mathrm{d}F_X(x) = \frac{1}{S_X(u)} \int_u^\infty \frac{S_X(x)}{x}\, \mathrm{d}x.$$

If $S_X(x) \approx L(x)x^{-\alpha}$ and $u$ is large, then $L(x)$ is approximately constant over the relevant range, so

$$e_{\log X}(\log u) \approx \frac{L(u)}{S_X(u)} \int_u^\infty x^{-(\alpha+1)}\, \mathrm{d}x \approx \frac{L(u)}{S_X(u)} \cdot \frac{u^{-\alpha}}{\alpha} \approx \frac{1}{\alpha} = \gamma.$$

Thus the mean excess function of the log-losses approaches the extreme value parameter $\gamma$.

**Definition 7.1** Let

$$x_{(1)} \le x_{(2)} \le \cdots \le x_{(n)}$$

be the ordered sample. Choosing a threshold at $d = x_{(n-j)}$, the **Hill estimator** of $\alpha$ is

$$\widehat{\alpha}_j^H = \left( \sum_{k=j+1}^{n} \frac{\log\big(x_{(k)}\big)}{n-j+1} - \log\big(x_{(j)}\big) \right)^{-1}.$$

The Hill estimator is a slight variant of the mean excess function of the log-losses, and it is consistent for $\alpha$ under suitable conditions. The choice of $j$ is crucial. If $j$ is too small, the estimate is based on very few data points and is highly variable. If $j$ is too large, the estimate is biased because it includes data from the central part of the distribution rather than just the tail. Therefore, one typically examines a **Hill plot** over a range of upper order statistics and looks for a stable region.

Once the threshold is fixed at $d = x_{(n-j)}$, the exceedance probability is estimated by

$$\mathbb{P}(X > d) \approx \frac{j}{n}.$$

Using the Pareto-type tail form, the Hill estimator of the survival function for $x > d$ becomes

$$\widehat{S}^H(x) = \frac{j}{n}\left(\frac{x}{x_{(n-j)}}\right)^{-\widehat{\alpha}_j^H}.$$

This provides a direct way to estimate extreme tail probabilities beyond the observed threshold.

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**Lecture 8 — Tuesday, November 03, 2015**

# 5. Copulas and Dependence

## Importance of Dependence

Portfolio loss distributions depend not only on the marginal behaviour of the individual risks, but also on their dependence structure. Expected loss, standard deviation, tail probabilities, VaR, and CTE can all change materially when dependence changes. Correlation alone does not capture this. It measures only linear dependence, so zero correlation does not imply independence, although independence does imply zero correlation.

If the correlation between $X$ and $Y$ is $\pm 1$, then $X$ and $Y$ have a linear relationship of the form $X = aY + b$. Away from this extreme case, however, the correlation can miss important dependence features. For example, $Y = X^2$ can exhibit strong deterministic nonlinear dependence even when the linear correlation is zero.

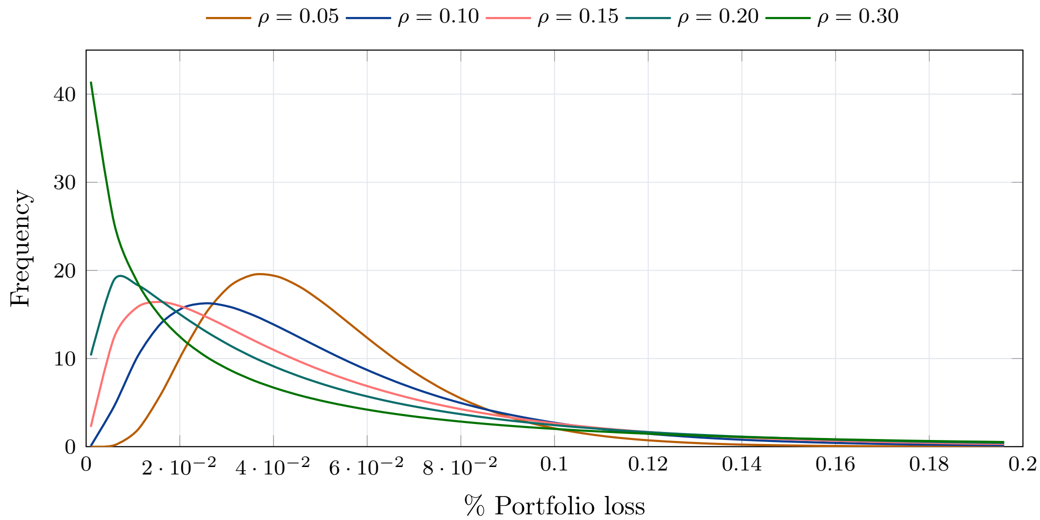



Figure 1: A Vasicek one-factor credit-model illustration with unconditional default probability $p = 5\%$. As the asset correlation $\rho$ increases, the distribution puts more mass near very small portfolio losses but also leaves a heavier right tail.

## Copulas and Sklar's Theorem

**Definition 8.1** A **copula** is a joint distribution function

$$C : [0,1]^d \to [0,1]$$

with uniform $(0,1)$ marginals.

By Sklar's theorem in Theorem 8.4, if $\boldsymbol{X} = (X_1, \ldots, X_d)$ has marginal distribution functions $F_1, \ldots, F_d$, then a copula links the marginals to the joint distribution by

$$F_{\boldsymbol{X}}(x_1, \ldots, x_d) = C(F_1(x_1), \ldots, F_d(x_d)).$$

This is useful because the marginals and the dependence can be modelled separately.

**Example 8.2** Suppose

$$L_A \sim N(1000, 300^2), \qquad L_B \sim \Gamma(4, 250),$$

and consider the limits $x_A = 1500$ and $x_B = 1600$. Then

$$\mathbb{P}(L_A \le 1500) = \Phi\left(\frac{1500 - 1000}{300}\right) = \Phi(1.6667) \approx 0.95221.$$

For the gamma risk,

$$\mathbb{P}(L_B \le 1600) = F_{\Gamma(4,250)}(1600) \approx 0.88108.$$

Under independence,

$$\mathbb{P}(L_A \le 1500,\ L_B \le 1600) = 0.95221(0.88108) \approx 0.8389.$$

If instead

$$C(u,v) = (u^{-2} + v^{-2} - 1)^{-1/2},$$

then

$$\mathbb{P}(L_A \le 1500,\ L_B \le 1600) = C(0.95221, 0.88108) \approx 0.84787.$$

For the stronger dependence model,

$$C(u,v) = (u^{-50} + v^{-50} - 1)^{-1/50},$$

the corresponding joint probability becomes

$$C(0.95221, 0.88108) \approx 0.88075.$$

The largest possible value of

$$\mathbb{P}(L_A \le x_A,\ L_B \le x_B)$$

is

$$\min\{\mathbb{P}(L_A \le x_A), \mathbb{P}(L_B \le x_B)\} = \min\{0.95221, 0.88108\} = 0.88108,$$

which occurs under **comonotonicity**. The smallest possible value is

$$\mathbb{P}(A \cap B) \ge \mathbb{P}(A) + \mathbb{P}(B) - 1 = 0.95221 + 0.88108 - 1 = 0.83329,$$

which is attained in the bivariate case by **countermonotonicity**.

In the notation of the example, the upper bound is attained when the two losses are increasing transformations of the same underlying uniform variable. Equivalently,

$$L_A = F_A^{-1}(F_B(L_B)).$$

The lower Fréchet bound in the bivariate case is attained by making one loss an increasing transformation of the opposite tail of the other:

$$L_A = F_A^{-1}\big(1 - F_B(L_B)\big).$$

**Proposition 8.3** To be a copula, a function must have uniform marginals, must be increasing in each coordinate, and must be **2-increasing**. The uniform marginal condition is

$$C(1, \dots, 1, u_i, 1, \dots, 1) = u_i$$

for each coordinate $i$, together with the condition that all rectangular probabilities are nonnegative. More explicitly, for vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ satisfying $a_j < b_j$ for each $j$,

$$\mathbb{P}(a_1 < U_1 < b_1, \dots, a_d < U_d < b_d) \geq 0,$$

which requires

$$\sum_{j_1=1}^{2} \cdots \sum_{j_d=1}^{2} (-1)^{j_1+\cdots+j_d} C(u_{1j_1}, \dots, u_{dj_d}) \geq 0, \qquad u_{i1} = a_i, \quad u_{i2} = b_i.$$

In the smooth case, a sufficient condition is

$$\frac{\partial^d}{\partial u_1 \cdots \partial u_d} C(u_1, \dots, u_d) \geq 0$$

for all $0 < u_j < 1$.

**Theorem 8.4 (Sklar's theorem)** For any joint distribution function $F$ with marginals $F_1, \dots, F_d$, there exists a copula $C$ such that

$$F(x_1, \dots, x_d) = C(F_1(x_1), \dots, F_d(x_d)).$$

Conversely, any choice of marginals together with any copula produces a valid joint distribution. If the marginals are continuous, then the copula is unique.

The basic idea behind Sklar's theorem is the probability integral transform. If the marginals are continuous, then

$$F_j(X_j) \sim \mathrm{Unif}(0, 1),$$

and hence

$$\begin{aligned} F_{\mathbf{X}}(x_1, \ldots, x_d) &= \mathbb{P}(X_1 \le x_1, \ldots, X_d \le x_d) \\ &= \mathbb{P}(F_1(X_1) \le F_1(x_1), \ldots, F_d(X_d) \le F_d(x_d)) \\ &= C(F_1(x_1), \ldots, F_d(x_d)). \end{aligned}$$

In this continuous case the copula may be written as

$$C(u_1, \ldots, u_d) = F_{\mathbf{X}}(F_1^{-1}(u_1), \ldots, F_d^{-1}(u_d)).$$

For simplicity, we focus on bivariate copulas going forward, but most of the ideas extend to higher dimensions.

**Proposition 8.5 (Invariance under increasing transformations)** Copulas are invariant under strictly increasing transformations of the marginals. If $C$ is the copula for $(X, Y)$, then it is also the copula for $(g(X), h(Y))$ whenever $g$ and $h$ are strictly increasing deterministic functions.

**Proof.** If $C$ is the copula for $(X, Y)$, then

$$C(u, v) = \mathbb{P}(X \le F_X^{-1}(u), Y \le F_Y^{-1}(v)).$$

If $g$ and $h$ are strictly increasing, then

$$\begin{aligned} C(u, v) &= \mathbb{P}(g(X) \le g(F_X^{-1}(u)), h(Y) \le h(F_Y^{-1}(v))) \\ &= \mathbb{P}(g(X) \le F_{g(X)}^{-1}(u), h(Y) \le F_{h(Y)}^{-1}(v)), \end{aligned}$$

which shows that the copula remains the same under strictly increasing transformations of the marginals. □

## Explicit and Implicit Bivariate Copulas

**Definition 8.6** Three fundamental bivariate copulas are the **independence copula**

$$C^{\perp}(u, v) = uv,$$

the **comonotonicity copula**

$$C^+(u,v) = \min(u,v),$$

and the **countermonotonicity copula**

$$C^-(u,v) = \max(u+v-1,0).$$

Important parametric copulas include the **Clayton copula**

$$C_\theta(u,v) = \big(u^{-\theta} + v^{-\theta} - 1\big)^{-1/\theta}, \qquad \theta > 0,$$

and the **Gumbel copula**

$$C_\theta(u,v) = \exp\Big(-\big((-\log u)^\theta + (-\log v)^\theta\big)^{1/\theta}\Big), \qquad \theta > 1.$$

One can show that the Clayton copula tends to the independence copula as $\theta \downarrow 0$ and to the comonotonicity copula as $\theta \to \infty$. The Gumbel copula tends to the independence copula as $\theta \downarrow 1$ and to the comonotonicity copula as $\theta \to \infty$.

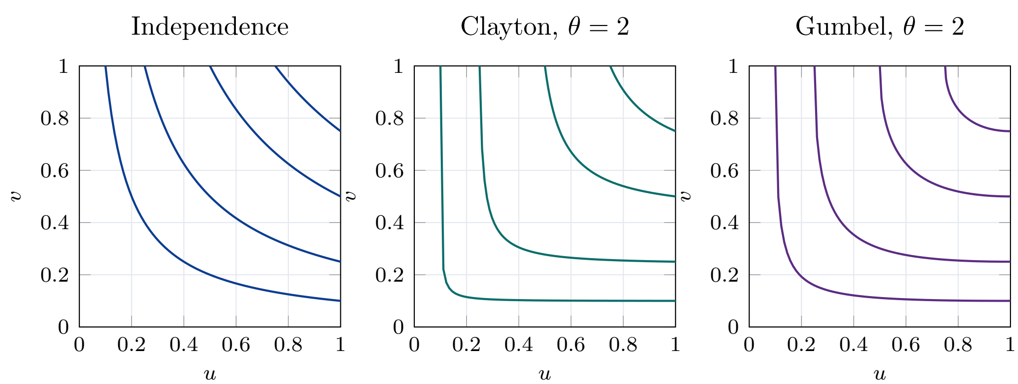



Figure 2: Level curves $C(u,v) = c$ for three bivariate copulas. The Clayton family concentrates lower-tail dependence, while the Gumbel family concentrates upper-tail dependence.

**Definition 8.7** Implicit copulas arise from multivariate distributions. The **Gaussian copula** is

$$C(u,v) = \Phi_\rho(\Phi^{-1}(u), \Phi^{-1}(v)), \qquad -1 \le \rho \le 1,$$

and the **Student $t$ copula** is

$$C(u,v) = F_{\nu,\rho}\big(F_\nu^{-1}(u), F_\nu^{-1}(v)\big).$$

Here $\Phi_\rho$ is the bivariate normal distribution function with correlation $\rho$, $F_{\nu,\rho}$ is the bivariate Student $t$ distribution function with degrees of freedom $\nu$ and correlation parameter $\rho$, and $F_\nu$ is the ordinary univariate Student $t$ distribution function.

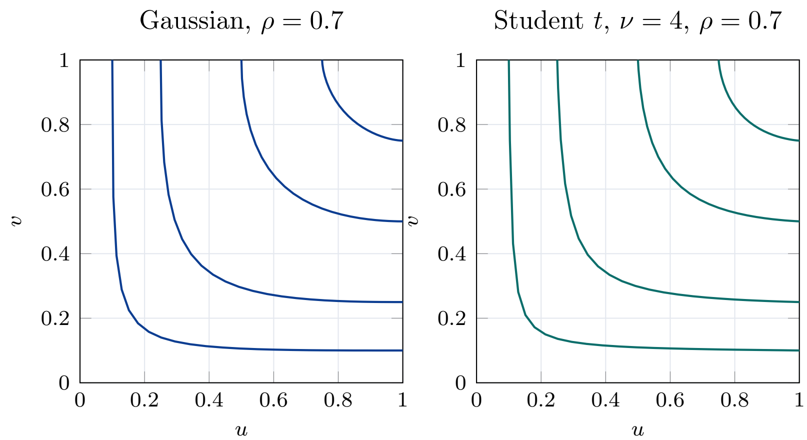



Figure 3: Level curves $C(u, v) = c$ for Gaussian and Student $t$ copulas with the same correlation parameter. The Student $t$ copula bends more sharply into the corners, reflecting symmetric tail clustering.

Figure 3 gives a comparable view for implicit copulas. Copulas with similar positive association can have very different tail behaviour: the Gaussian copula has no tail clustering in the limit, the Student $t$ copula has symmetric tail clustering, the Gumbel copula emphasizes the upper tail, and the Clayton copula emphasizes the lower tail.

**Definition 8.8** If $C$ is the copula of $(U, V)$, then the associated **joint survival function** is

$$\overline{C}(u, v) = \mathbb{P}(U > u, V > v) = 1 - u - v + C(u, v).$$

For losses $X$ and $Y$ with $F_X(x) = u$ and $F_Y(y) = v$, this is also

$$\mathbb{P}(X > x, Y > y) = \overline{C}(u, v).$$

The **survival copula**, namely the copula of $(1 - F_X(X), 1 - F_Y(Y))$, satisfies

$$\widehat{C}(u, v) = u + v - 1 + C(1 - u, 1 - v).$$

**Proposition 8.9 (Fréchet-Hoeffding bounds)** For any $d$-dimensional copula,

$$\max\{u_1 + \cdots + u_d - (d-1), 0\} \leq C(u_1, \ldots, u_d) \leq \min\{u_1, \ldots, u_d\}.$$

The upper bound is the comonotonicity copula. In dimensions larger than two, the lower bound is not a copula.

**Definition 8.10** **Radial symmetry** means that $U$ and $1 - U$ have the same dependence structure, or equivalently that

$$C(u, v) = \widehat{C}(u, v) = u + v - 1 + C(1 - u, 1 - v).$$

**Exchangeability** means that the variables can be swapped without changing the copula:

$$C(u, v) = C(v, u).$$

## Measures of Dependence

**Pearson correlation** measures linear association. It can be misleading when dependence is nonlinear, for example when $Y = X^2$, and some marginal distributions make certain Pearson correlations unattainable. Rank-based measures are often preferred because they are functions of the copula rather than of the marginal scales. If

$$U = F_X(X), \qquad V = F_Y(Y),$$

then the following two rank-based dependence measures are standard.

**Definition 8.11** **Spearman's rho** is

$$\rho_S(X, Y) = \rho(U, V) = \frac{\mathbb{E}[UV] - \mathbb{E}[U]\mathbb{E}[V]}{\sqrt{\operatorname{Var}(U)\operatorname{Var}(V)}} = \frac{\mathbb{E}[UV] - 1/4}{1/12} = 12\mathbb{E}[UV] - 3.$$

Comonotonic risks have $\rho_S = 1$, while countermonotonic risks have $\rho_S = -1$.

**Kendall's tau** is

$$\tau_K = \mathbb{E}\left[\operatorname{sign}((X - X^*)(Y - Y^*))\right] = 4\mathbb{E}\left[C(U, V)\right] - 1,$$

where $(X^*, Y^*)$ is an independent copy of $(X, Y)$. It measures concordance: the product $(X - X^*)(Y - Y^*)$ tends to be positive when larger values of $X$ are associated with larger values of $Y$, and tends to be negative when larger values of $X$ are associated with smaller values of $Y$. Here

$$\operatorname{sign}(z) = \begin{cases} -1, & z < 0, \\ 0, & z = 0, \\ 1, & z > 0. \end{cases}$$

Spearman's rho and Kendall's tau both measure **concordance**, and are invariant under strictly increasing transformations of the marginals.

For data without ties, Spearman's rho can be estimated from ranks. Let $r_{x,i}$ and $r_{y,i}$ be the ranks of the $i$th observations of $X$ and $Y$, using a consistent rank convention. With

$$\overline{r}_x = \overline{r}_y = \frac{n+1}{2}, \qquad s_{r_x}^2 = s_{r_y}^2 = \frac{n(n+1)}{12},$$

we have

$$r_S = \frac{1}{n-1} \sum_{i=1}^{n} \left( \frac{r_{x,i} - \overline{r}_x}{s_{r_x}} \right) \left( \frac{r_{y,i} - \overline{r}_y}{s_{r_y}} \right) = 12 \frac{\sum_{i=1}^{n} r_{x,i} r_{y,i}}{n(n^2-1)} - 3\frac{n+1}{n-1}.$$

Equivalently, if $d_i = r_{x,i} - r_{y,i}$, then

$$r_S = 1 - \frac{6 \sum_{i=1}^{n} d_i^2}{n(n^2-1)}.$$

If there are ties, the tied observations are assigned their average rank. For example, if three observations would occupy ranks $j$, $j+1$, and $j+2$, each is assigned rank $j+1$.

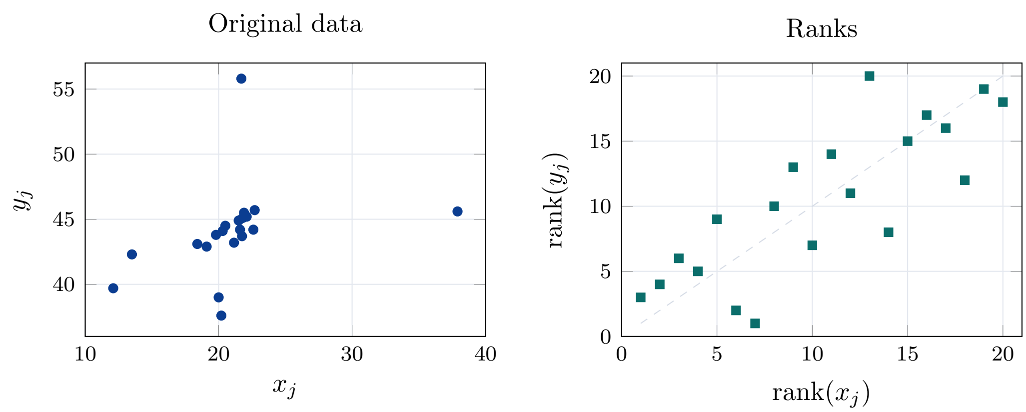
Kendall's tau is estimated by

$$t_K = \frac{2}{n(n-1)} \sum_{i=2}^{n} \sum_{j=1}^{i-1} \operatorname{sign}\big((x_i - x_j)(y_i - y_j)\big),$$

assuming no ties. If there are ties, the formula can be modified.

**Example 8.12** Consider the following data:

| $j$ | $x_j$ | $y_j$ | rank($x_j$) | rank($y_j$) |
|---|---|---|---|---|
| 1 | 21.60 | 44.20 | 12 | 11 |
| 2 | 22.60 | 44.20 | 18 | 12 |
| 3 | 21.80 | 45.10 | 15 | 15 |
| 4 | 21.70 | 55.80 | 13 | 20 |
| 5 | 22.10 | 45.20 | 17 | 16 |
| 6 | 21.75 | 43.70 | 14 | 8 |
| 7 | 21.50 | 44.90 | 11 | 14 |
| 8 | 21.90 | 45.50 | 16 | 17 |
| 9 | 22.70 | 45.70 | 19 | 19 |
| 10 | 37.90 | 45.60 | 20 | 18 |
| 11 | 21.15 | 43.20 | 10 | 7 |
| 12 | 20.50 | 44.50 | 9 | 13 |
| 13 | 20.30 | 44.10 | 8 | 10 |
| 14 | 19.80 | 43.80 | 5 | 9 |
| 15 | 19.10 | 42.90 | 4 | 5 |
| 16 | 18.40 | 43.10 | 3 | 6 |
| 17 | 13.50 | 42.30 | 2 | 4 |
| 18 | 12.10 | 39.70 | 1 | 3 |
| 19 | 20.20 | 37.60 | 7 | 1 |
| 20 | 20.00 | 39.00 | 6 | 2 |




For these data,

$$r = \frac{1}{n-1}\sum_{i=1}^{n}\left(\frac{x_i-\overline{x}}{s_x}\right)\left(\frac{y_i-\overline{y}}{s_y}\right) \approx 0.33, \qquad r_S \approx 0.81, \qquad t_K \approx 0.60.$$

This shows that rank dependence can be much stronger than linear dependence.

## Tail Dependence

**Tail dependence** is especially important in risk management. The practical questions are whether bad outcomes tend to occur independently, and whether dependence becomes stronger in extreme cases. Tail dependence is defined using quantiles so that it depends on the copula rather than on the marginal distributions:

$$C(u,v) = \mathbb{P}(X \le Q_u(X),\, Y \le Q_v(Y)).$$

**Definition 8.13** For the bivariate case, define

$$\lambda_U(q) = \mathbb{P}(X > Q_q(X) \mid Y > Q_q(Y)), \qquad \lambda_L(q) = \mathbb{P}(X \le Q_q(X) \mid Y \le Q_q(Y)).$$

If the limits exist, the **upper** and **lower tail dependence coefficients** are

$$\lambda_U = \lim_{q\uparrow 1} \lambda_U(q), \qquad \lambda_L = \lim_{q\downarrow 0} \lambda_L(q).$$

**Proposition 8.14** In terms of the copula,

$$\lambda_U = \lim_{q\uparrow 1} \frac{1-2q+C(q,q)}{1-q}, \qquad \lambda_L = \lim_{q\downarrow 0} \frac{C(q,q)}{q}.$$

The upper-tail expression may also be written using the survival probability:

$$\lambda_U = \lim_{q\uparrow 1} \frac{\overline{C}(q,q)}{1-q} = \lim_{q\uparrow 1} \frac{1-2q+C(q,q)}{1-q}.$$

By L'Hopital's rule, when the relevant derivative limit exists,

$$\lambda_U = 2 - \lim_{q\uparrow 1} \frac{\mathrm{d}}{\mathrm{d}q} C(q,q), \qquad \lambda_L = \lim_{q\downarrow 0} \frac{\mathrm{d}}{\mathrm{d}q} C(q,q).$$

The coefficients satisfy $0 \le \lambda_U, \lambda_L \le 1$. If $\lambda_U > 0$, then $X$ and $Y$ have upper-tail dependence; if $\lambda_L > 0$, then they have lower-tail dependence. A copula may have any combination of upper- and lower-tail dependence. The Gaussian copula has zero tail dependence, while the Student $t$ copula has positive tail dependence.

## Lecture 9 — Tuesday, November 10, 2015

### Archimedean Copulas

**Definition 9.1** An **Archimedean copula** is built from a **generator function**

$$\phi : [0,1] \to [0,\infty], \qquad \phi(1) = 0, \qquad \phi(0) = \infty,$$

that is continuous and strictly decreasing. In the bivariate case, convexity of $\phi$ is also required (a more onerous condition called **$d$-monotonicity** is required for higher dimensions). The associated copula is

$$C(u_1, \ldots, u_n) = \phi^{-1}\big(\phi(u_1) + \cdots + \phi(u_n)\big).$$

**Proposition 9.2** For Archimedean copulas, Kendall's tau can be expressed directly in terms of the generator:

$$\tau_K = 1 + 4 \int_0^1 \frac{\phi(u)}{\phi'(u)}\, \mathrm{d}u.$$

The upper and lower tail dependence coefficients can likewise be written in terms of limits involving $\phi^{-1}$:

$$\lambda_U = \begin{cases} 2 - 2 \lim\limits_{t \downarrow 0} \dfrac{(\phi^{-1})'(2t)}{(\phi^{-1})'(t)}, & \text{if } \lim\limits_{t \downarrow 0} (\phi^{-1})'(t) = -\infty, \\ 0, & \text{otherwise,} \end{cases}$$

and

$$\lambda_L = \begin{cases} 2 \lim\limits_{t \to \infty} \dfrac{(\phi^{-1})'(2t)}{(\phi^{-1})'(t)}, & \text{if } \lim\limits_{t \to \infty} (\phi^{-1})'(t) = 0, \\ 0, & \text{otherwise.} \end{cases}$$

**Example 9.3** For the Clayton copula with parameter $\theta > 0$, the generator is

$$\phi(u) = \frac{u^{-\theta} - 1}{\theta}.$$

Compute $\phi^{-1}$, $\tau_K$, $\lambda_U$, and $\lambda_L$.

**Solution.** We have

$$\phi^{-1}(t) = (1+\theta t)^{-1/\theta}.$$

Then

$$\tau_K = 1 + 4\int_0^1 \frac{\phi(u)}{\phi'(u)}\,\mathrm{d}u = 1 + 4\int_0^1 \frac{(u^{-\theta}-1)/\theta}{-u^{-\theta-1}}\,\mathrm{d}u = 1 - \frac{4}{\theta}\int_0^1 (u - u^{\theta+1})\,\mathrm{d}u.$$

Now

$$\int_0^1 (u - u^{\theta+1})\,\mathrm{d}u = \int_0^1 u\,\mathrm{d}u - \int_0^1 u^{\theta+1}\,\mathrm{d}u = \frac{1}{2} - \frac{1}{\theta+2} = \frac{\theta}{2(\theta+2)}.$$

Therefore

$$\tau_K = 1 - \frac{4}{\theta}\cdot\frac{\theta}{2(\theta+2)} = 1 - \frac{2}{\theta+2} = \frac{\theta}{\theta+2}.$$

Next,

$$\lambda_U = 2 - 2\lim_{t\downarrow 0}\frac{(\phi^{-1})'(2t)}{(\phi^{-1})'(t)}$$

and since

$$(\phi^{-1})'(t) = -(1+\theta t)^{-1/\theta-1},$$

we obtain

$$\frac{(\phi^{-1})'(2t)}{(\phi^{-1})'(t)} = \frac{-(1+2\theta t)^{-1/\theta-1}}{-(1+\theta t)^{-1/\theta-1}} = \left(\frac{1+\theta t}{1+2\theta t}\right)^{1/\theta+1}.$$

As $t\downarrow 0$, this ratio tends to 1, so

$$\lambda_U = 2 - 2\cdot 1 = 0,$$

and

$$\lambda_L = 2\lim_{t\to\infty}\frac{(\phi^{-1})'(2t)}{(\phi^{-1})'(t)} = 2\lim_{t\to\infty}\left(\frac{1+\theta t}{1+2\theta t}\right)^{1/\theta+1}.$$

Dividing numerator and denominator inside the ratio by $t$ gives

$$\lambda_L = 2\left(\frac{\theta}{2\theta}\right)^{1/\theta+1} = 2\left(\frac{1}{2}\right)^{1/\theta+1} = 2^{-1/\theta}.$$

□

**Example 9.4** For the Gumbel copula with parameter $\theta > 1$, the generator is

$$\phi(u) = (-\log u)^{\theta}.$$

Compute $\phi^{-1}$, $\tau_K$, $\lambda_U$, and $\lambda_L$.

**Solution.** We have
$$\phi^{-1}(t) = \exp\left(-t^{1/\theta}\right).$$
Then
$$\tau_K = 1 + 4\int_0^1 \frac{\phi(u)}{\phi'(u)}\,\mathrm{d}u = 1 + 4\int_0^1 \frac{(-\log u)^\theta}{-\theta(-\log u)^{\theta-1}/u}\,\mathrm{d}u = 1 - \frac{4}{\theta}\int_0^1 u(-\log u)\,\mathrm{d}u = 1 - \frac{1}{\theta}.$$
Next,
$$\lambda_U = 2 - 2\lim_{t\downarrow 0}\frac{(\phi^{-1})'(2t)}{(\phi^{-1})'(t)} = 2 - 2\lim_{t\downarrow 0}\frac{-2^{1/\theta}\exp\left(-2t^{1/\theta}\right)}{-\exp\left(-t^{1/\theta}\right)} = 2 - 2^{1/\theta},$$
and
$$\lambda_L = 2\lim_{t\to\infty}\frac{(\phi^{-1})'(2t)}{(\phi^{-1})'(t)} = 2\lim_{t\to\infty}\frac{-2^{1/\theta}\exp\left(-2t^{1/\theta}\right)}{-\exp\left(-t^{1/\theta}\right)} = 0.$$
□

## Fitting Copulas

One approach is to maximize the likelihood simultaneously over the marginal parameters and the copula parameters. A common step-by-step alternative is to obtain pseudo-maximum likelihood estimates from the marginals, fit the copula, and then iterate this procedure.

If $C$ is an absolutely continuous bivariate copula, its density is
$$c(u,v) := \frac{\partial^2}{\partial u\,\partial v}C(u,v).$$

**Proposition 9.5** If $X$ and $Y$ are absolutely continuous, then the joint density of $(X,Y)$ can be factorized as
$$f_{X,Y}(x,y) = f_X(x)f_Y(y)c(F_X(x),F_Y(y)).$$

**Proof.** By Sklar's theorem (Theorem 8.4), the joint distribution function of $(X,Y)$ can be written as
$$F_{X,Y}(x,y) = C(F_X(x),F_Y(y)).$$
Because $X$ and $Y$ are absolutely continuous, the marginal distribution functions are differentiable almost everywhere with derivatives $f_X$ and $f_Y$. Since the copula is absolutely continuous, its mixed partial derivative exists almost everywhere and equals the copula density $c$. Differentiating

first with respect to $x$ gives

$$\frac{\partial}{\partial x} F_{X,Y}(x, y) = \frac{\partial C}{\partial u}(F_X(x), F_Y(y))\, f_X(x).$$

Now differentiate with respect to $y$. Since $f_X(x)$ does not depend on $y$, we obtain

$$f_{X,Y}(x, y) = \frac{\partial^2 C}{\partial v\, \partial u}(F_X(x), F_Y(y))\, f_Y(y) f_X(x).$$

By definition,

$$\frac{\partial^2 C}{\partial u\, \partial v}(u, v) = c(u, v),$$

so

$$f_{X,Y}(x, y) = f_X(x) f_Y(y) c(F_X(x), F_Y(y)).$$

□

Given data $(x_1, y_1), \ldots, (x_n, y_n)$, the log-likelihood of a bivariate copula model can be written as

$$\ell = \sum_{i=1}^{n} \log f_X(x_i) \ + \sum_{i=1}^{n} \log f_Y(y_i) \ + \sum_{i=1}^{n} \log c(F_X(x_i), F_Y(y_i)).$$

It is natural to decompose this as

$$\ell = \ell_M + \ell_C,$$

where $\ell_M$ is the marginal contribution and $\ell_C$ is the copula contribution.

For a full parametric model, one maximizes the log-likelihood over both the marginal and copula parameters. In practice this is usually done with software, and one must be careful about local maxima. If the main interest is the copula itself, a semi-parametric approach replaces the unknown marginals by empirical estimates.

**Definition 9.6** The **pseudo-observations** associated with data $x_{ij}$ are

$$u_{ij} = \frac{\text{rank}(x_{ij})}{n + 1},$$

where the rank is computed within each margin. These transform the observed marginal scales to approximate uniform scores.

After forming pseudo-observations, one maximizes the **copula log-likelihood**

$$\ell_C = \sum_{j=1}^{n} \log c(u_{1j}, \ldots, u_{dj}).$$

For one-parameter copulas, one may also estimate Kendall's tau first and then recover the copula parameter from the relationship $\theta = h(\tau_K)$ if $h$ is known.

## Goodness of Fit and Simulation

Chi-squared tests can be used in principle, but sparse cells often make them unreliable. Instead, one can use the following diagnostic based on the probability integral transform.

**Proposition 9.7** If $U = F_X(X)$, $V = F_Y(Y)$, and $C(u, v)$ is the correct copula, then conditional transforms such as

$$F_{U|V}(U \mid V = v) \qquad \text{and} \qquad F_{V|U}(V \mid U = u)$$

have the $\text{Unif}(0, 1)$ distribution.

**Proof.** Consider the first transform. For any fixed value $v$ such that the conditional distribution of $U$ given $V = v$ is continuous, the random variable $U \mid V = v$ has conditional distribution function $F_{U|V}(\cdot \mid v)$. Therefore, by the probability integral transform applied conditionally,

$$F_{U|V}(U \mid V = v) \mid V = v \sim \text{Unif}(0, 1).$$

Equivalently, for $0 \le t \le 1$,

$$\mathbb{P}\big(F_{U|V}(U \mid V = v) \le t \mid V = v\big) = t.$$

Since this conditional distribution does not depend on $v$, averaging over $V$ gives

$$\mathbb{P}\big(F_{U|V}(U \mid V) \le t\big) = \mathbb{E}\left[\mathbb{P}\big(F_{U|V}(U \mid V) \le t \mid V\big)\right] = \mathbb{E}[t] = t.$$

Hence $F_{U|V}(U \mid V)$ has the $\text{Unif}(0, 1)$ distribution.

The proof for $F_{V|U}(V \mid U)$ is identical, with the roles of $U$ and $V$ reversed. Thus both conditional transforms are uniformly distributed when the copula model is correct. $\square$

From Section 9, we can check the fit of a copula model by plotting the conditional transforms against the uniform distribution (for example with a P-P plot a Q-Q plot), or by performing a distributional hypothesis test such as the Kolmogorov-Smirnov test.

Simulation is used to generate samples from joint distributions, compute risk measures, and test strategies. We examine Gaussian copulas, Student $t$ copulas, and Archimedean copulas.

To simulate a Gaussian copula with correlation matrix $\mathbf{\Sigma}$, first simulate

$$\boldsymbol{V} = (V_1, \ldots, V_n), \qquad V_1, \ldots, V_n \overset{\text{iid}}{\sim} \text{Unif}(0,1), \qquad Z_k = \Phi^{-1}(V_k),$$

so that $\boldsymbol{Z} = (Z_1, \ldots, Z_n)$ is a standard normal random vector. Then choose a matrix $\mathbf{A}$ such that

$$\mathbf{A}\mathbf{A}^T = \mathbf{\Sigma},$$

and set

$$\boldsymbol{X} = \mathbf{A}\boldsymbol{Z}.$$

The matrix $\mathbf{A}$ is often obtained by Cholesky decomposition, in which case $\mathbf{A}$ is lower triangular. Another approach is spectral decomposition: if

$$\mathbf{\Sigma} = \mathbf{O}\mathbf{D}\mathbf{O}^T$$

with $\mathbf{O}$ orthonormal and $\mathbf{D}$ diagonal with nonnegative entries, then one may take the symmetric square root

$$\mathbf{A} = \mathbf{O}\mathbf{D}^{1/2}\mathbf{O}^T,$$

so that

$$\mathbf{A}\mathbf{A}^T = \mathbf{O}\mathbf{D}^{1/2}\mathbf{O}^T\mathbf{O}\mathbf{D}^{1/2}\mathbf{O}^T = \mathbf{O}\mathbf{D}\mathbf{O}^T = \mathbf{\Sigma}.$$

Finally define

$$U_k = \Phi(X_k), \qquad k = 1, \ldots, n.$$

Then $\boldsymbol{U} = (U_1, \ldots, U_n)$ has the desired Gaussian copula.

To simulate a Student $t$ copula with correlation matrix $\mathbf{\Sigma}$ and degrees of freedom $\nu$, first simulate a correlated normal vector $\boldsymbol{X}$ with correlation matrix $\mathbf{\Sigma}$. Next simulate

$$R \sim \chi^2_\nu$$

independently, for example by the probability integral transform, and set

$$\boldsymbol{Y} = \boldsymbol{X}\sqrt{\frac{\nu}{R}}.$$

Then $\boldsymbol{Y}$ has a multivariate Student $t$ distribution with the required degrees of freedom and correlation matrix. Define

$$U_k = F_\nu(Y_k), \qquad k = 1, \dots, n,$$

where $F_\nu$ is the univariate Student $t$ distribution function. This gives the required $t$ copula sample.

For Archimedean copulas, one can use the following approach. First simulate

$$W \sim \text{Unif}(0,1), \qquad V_1, \dots, V_n \overset{\text{iid}}{\sim} \text{Unif}(0,1),$$

and set

$$U_k = \phi^{-1}\left(-\log V_k / W\right), \qquad k = 1, \dots, n.$$

Then $\boldsymbol{U} = (U_1, \dots, U_n)$ has the required Archimedean copula.

More generally, if one wants a multivariate random vector $\boldsymbol{X} = (X_1, \dots, X_n)$ with copula $C$ and marginals $F_1, \dots, F_n$, one first simulates

$$\boldsymbol{U} = (U_1, \dots, U_n) \sim C$$

and then sets

$$X_k = F_k^{-1}(U_k), \qquad k = 1, \dots, n.$$

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**Lecture 10 — Tuesday, November 17, 2015**

# 6. Models for Market Risk and Other Economic Risks

Losses from different sources may be dependent because they are driven by common economic conditions such as inflation, stock prices, interest rates, unemployment, or broader macroeconomic trends. Econometric models are used to capture this structural dependence, but they bring their own risks, including model risk, basis risk, people risk from insufficient competency, and regulatory risk.

Common univariate econometric models include stock price models for market risk and embedded options, interest rate models for long-term liabilities and annuities, inflation models for deferred liabilities and projected future costs, and macroeconomic models for business cycles. These single-series models become building blocks for multivariate economic scenario generators.

Deterministic simulations correspond to scenario or stress testing. They can be very detailed, but they do not supply a probabilistic interpretation. Stochastic simulations generate many sample paths for each variable of interest, treat each path as equally likely, and are used for risk measures and economic capital.

An **economic scenario generator** models the joint distribution of several economic series, typically including stock returns, interest rates, and inflation, and often also credit spreads, property indices, exchange rates, unemployment, or GDP growth. It is used to project asset and liability cash flows, assess profitability and risk, explore pricing, test risk transfer strategies, and evaluate the range of outcomes, including stress scenarios.

## Financial Return Stylized Facts

Financial return series exhibit several persistent empirical features, often called **stylized facts**. The usual focus is on changes or returns measured over horizons such as daily, weekly, or monthly. Very high-frequency data, such as tick data, have their own stylized facts. At lower frequencies it may be more difficult to identify stylized facts because there are fewer observations and stationarity is harder to defend.

Usually the returns under discussion are log-returns. They are not iid, though serial correlation in the returns themselves is usually small:

$$\mathrm{Corr}(R_t, R_{t+1}) \approx 0.$$

Absolute or squared returns show strong serial correlation. Short-horizon conditional expected returns are close to zero,

$$\mathbb{E}[R_t \mid R_{t-h},\, h \geq 1] \approx 0,$$

while volatility changes over time and tends to cluster. Conditional standard deviations therefore change continually in a partly predictable way. Extreme returns appear in bunches, and return distributions are often heavy-tailed.

If $R_t$ denotes a return with mean $\mu$ and standard deviation $\sigma$, then **kurtosis** is

$$\kappa = \frac{\mathbb{E}\left[(R_t - \mu)^4\right]}{\sigma^4},$$

and **excess kurtosis** is

$$\kappa_{\mathrm{ex}} = \kappa - 3.$$

Fat-tailed distributions have positive excess kurtosis (i.e., they are **leptokurtic**), while thin-tailed distributions have negative excess kurtosis (i.e., they are **platykurtic**). The normal distribution has zero excess kurtosis (i.e., it is **mesokurtic**). Return series tend to be leptokurtic, with excess kurtosis often in the range of 5 to 10 or even higher.

As the return horizon increases, for example from monthly to annual returns, these effects generally become less pronounced. There is less visible volatility clustering, the data look more like iid observations, and the tails look less heavy.

Multivariate return series also show stylized facts. Contemporaneous returns may be correlated, correlations vary over time, absolute returns show cross-correlation, and extreme returns in one series often coincide with extreme returns in others.

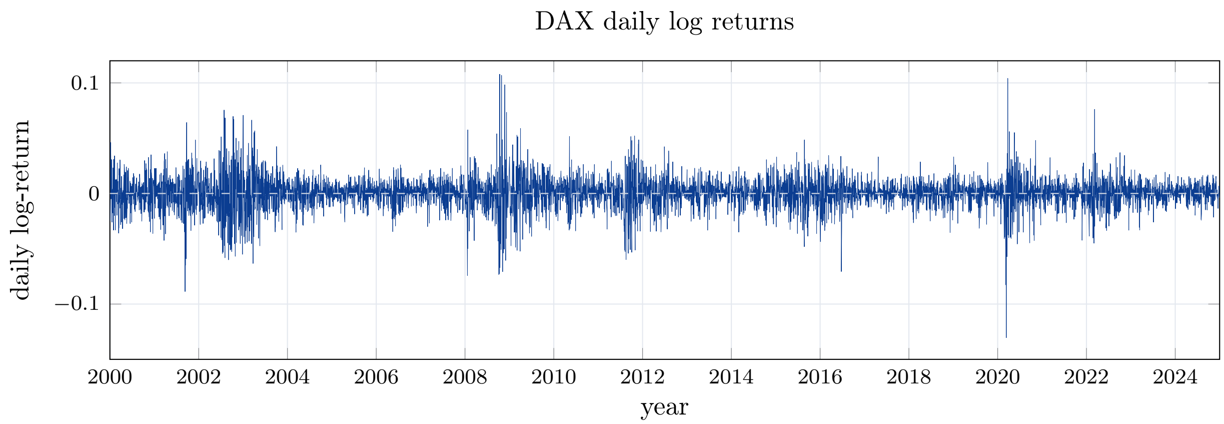


Figure 4: Daily log-returns for the DAX index, computed from Yahoo Finance adjusted closes from 2000 through 2024. The time series shows the usual volatility clustering and bunching of large returns.

**Editorial Note** The course (and more specifically, the *Quantitative Risk Management* textbook) originally presented stylized facts for the above return series over the late 80s and early 90s, capturing important events such as Black Monday, Black Friday, and the collapse of the Soviet Union. I have updated the data to show more recent returns, since we have seen several important events since then, including the dot-com bubble, the global financial crisis, and the COVID-19 pandemic. The stylized facts are still very much present in the data.

In many situations in finance, one assumes that variables are normally distributed (for example when using the independent lognormal model for stock returns). It is therefore important to check whether this assumption is reasonable. The Jarque-Bera test is a commonly used test of normality that is based on the sample skewness and excess kurtosis.

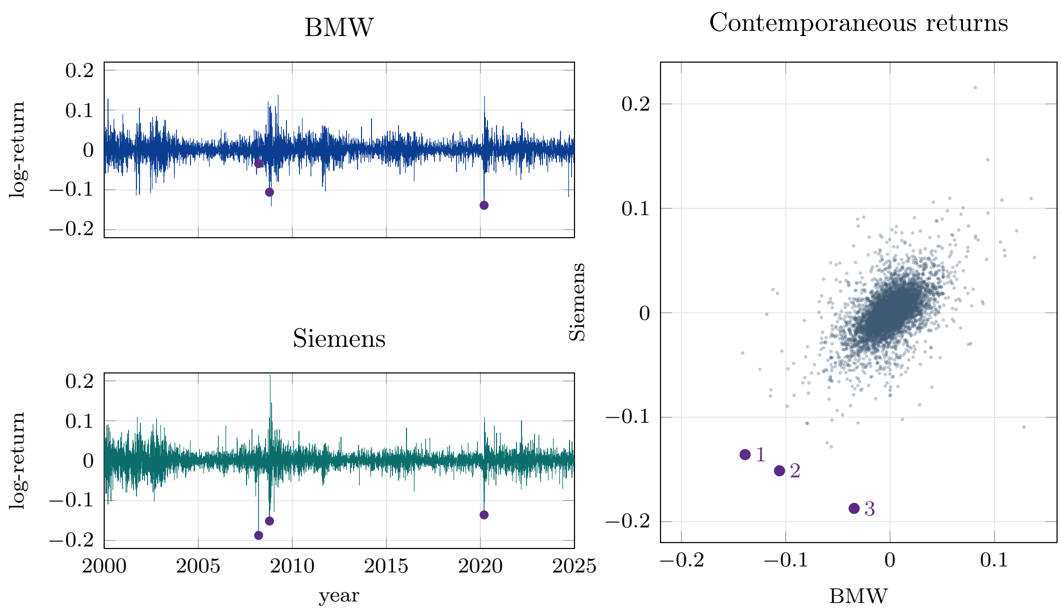


Figure 5: Daily log-returns for BMW and Siemens, computed from Yahoo Finance adjusted closes from 2000 through 2024. The scatterplot shows contemporaneous positive association and joint lower-tail observations; the three highlighted dates are March 12, 2020, October 15, 2008, and March 17, 2008.

**Definition 10.1** The **Jarque-Bera statistic** for a sample of size $n$ is

$$JB = \frac{n}{6}\left(g_1^2 + \frac{g_2^2}{4}\right),$$

where

$$g_1 = \frac{(1/n)\sum_{i=1}^n (X_i - \overline{X})^3}{\left((1/n)\sum_{i=1}^n (X_i - \overline{X})^2\right)^{3/2}}$$

is the sample skewness coefficient and

$$g_2 = \frac{(1/n)\sum_{i=1}^n (X_i - \overline{X})^4}{\left((1/n)\sum_{i=1}^n (X_i - \overline{X})^2\right)^{2}} - 3$$

is the sample excess kurtosis. Under the null hypothesis of normality, $JB$ is asymptotically $\chi_2^2$, so large values argue against normality.

## Stock Price Models

For real-world loss modelling, one uses real-world probabilities rather than risk-neutral probabilities. If $S_t$ is the asset price, the log-return is

$$Y_t = \log\left(\frac{S_{t+1}}{S_t}\right).$$

Risk-neutral probabilities are used for pricing and hedging, not for projecting real-world losses. Before choosing a Gaussian or Brownian model for $\{Y_t\}$, one should check whether it fits the historical data.

Consider the S&P 500 monthly log-returns below. We use total return, so dividends are assumed to be reinvested. The important features are fat tails, especially in the left tail, nonconstant volatility, volatility clustering, more downside than upside during high-volatility periods, and some — but not much — autocorrelation.

**Definition 10.2** In the **independent lognormal model**, the log-returns $\{Y_t\}$ are iid normal random variables with mean $\mu$ and variance $\sigma^2$:

$$Y_t \overset{\text{iid}}{\sim} N(\mu, \sigma^2), \qquad Y_t = \mu + \sigma\varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} N(0,1),$$

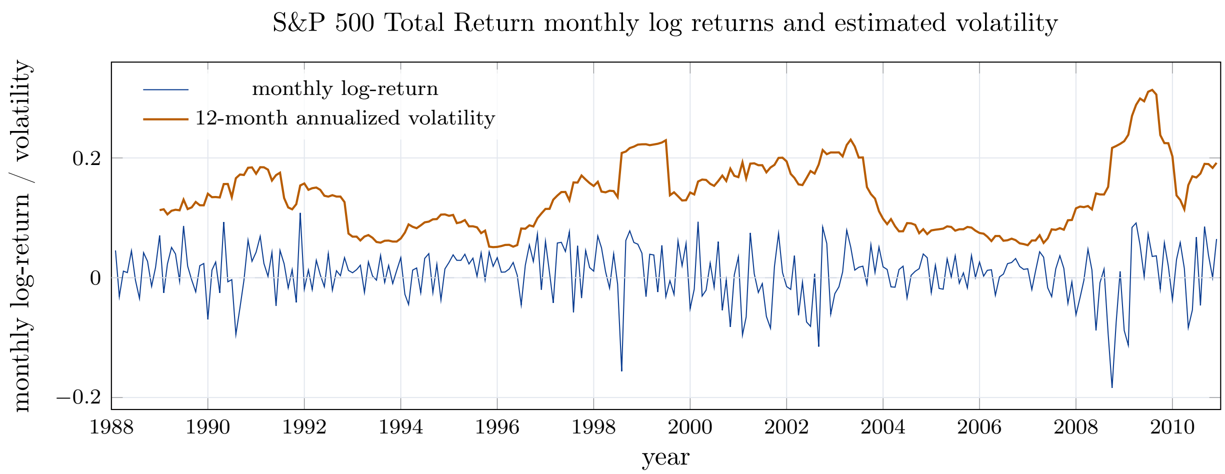


Figure 6: Monthly log-returns for the S&P 500 Total Return Index and a rolling 12-month annualized volatility estimate, computed from Yahoo Finance adjusted closes from 1988 through 2010.

so

$$S_{t+1} = S_t e^{Y_t} = S_t e^{\mu + \sigma \varepsilon_t}.$$

This is the discrete-time version of geometric Brownian motion, with parameter vector

$$\boldsymbol{\theta} = (\mu, \sigma).$$

The same model generates lognormal annual, monthly, weekly, and daily data, and it is the model behind the Black-Scholes option-pricing formula. It is analytically convenient and tractable, but it produces thin tails, fixed volatility, no volatility clustering, and essentially no autocorrelation structure. The Q-Q diagnostic in Figure 7 shows a much heavier empirical left tail than the normal model predicts.

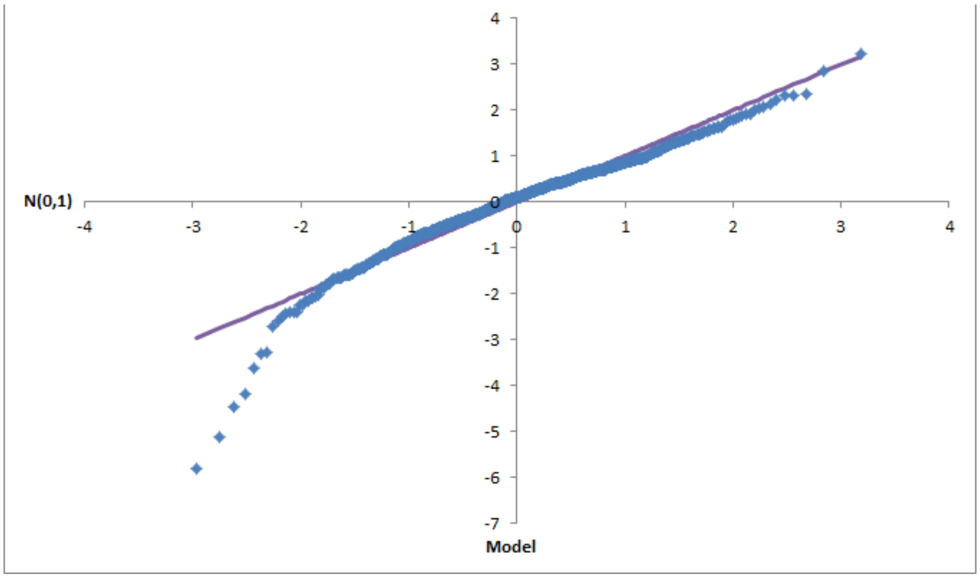


Figure 7: Q-Q plot for the independent lognormal model. The empirical left tail is much heavier than the fitted normal model predicts.

The **GARCH**$(1,1)$ **model** keeps a Gaussian innovation but allows time-varying conditional variance:

$$Y_t = \mu + \sigma_t \varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} N(0,1),$$

with

$$\sigma_t^2 = a_0 + a_1 (Y_{t-1} - \mu)^2 + \beta \sigma_{t-1}^2, \qquad a_0, a_1 > 0, \qquad a_1 + \beta < 1.$$

The parameter $\beta$ drives volatility persistence. The parameter set is

$$\Theta = \{\mu, a_0, a_1, \beta\}.$$

The model itself has no autocorrelation in returns, although it can be combined with an AR(1) mean equation. More generally, GARCH$(p,q)$ refers to using lagged return-shock terms through lag $p$ and lagged variance terms through lag $q$. GARCH is still a single-factor model. It captures stochastic volatility and volatility clustering and gives heavier tails than the independent lognormal model or a basic ARCH model, though it does not capture the association between high volatility and low mean returns. The GARCH Q-Q plot in Figure 8 improves the centre of the distribution but still leaves tail discrepancies.

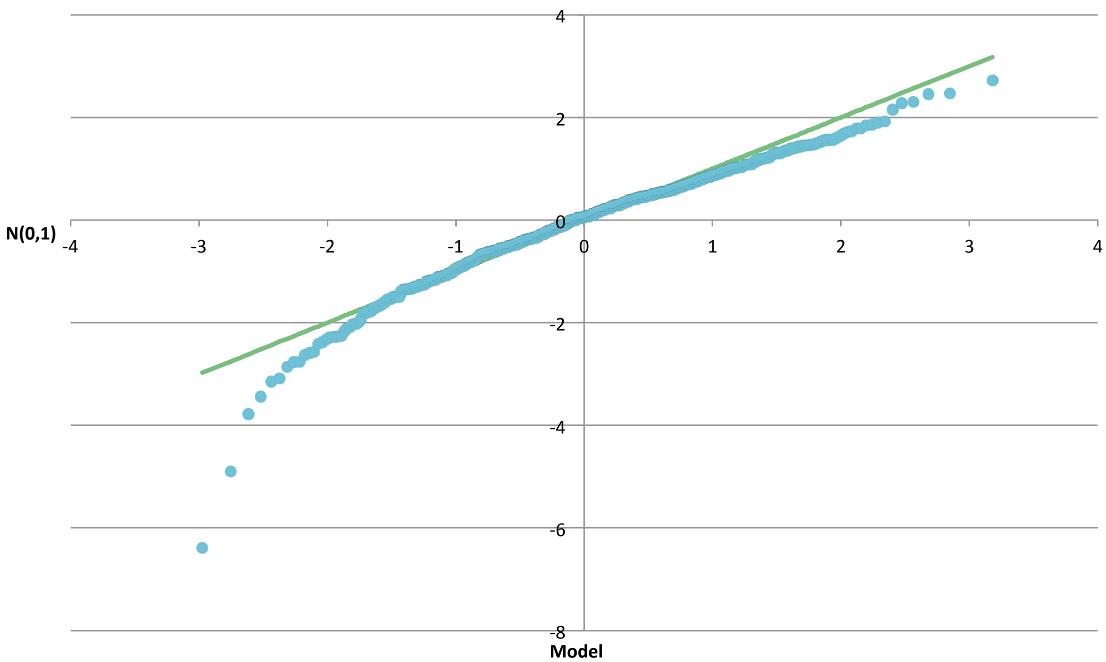


Figure 8: Q-Q plot for the fitted GARCH$(1,1)$ model. The centre is closer to the normal benchmark than in the independent lognormal diagnostic, but tail discrepancies remain.

A simple **regime-switching lognormal model** assumes that the return distribution depends on an unobserved regime $\rho_t$. In regime $k$,

$$Y_t \mid \{\rho_t = k\} \sim N(\mu_k, \sigma_k^2),$$

and the regime follows a Markov chain with transition probabilities

$$\mathbb{P}(\rho_t = k \mid \rho_{t-1} = j) = p_{jk}.$$

With two regimes, the parameter set is

$$\Theta = \{\mu_1, \sigma_1, \mu_2, \sigma_2, p_{12}, p_{21}\}.$$

The probabilities $p_{11}$ and $p_{22}$ are not listed separately because each row of the transition matrix sums to one, so $p_{11} = 1 - p_{12}$ and $p_{22} = 1 - p_{21}$. Models with $K$ regimes are possible, though two regimes are often enough; with three regimes there are twelve parameters. This model captures stochastic volatility in a simple way, volatility clustering, some autocorrelation, fat tails, and the tendency for high-volatility periods to be associated with lower returns. Its cost is greater complexity: it is a two-factor model, since both returns and regimes evolve stochastically, and residuals are not clearly defined. The RSLN-2 residual Q-Q plot in Figure 9 is much closer to a straight line than the ILN diagnostic.

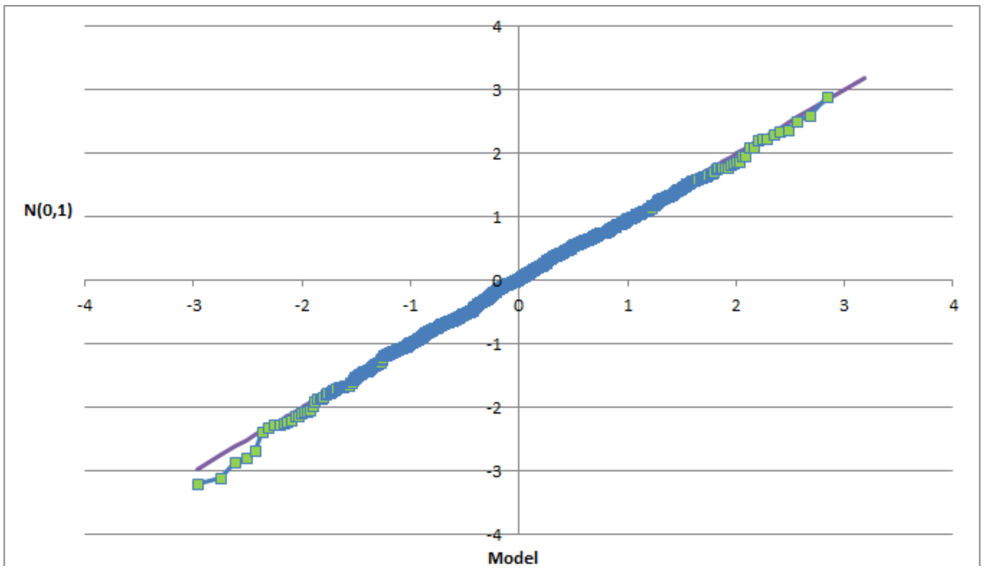


Figure 9: Q-Q plot for the RSLN-2 residual diagnostic. The residuals track the normal benchmark much more closely than the independent lognormal diagnostic.

Other widely used models include regime-switching AR, ARCH, and GARCH models, as well as **stochastic volatility models** of the form

$$Y_t = \mu + \sigma_t \varepsilon_t^Y, \qquad \log \sigma_t = \nu_t = m + a(\nu_{t-1} - m) + \sigma_\nu \varepsilon_t^\nu,$$

where the innovation vector

$$\boldsymbol{\varepsilon}_t = \begin{pmatrix} \varepsilon_t^Y \\ \varepsilon_t^\nu \end{pmatrix} \sim N_2(\mathbf{0}, \boldsymbol{\Sigma}), \qquad \boldsymbol{\Sigma} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

The correlation parameter allows return shocks and volatility shocks to move together.

## Economic Scenario Generators

**Economic scenario generators** (**ESGs**) are commonly used in insurance, pensions, and banking. In insurance, they are important for Own Risk and Solvency Assessment (ORSA), including the US solvency framework and Solvency II. An economic scenario generator may be deterministic or stochastic, and design choices include the model frequency, projection horizon, included variables, multivariate structure, and whether the emphasis is on best-estimate behaviour or tail risk.

Real-world models are used for projecting actual cash flows, profit testing, setting reserves, calculating risk measures, and testing risk mitigation strategies. Risk-neutral or market-consistent models are used for pricing and hedging financial guarantees and derivatives.

A good economic scenario generator should have sound economic properties, internal consistency, realistic ranges and relationships, an adequate fit to data, and acceptable computational efficiency. It should also be tractable enough to work as a module inside a larger asset-liability model. Sound economic properties include valid short-term relationships, such as no free lunches, the principle that extra expected return should come with extra risk, interest-rate and inflation differentials affecting exchange rates, and dependence in extreme paths. They also include feasible long-term relationships, where economic theory gives some guidance about causes and effects over medium and long horizons. Calibration requires judgment as well as statistical and economic theory, especially because too little history gives unstable estimates and too much history may ignore structural change. Testing should include in-sample and out-of-sample validation, robustness across calibration periods, and sensitivity to parameters.

## Model Risk and Model Governance

Every quantitative risk management calculation depends on a model, and every model is a simplification of reality. A model is a mathematical representation of some aspect of the real world, used to analyze a complex problem so that decisions can be made. Examples include a fixed interest assumption such as 6% per year, CAPM, Markov regime-switching models, compound Poisson loss models with lognormal severities, stock-price dynamics such as

$$\mathrm{d}S_t = \mu S_t \,\mathrm{d}t + \sigma S_t \,\mathrm{d}W_t,$$

service tables for pension plans, and combinations of these components.

Some objects used with models are not themselves models. The Black-Scholes option price, VaR, CTE, and Monte Carlo simulation are outputs, risk measures, or computational methods rather than full representations of the underlying real-world mechanism.

The two common mistakes are to trust a model too much or to dismiss it too quickly. A sensible modelling process starts from the problem requirements, explores the data, formulates a modelling hypothesis, calibrates to appropriate data, tests goodness of fit, compares or adapts alternatives, and then selects the model that is fit for purpose. Implementation also requires testing and development protocols, review, calibration procedures, interface design, output-format decisions, documentation, and controls. Minor models are often internal to the company, while major valuation models may be purchased from software firms. **Model risk** is the risk of loss from using a model that is not fit for purpose, and it can be mitigated by good model governance. **Model governance** includes policies and procedures for model development, testing, implementation, use, and review. It also includes controls such as independent validation, documentation standards, and regular review cycles.

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**Lecture 11 — Tuesday, November 24, 2015**

# 7. Capital, Credit Risk, and Regulation

## Capital and Balance Sheets

In a financial institution, **capital** is the amount held or required to be held to underpin the risk of loss in value of exposures, businesses, and other activities, so as to protect depositors and general creditors against loss. According to OSFI, it is a source of financial support that protects the institution against unexpected losses and therefore contributes to safety and soundness.

On the balance sheet, capital is the surplus of assets over liabilities. Assets include current assets, fixed assets, and other assets. Liabilities include current liabilities and fixed liabilities, and the residual claim belongs to owners' equity.

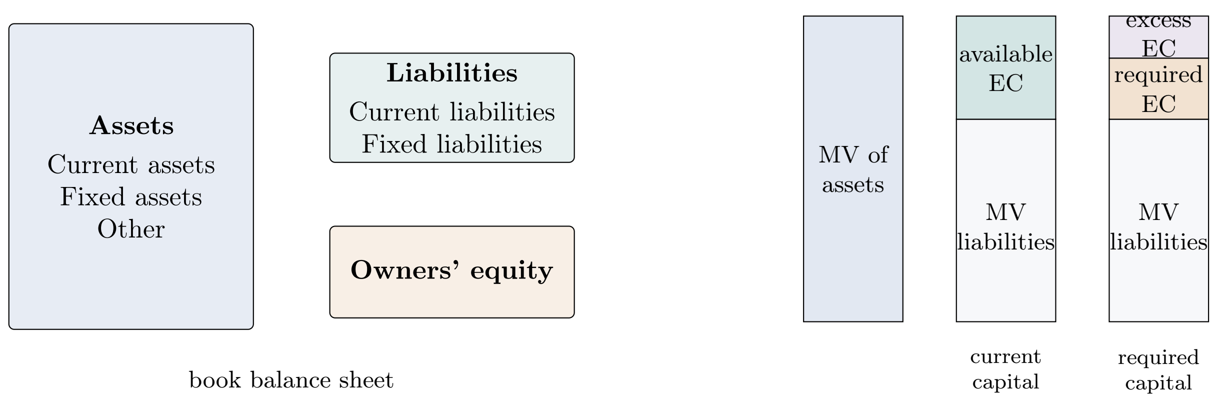


Figure 10: Capital as balance-sheet surplus and as economic capital measured using market or market-consistent values.

**Economic capital** is the amount of capital a financial institution thinks it should have. **Regulatory capital** is the amount of capital the regulator requires the institution to have.

Economic capital uses market or market-consistent valuation of assets and liabilities. It is natural to distinguish **available economic capital**, which is the current market-value excess of assets over liabilities, from **required economic capital**, which is the amount needed to support the firm's risks at a specified confidence level.

Capital is not the same thing as reserves. **Reserves** are liquid, high-quality assets, such as precious metals, deposits at a central bank, or government bonds. A **reserve requirement** is a constraint on assets based on existing liabilities. Capital instead refers to the sources of funding on the right-hand side of the balance sheet. A **capital requirement** is a constraint on sources of funds based on the risks of existing assets.

Capital acts as a buffer against large unexpected losses, protects depositors and other claimants, and provides confidence to external investors and rating agencies about the financial health and viability of the firm. **Capital management** is the ongoing process of raising and maintaining capital at levels sufficient to support planned operations. It also involves allocating capital to recognize the level of risk in the firm's various activities.

## Bank Equity

High equity requirements are often criticized on the grounds that equity is expensive and that stronger capital requirements would affect credit markets adversely. The counterargument is that the required return on equity falls as leverage falls; tax subsidies, implicit guarantees, and other policies that favour debt are distortions; and the claim that high debt provides discipline has little empirical support.

Several common arguments confuse capital requirements with reserve requirements. Increased equity does not force a bank to set aside funds that could otherwise be lent, because lending can be funded by equity. Increased equity requirements do not mechanically raise funding costs merely because the cost of equity exceeds the cost of debt, since the required return on equity decreases when the debt-to-equity ratio decreases. A higher return on equity at high leverage reflects a higher risk premium, not free value creation.

Debt tax shields and subsidies can make debt privately attractive, but that is an argument about distortions rather than an argument for excessive leverage. Debt may create some market discipline for managers, but high leverage also gives incentives to take risk and relies on a fragile capital structure with substantial short-term debt. Discipline can also come from equity holders and regulation. A further objection is that higher capital requirements may push banking activities into less-regulated markets, but many shadow-banking activities still relied on backstops from regulated institutions.

A Modigliani-Miller style relation for the required return on equity is

$$R_S = R_0 + \frac{B}{S}(1 - T_C)(R_0 - R_B),$$

where $R_S$ is the required return on equity, $R_0$ is the unlevered required return, $B/S$ is the debt-to-equity ratio, $T_C$ is the corporate tax rate, and $R_B$ is the return on debt. The formula illustrates why required return on equity changes with leverage.

## Economic Capital

Required **economic capital** for a firm as a whole is the notional amount of capital needed to withstand an extreme but rare loss event after taking into account all interactions among the risks in the portfolio. Economic capital may need to reflect interest-rate risk, currency risk, credit risk, market risk, operational risk, and other material risks. It is often assessed using a top-down risk measure in the sense of Definition 3.1 applied to aggregate firm losses.

One common approach is to choose a very high confidence level, perhaps associated with a target credit rating, and then define economic capital as the VaR from Definition 3.2 at that level, possibly less expected loss. Historical annual survival rates for AA-rated companies have been approximately 99.98%. Setting VaR at this confidence level corresponds to a one-in-5000-year loss. Other approaches introduce subjective information through Bayesian methods or use a more realistic confidence level together with a safety factor, as in the Basel market-risk charge.

If the firm's actual equity is below required economic capital, the firm may be taking unwarranted or ill-advised risks. If equity capital is more than enough to cover economic capital, the firm is considered more resilient to shocks and better able to take advantage of growth opportunities. Economic capital therefore sends an important signal to capital market investors, including bondholders and shareholders. These disclosures are connected to Pillar III under the Basel Accord.

Under Pillar II of Basel, banks are expected to maintain an **internal capital adequacy assessment process** (**ICAAP**). Principle 1 says that banks should have a process for assessing overall capital adequacy in relation to their risk profile and a strategy for maintaining capital levels. The ICAAP depends on the nature, size, and complexity of the bank. Advanced banks are expected to use a bottom-up approach with analytical tools.

The ICAAP components include an explicit board-approved capital policy, including objectives, time horizon, and limits; a capital planning process with responsibilities assigned across the capital adequacy assessment process; a process to identify all risks inherent in the bank's portfolios; a system to measure all identified risks; and regular, adequate internal reporting to the board and senior management.

## Regulatory Capital

**Regulatory capital** rules may be factor- or formula-based, as in United States risk-based capital requirements for insurance; scenario-based, as in some Canadian life-insurance requirements; or based on economic-capital ideas, as in Solvency II and Basel II/III. The goals of capital regulation include stability of the financial system, macroprudential regulation, adequate capital levels at individual institutions, transparency, comparability, implementation across a wide variety of institutions, and avoidance of procyclicality in capital requirements.

**Procyclicality** arises because losses in recessions erode bank capital while risk-based capital requirements, such as those in Basel II, often rise at the same time. If banks cannot quickly raise sufficient new capital, their lending capacity falls and a credit crunch may follow. Relaxing requirements in bad times may reduce the contractionary effect on credit supply, but it may also increase bank failure probabilities precisely when loan defaults are high. Basel III is a compromise: it reinforces the quality and quantity of minimum required capital while also creating mandatory buffers, including a capital conservation buffer and a countercyclical buffer, to be built up in good times and released in bad times.

For the trading book under Basel III, different risk measures use different confidence levels and horizons:

| **Risk** | **Risk measure** | **Confidence level and horizon** |
|---|---|---|
| Market risk | VaR | 99%, 10 days |
| Incremental risk charge | VaR | 99.9%, 1 year |
| Credit risk and counterparty credit risk | VaR | 99.9%, 1 year |
| Credit valuation adjustment | VaR | 99%, 10 days. |

The trading portfolio contains market risk and credit risk. Market risk includes equities, commodities, bonds and loans, over-the-counter derivatives, and credit derivatives such as credit default swaps and collateralized debt obligations. Portfolio VaR over 10 days at 99% includes spread risk, also called specific risk. The stressed VaR component uses historical data from a period of significant financial stress, such as 2007–2008.

The market risk charge is split into ordinary and stressed components:

$$C = C_{MV} + C_{SV},$$

where

$$C_{MV} = \max\{\mathrm{VaR}_{t-1}, m_b\overline{\mathrm{VaR}}\}$$

and

$$C_{SV} = \max\{\mathrm{sVaR}_{t-1}, m\overline{\mathrm{sVaR}}\}.$$

Here $\overline{\mathrm{VaR}}$ and $\overline{\mathrm{sVaR}}$ denote average ordinary and stressed VaR values used in the regulatory calculation, and the stressed component is calibrated to a period of significant historical stress.

## Trading-Book Credit Risk

The **incremental risk charge** (**IRC**) captures default and migration risks in trading-book positions that are incremental to the risks captured by market VaR. It applies to trading-book positions in unsecuritized credit products and is measured over a one-year capital horizon under a constant-risk assumption.

The revisions to the Basel II market-risk framework finalized in July 2009 required banks to develop a methodology for calculating the incremental risk charge. The **capital horizon** is the time over which credit events are measured, usually one year. The **liquidity horizon** is the frequency at which the portfolio or transaction is rebalanced to a target level of risk, such as three months, six months, or one year.

Basel guidance says that liquidity horizons should reflect the time required to sell the position or hedge all material risks in a stressed market. They should be long enough that selling or hedging does not materially affect market prices. The floor is three months. Non-investment-grade positions are expected to have longer liquidity horizons than investment-grade positions, and concentrated positions are expected to have longer liquidity horizons.

Under a **constant position assumption**, a deteriorated name remains in the portfolio for the whole capital horizon. Under a **constant risk assumption**, positions whose credit characteristics improve or deteriorate are replaced after the liquidity horizon so that the credit characteristics are equivalent to those at the start of the liquidity horizon. Liquidity-horizon default probabilities can be estimated from ratings data, KMV or other market-implied sources, or roots of transition matrices.

**Example 11.1** Suppose an investment-grade name can be investment grade, non-investment grade, or default after one period, with transition probabilities from investment grade of 0.2, 0.7, and 0.1, and from non-investment grade of 0.1, 0.1, and 0.8. Over two periods, the default probability under constant position is

$$0.1(1) + 0.7(0.8) + 0.2(0.1) = 0.68.$$

Under constant risk, once the name survives the first period it is replaced by a fresh investment-grade exposure, so the two-period default probability is

$$0.1(1) + 0.9(0.1) = 0.19.$$

For constant risk calculations, the PD scaling factor is

$$\text{PD scaling factor} = \frac{\text{PD with rebalancing}}{\text{PD buy-and-hold}}.$$

For a one-month liquidity horizon and one-year capital horizon, the following scaling factors are used:

| **Rating** | **Moody's** | **MKMV** |
|---|---|---|
| All investment grade | 0.18 | 0.16 |
| Ba | 0.34 | 0.23 |
| B | 0.60 | 0.35 |
| Caa–C | 1.30 | 1.92. |

**Trading book credit risk** can be divided into counterparty credit risk, issuer or borrower risk, and structured credit risk. **Counterparty credit risk** arises in derivatives, including credit derivatives, and is affected by collateral, guarantees, and other mitigation. **Issuer or borrower risk** arises in bonds, loans, and credit default swaps. **Structured credit** combines issuer or borrower risk with counterparty risk. Basel III capital for trading book credit risk includes CreditVaR for counterparty credit risk, the incremental risk charge for default and migration risk in bonds, loans, and credit default swaps, securitization and correlation trading capital, and CVA VaR.

For Basel credit risk capital, the regulatory formula has the same broad structure as the asymptotic single-factor model, with heterogeneous exposures and default and migration risk:

$$\text{Capital} = \sum_j \text{LGD}_j\,\text{EAD}_j \left[\Phi\left(\frac{\Phi^{-1}(\text{PD}_j) - \sqrt{\rho_j}\,\Phi^{-1}(0.001)}{\sqrt{1-\rho_j}}\right) - \text{PD}_j\right] \text{MF}(M_j, \text{PD}_j).$$

The calculation of risk-weighted assets relies on four quantitative risk components: probability of default, exposure at default, loss given default, and maturity. The asset-correlation parameter is another model parameter set by the Accord. We will discuss the Basel credit risk formula in more detail later in the course.

## Capital Allocation and Performance

Suppose an insurer's aggregate loss from $n$ business lines is

$$L = \sum_{i=1}^{n} L_i,$$

so the economic capital for the firm as a whole is

$$C = \rho(L),$$

where $\rho$ is the chosen risk measure; see Definition 3.1. Economic capital can be allocated to individual business lines for profitability analysis, pricing, risk budgeting, strategic planning, and incentives.

**Definition 11.2** For a portfolio with losses

$$L(\boldsymbol{w}) = \sum_{n=1}^{N} w_n L_n$$

and portfolio risk

$$R(\boldsymbol{w}) = \rho\left(\sum_{n=1}^{N} w_n L_n\right),$$

the **stand-alone contribution** of instrument or line $n$ is

$$C_n^{SA} = \rho(w_n L_n)$$

the **incremental contribution** is

$$C_n^I = \rho(L) - \rho\left(\sum_{k \neq n} w_k L_k\right)$$

and the **marginal contribution** (or **Euler contribution**) is

$$C_n^M = w_n \frac{\partial \rho(L)}{\partial w_n}$$

when the derivative exists.

Stand-alone contributions and incremental contributions are generally not additive. Marginal contributions are used because, under appropriate homogeneity and regularity assumptions, they are additive.

**Definition 11.3** A function $f$ is **positive homogeneous of degree** $k$ if

$$f(\alpha \boldsymbol{x}) = \alpha^k f(\boldsymbol{x})$$

for all $\alpha > 0$.

**Theorem 11.4 (Euler's theorem)** If $f$ is continuously differentiable, then $f$ is positive homogeneous of degree $k$ if and only if

$$\boldsymbol{x} \cdot \nabla f(\boldsymbol{x}) = \sum_{n=1}^{N} x_n \frac{\partial f}{\partial x_n}(\boldsymbol{x}) = k f(\boldsymbol{x}).$$

Suppose the risk measure $\rho$ is positive homogeneous. Set

$$f(\boldsymbol{w}) = \rho(L_{\boldsymbol{w}}) = \rho\left(\sum_{n=1}^{N} w_n L_n\right).$$

Then $f$ is positive homogeneous of degree one, and Euler's theorem gives

$$\rho(L_{\boldsymbol{w}}) = f(\boldsymbol{w}) = \boldsymbol{w} \cdot \nabla f(\boldsymbol{w}) = \sum_{n=1}^{N} w_n \frac{\partial \rho(L_{\boldsymbol{w}})}{\partial w_n} = \sum_{n=1}^{N} C_n^M.$$

This additivity is one of the main reasons Euler allocation is used in practice. Applications include risk management, hedging, risk-adjusted performance measurement, and macroprudential financial regulation.

For important risk measures, the marginal contributions take explicit forms:

$$C_n^{\mathrm{VaR}} = w_n \mathbb{E}\left[L_n \mid L = \mathrm{VaR}_\alpha(L)\right],$$

$$C_n^{\text{CVaR}} = w_n \mathbb{E}\left[L_n \mid L \geq \text{VaR}_\alpha(L)\right],$$

and

$$C_n^{\sigma} = w_n \frac{\text{Cov}(L_n, L)}{\sigma(L)}.$$

Traditional performance measures such as return on assets and return on equity ignore the amount of risk taken to achieve the return. For example, if assets are $A = 100$, liabilities are 84, equity is $\text{Eq} = 16$, and net income is $I = 4.0$, then

$$\text{ROA} = \frac{4.0}{100} = 0.04, \qquad \text{ROE} = \frac{4.0}{16} = 0.25.$$

The return looks attractive, but the calculation does not say how much risk was taken.

Risk-adjusted performance measurement either adjusts the numerator, for example by deducting expected losses or focusing only on variation arising from random factors, or adjusts the denominator by comparing return to required economic capital. A generic performance measure has the form

$$\text{Performance} = \frac{\text{reward}}{\text{risk}}.$$

For example, the Sharpe ratio is

$$\text{SR} = \frac{\mu - r}{\sigma}.$$

Using capital in the denominator allows comparison across business units inside a financial institution.

A general **risk-adjusted return on capital** (**RAROC**) style measure is

$$\text{RAROC} = \frac{\text{revenues} - \text{costs} - \text{expected losses}}{\text{capital}}.$$

RAROC can be used to assess capital requirements by business line, product, or customer; measure risk consistently across business units; promote fair and consistent risk-adjusted performance measures; and improve the efficiency of the risk-reward tradeoff. Expected losses should capture losses arising from all sources, including credit, market, and operational risk. The denominator may be stand-alone capital or the unit's contribution to overall firm capital. When RAROC is measured in advance, expected revenues, losses, and costs are used. For performance evaluation, realized values may be used in the numerator.

**Example 11.5** Suppose assets are 100, liabilities are 84, so equity and available economic capital are 16. If net income over the year is 4.0, required economic capital is 12, and risk-adjusted income is $4.0 - 1.5 = 2.5$, then

$$\text{RAROC} = \frac{2.5}{16} = 0.15625 \approx 15.6\%,$$

$$\text{RARORAC} = \frac{2.5}{12} = 0.2083 \approx 20.8\%,$$

and

$$\text{RORAC} = \frac{4.0}{12} = 0.3333 \approx 33.3\%.$$

RAROC measures return against the capital actually supplied by shareholders, while RARORAC measures return against required capital and is therefore useful for internal risk budgeting and business-unit comparison.

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### Lecture 12 — Thursday, December 03, 2015

## Credit Risk Models

**Credit risk** is the risk that the value of a portfolio changes because of unexpected changes in the credit quality of issuers or trading counterparties. It includes losses caused by default and losses caused by changes in credit quality before default. **Issuer risk** is the risk arising from the credit quality of issuers of securities such as bonds. **Counterparty risk** is the risk that the counterparty to a derivative contract or similar agreement defaults.

Credit risk models have two main uses. The first is credit risk management: determining the loss distribution of a loan or bond portfolio over a fixed time horizon, computing risk measures, and allocating risk capital. The second is the analysis and pricing of credit-risky securities, such as credit default swaps.

A **static credit model** builds a future loss distribution from the market factors observed today. Such a model is built from a finite set of random variables. A **dynamic credit model** describes the future evolution of the state of the credit market, usually through continuous-time stochastic processes.

A **structural model**, also called a **firm-value model**, models default as a function of the value of the firm's assets and liabilities. A **reduced-form model** specifies a mathematical model for default times without modelling the firm-value mechanism that produces default.

Credit risk modelling is difficult for three reasons. First, public information and data are often limited, making calibration hard and creating asymmetric-information problems such as moral hazard and adverse selection. Second, credit loss distributions are highly skewed: small gains occur frequently, but occasional large losses determine the tail. Third, default events are dependent, and simultaneous defaults have a major effect on portfolio tail risk.

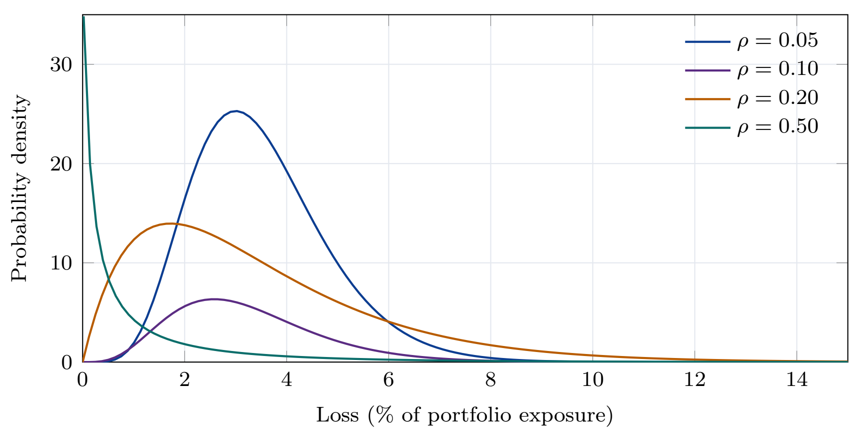


Figure 11: Higher default dependence shifts asymptotic credit losses toward heavier right tails, even when the individual default probabilities are unchanged.

## Credit Ratings and Recovery

**Credit ratings** are assessments of credit quality produced by rating agencies such as Moody's and S&P. They provide information about default probability and potential recovery, and they affect who is able or willing to hold a bond issue.

Issuers pay rating agencies to have issues rated. Rated issues are more attractive to bond traders, and some investors are restricted to bonds above a given rating. Almost all bonds from the same issuer usually have the same rating. Ratings tend to exhibit momentum, and agencies tend to rate **through the cycle**: in other words, they are relatively stable over time. During an economic downturn, defaults are more likely at least in the short term, but ratings often do not move immediately. As a result, bonds with the same rating can show a higher historical default frequency in recessions than in expansions. Stability is valuable to issuers and traders because a rating change, especially a downgrade, is a major event.

| Moody's | S&P | Category |
|---|---|---|
| Aaa | AAA | Investment grade |
| Aa | AA | Investment grade |
| A | A | Investment grade |
| Baa | BBB | Investment grade |
| Ba | BB | Speculative, high yield, or junk |
| B | B | Speculative, high yield, or junk |
| Caa | CCC | Speculative, high yield, or junk |
| D | D | In default |

Many bank borrowers are unrated externally, so banks also maintain internal rating systems and estimate both probability of default and recovery rate themselves.

If an obligor defaults, the lender usually recovers part of the exposure through a debt restructuring, through new securities, or through liquidation. The **recovery rate** RR is the proportion of the ideally market-valued contract that remains after default. The **loss given default** is

$$\mathrm{LGD} = 1 - \mathrm{RR}\,.$$

Historical recovery rates vary substantially by seniority. A Moody's study of US defaults found the following average recovery rates by class:

| Class | Average recovery rate |
|---|---|
| Senior secured | 57.4% |
| Senior unsecured | 44.9% |
| Senior subordinated | 39.1% |
| Subordinated | 32.0% |
| Junior subordinated | 28.9% |

Moody's also found that recovery rates and default rates are negatively related. One empirical fit (based on historical data from 1983-2004) is

$$\text{Average Recovery Rate} = 0.52 - 6.9 \cdot \text{Average Default Rate}.$$

Thus recoveries tend to be smaller precisely when defaults are more common. This is bad for banks: when firms are doing badly, they are more likely to default, and the collateral posted by those firms is often worth less at the same time.

## Merton's Structural Model

**Merton's model** applies the Black-Scholes argument to credit risk. A firm has a single liability with face value $L$ due at time $T$. The firm pays no dividends, does not issue new debt after time 0, and has asset value $A_t$ satisfying the geometric Brownian motion

$$\mathrm{d}A_t = \mu A_t\,\mathrm{d}t + \sigma A_t\,\mathrm{d}Z_t,$$

so

$$A_t = A_0 \exp\left(\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma Z_t\right).$$

Default can occur only at time $T$. If $A_T \geq L$, the debt is repaid, bondholders receive $L$, and shareholders receive $A_T - L$. If $A_T < L$, the firm is liquidated, bondholders receive $A_T$, and shareholders receive 0. Therefore

$$E_T = \max(A_T - L, 0).$$

Equity is a call option on the firm's assets with strike $L$ and maturity $T$. The debt payoff is

$$D_T = A_T - \max(A_T - L, 0) = \min(A_T, L) = L - \max(L - A_T, 0),$$

so bondholders own the firm but have sold a call option to shareholders; equivalently, risky debt is a default-free bond minus a put option on the firm's assets.

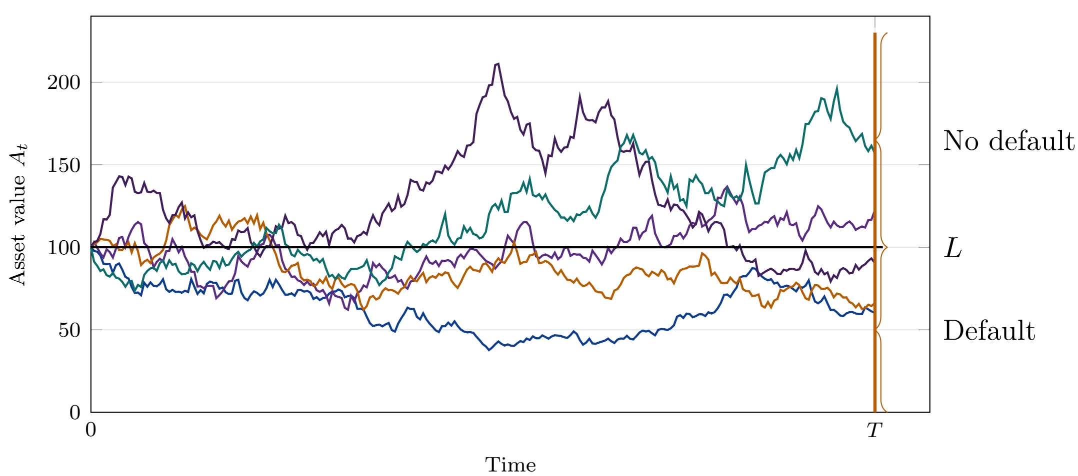


Figure 12: In Merton's model, default occurs at maturity if the terminal asset value falls below the liability threshold.

**Proposition 12.1 (Pricing in Merton's model)** Suppose continuous trading is possible, the risk-free rate is $r$, the asset value $A_t$ is independent of the face value $L$ and of how the firm is financed, and $A_t$ can be traded. A security with payoff $h(A_T)$ at time $T$ has time-$t$ price

$$\mathbb{E}^{\mathbb{Q}}\left[e^{-r(T-t)}h(A_T) \mid \mathcal{F}_t\right],$$

where under $\mathbb{Q}$ the drift $\mu$ is replaced by $r$:

$$A_T = A_t \exp\left(\left(r - \frac{\sigma^2}{2}\right)(T-t) + \sigma(W_T - W_t)\right).$$

The equity value is the Black-Scholes call value

$$E_t = A_t\Phi(d_1) - e^{-r(T-t)}L\Phi(d_2),$$

where

$$d_1 = \frac{\log(A_t/L) + (r+\sigma^2/2)(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 = d_1 - \sigma\sqrt{T-t}.$$

This risk-neutral pricing argument is mathematically convenient, but it depends on the strong assumption that the firm's assets can be traded in a way that supports replication. The key structural ideas are that the firm's assets and liabilities are modelled explicitly, and default occurs when assets fall below a liability threshold.

**Example 12.2 (Credit spread in Merton's model)** Let $P_1(T)$ be the price of a zero-coupon bond with face value 1 issued by the firm. If $c(0,T)$ is the credit spread over $[0,T]$, then

$$P_1(T) = e^{-(r+c(0,T))T}, \qquad c(0,T) = -\frac{\log P_1(T)}{T} - r.$$

For a bond with face value $L$, the time-0 debt value is

$$D_0 = A_0 - E_0.$$

Using the put representation of risky debt,

$$D_0 = Le^{-rT}\Phi(d_2) + A_0\Phi(-d_1),$$

where

$$d_1 = \frac{\log(A_0/L) + (r+\sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

Therefore

$$P_1(T) = \frac{D_0}{L}$$

and the spread $c(0,T)$ increases with volatility $\sigma$ and maturity $T$. The reason is that the value of the embedded put option increases with $\sigma$ and $T$, reducing the value of risky debt.

**Proposition 12.3** Under Merton's model, the real-world default probability is

$$\mathbb{P}(A_T \le L) = \Phi\left(\frac{\log(L/A_0) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}}\right),$$

while the risk-neutral default probability replaces $\mu$ by $r$:

$$\mathbb{Q}(A_T \le L) = \Phi\left(\frac{\log(L/A_0) - (r - \sigma^2/2)T}{\sigma\sqrt{T}}\right).$$

Default probability therefore increases with leverage, increases with volatility when $A_0 > L$, and decreases with asset value and drift.

**Example 12.4** A firm's asset value is modelled using Merton's model. Suppose $V_0 =$ 1,000,000, the asset return rate is $\mu_V = 0.1$, and the firm has issued a three-year zero-coupon bond with face value $B =$ 500,000.

1. Find the volatility $\sigma_V$ of the firm's asset return if its probability of default is 0.06.
2. Determine the expected fraction of money recovered by the lender when the firm defaults, conditional on default.

Use the fact that if $X \sim N(a,b)$, then

$$\mathbb{E}\left[e^X \mathbb{1}_{\{e^X < c\}}\right] = \exp(a + b/2)\,\Phi\left(\frac{\log(c) - a - b}{\sqrt{b}}\right).$$

**Solution.** Let $A_3$ denote the asset value at time 3. Under Merton's model,

$$A_3 = V_0 \exp\left(\left(\mu_V - \frac{\sigma_V^2}{2}\right)3 + \sigma_V\sqrt{3}\,Z\right), \qquad Z \sim N(0,1).$$

For (1), default occurs when $A_3 < B$, so

$$0.06 = \mathbb{P}(A_3 < B) = \Phi\left(\frac{\log(B/V_0) - \left(\mu_V - \sigma_V^2/2\right)3}{\sigma_V\sqrt{3}}\right).$$

Let $z_{0.06} = \Phi^{-1}(0.06) \approx -1.5548$. Then $\sigma_V$ solves

$$\frac{\log(500000/1000000) - \left(0.1 - \sigma_V^2/2\right)3}{\sigma_V\sqrt{3}} = z_{0.06},$$

that is,

$$1.5\sigma_V^2 + 2.693\sigma_V - 0.9931 = 0.$$

Taking the positive root gives

$$\sigma_V \approx 0.3139.$$

For (2), the lender receives $A_3$ in default, so the conditional recovery fraction is

$$\mathbb{E}\left[\frac{A_3}{B}\middle| A_3 < B\right] = \frac{1}{B\mathbb{P}(A_3 < B)}\mathbb{E}\left[A_3 \mathbb{1}_{\{A_3<B\}}\right].$$

Set $X = \log A_3$. Then $X \sim N(a, b)$ with

$$a = \log V_0 + \left(\mu_V - \frac{\sigma_V^2}{2}\right)3, \qquad b = 3\sigma_V^2.$$

Using the given identity with $c = B$, we obtain

$$\mathbb{E}\left[A_3 \mathbb{1}_{\{A_3<B\}}\right] = \exp(a + b/2)\Phi\left(\frac{\log B - a - b}{\sqrt{b}}\right).$$

Therefore

$$\mathbb{E}\left[\frac{A_3}{B}\middle| A_3 < B\right] = \frac{\exp(a + b/2)}{500000 \cdot 0.06}\Phi\left(\frac{\log(500000) - a - b}{\sqrt{b}}\right).$$

Substituting $\sigma_V \approx 0.3139$ gives

$$\mathbb{E}\left[\frac{A_3}{B}\middle| A_3 < B\right] \approx 0.8068.$$

So the expected recovery fraction conditional on default is about 80.7%. □

**Example 12.5** A firm's asset value is modelled using Merton's model. Suppose $V_0 = 750{,}000$, $\mu_V = 0.08$, $\sigma_V = 0.25$, and the firm issues a two-year bond with face value 500,000 paying an annual coupon of 6%. The risk-free rate is 4% annually.

1. What is the probability of default at time 1?
2. Given that the firm's value at time 1 is 775,000, what is the probability of default at time 2?
3. What is the value of the bond at time 0?
4. What is the firm's credit spread?

**Solution.** We use the natural discrete-payment extension of Merton's model: default can occur at coupon dates, and at time 1 shareholders continue only if the continuation value of equity is at least the coupon due. The coupon is

$$C = 0.06 \times 500000 = 30000,$$

so after the time-1 coupon is paid, the remaining one-year liability is

$$K_2 = 500000 + 30000 = 530000.$$

At time 1, if the firm continues, equity is a one-year call option on the asset value with strike 530000. Hence the continuation equity is

$$E_1^{\text{cont}}(a) = a\Phi(\widetilde{d}_1(a)) - 530000e^{-0.04}\Phi(\widetilde{d}_2(a)),$$

where

$$\widetilde{d}_1(a) = \frac{\log(a/530000) + (0.04 + 0.25^2/2)}{0.25}, \qquad \widetilde{d}_2(a) = \widetilde{d}_1(a) - 0.25.$$

Default at time 1 occurs when $E_1^{\text{cont}}(A_1) < 30000$. Solving

$$E_1^{\text{cont}}(a^*) = 30000$$

gives the default threshold

$$a^* \approx 466246.$$

For (1), under the physical measure,

$$\log A_1 \sim N\left(\log 750000 + \left(0.08 - \frac{0.25^2}{2}\right),\ 0.25^2\right),$$

so

$$\mathbb{P}(\tau = 1) = \mathbb{P}(A_1 < a^*) = \Phi\left(\frac{\log(a^*/750000) - \big(0.08 - 0.25^2/2\big)}{0.25}\right) \approx 0.0180.$$

For (2), since $775000 > a^*$, the firm survives to time 1. Default at time 2 then occurs iff $A_2 < 530000$. Conditional on $A_1 = 775000$,

$$\log A_2 \mid A_1 = 775000 \sim N\left(\log 775000 + \left(0.08 - \frac{0.25^2}{2}\right),\ 0.25^2\right),$$

so

$$\mathbb{P}(\tau = 2 \mid A_1 = 775000) = \Phi\left(\frac{\log(530000/775000) - \big(0.08 - 0.25^2/2\big)}{0.25}\right) \approx 0.0432.$$

For (3), under the risk-neutral measure we replace $\mu_V$ by $r = 0.04$. Let

$$D_1(a) = 530000e^{-0.04}\Phi(\widetilde{d}_2(a)) + a\Phi(-\widetilde{d}_1(a))$$

be the time-1 value of the remaining risky zero-coupon debt when the asset value is $a$. Therefore

$$D_0 = e^{-0.04}\mathbb{E}^{\mathbb{Q}}\big[A_1 \mathbb{1}_{\{A_1 < a^*\}} + \big(30000 + D_1(A_1)\big)\mathbb{1}_{\{A_1 \geq a^*\}}\big]\,.$$

Evaluating this expectation numerically gives

$$D_0 \approx 506076.$$

For (4), if $c$ is the constant credit spread over the flat risk-free curve, then the bond price also satisfies

$$506076 = 30000e^{-(0.04+c)} + 530000e^{-2(0.04+c)}.$$

Solving for $c$ gives

$$c \approx 0.0121.$$

So the credit spread is about 1.21% per year, or about 121 basis points. □

Two common extensions of Merton's model are first-passage models and richer asset processes. In a first-passage model, default occurs the first time the asset value falls below a liability boundary rather than only at time $T$. Richer stochastic processes can allow jumps in asset prices, producing surprise defaults.

The **KMV model** builds on the Merton model. The key quantity is the **expected default frequency** EDF, the one-year default probability. It is modelled as

$$\mathrm{EDF} = f(\mathrm{DtD}),$$

where $f$ is estimated empirically and the **distance to default** is

$$\mathrm{DtD} = \frac{\log(A_0/L) + (\mu - \sigma^2/2)T}{\sigma\sqrt{T}},$$

with $L$ replaced by a firm-specific liability threshold.

In practice the firm's asset value $A_0$ is not directly observable, so it is inferred from the firm's equity value using iterative techniques. The KMV method replaces the normal tail expression in Merton's default probability with an empirically estimated decreasing function. The model has been widely used in industry.

## Credit Migration and Threshold Models

A **credit migration model** assumes that there is a finite number of credit states, such as AAA, AA, A, BBB, BB, B, CCC, and default. Each firm starts in one state and may move between states over the horizon. The model requires a transition matrix $\mathbf{P}$ whose entry $p_{ij}$ gives the probability of moving from rating $i$ to rating $j$ over the horizon. Markov-chain methods can then be used to compute default probabilities over longer horizons.

Structural default can be written in threshold form. From Merton's model, default occurs when $A_T < L$, equivalently when

$$Z < \frac{\log(L/A_0) - (\mu - \sigma^2/2)T}{\sigma\sqrt{T}},$$

where $Z$ is a standard normal random variable. If a default probability PD is estimated directly, the threshold can instead be calibrated by

$$Z < \Phi^{-1}(\mathrm{PD}).$$

So far, we have analyzed the probability of default for a single firm. However, the risk of a portfolio depends on more than just the individual default probabilities. To model a portfolio of firms, we need to model the *dependence* among defaults. Dependence between the defaults of different firms is vitally important. A natural way to understand it is through a **general threshold model**, which allows for multiple credit states and dependence among firms through a copula.

Consider $m$ obligors. Let $\boldsymbol{X} = (X_1, \ldots, X_m)$ be an $m$-dimensional vector of creditworthiness variables. For obligor $i$, let

$$d_{i,1} < d_{i,2} < \cdots < d_{i,n}$$

be thresholds, collected in an $m \times n$ matrix $\mathbf{D}$. Set $d_{i,0} = -\infty$ and $d_{i,n+1} = \infty$. The state $S_i$ of firm $i$ is defined by

$$S_i = j \quad \Longleftrightarrow \quad d_{i,j} < X_i \le d_{i,j+1}, \qquad j = 0, \ldots, n.$$

The state $S_i = 0$ represents default. The default indicator is

$$Y_i = \mathbb{1}_{\{X_i \le d_{i,1}\}},$$

and the marginal default probability is

$$\bar{p}_i = \mathbb{P}(X_i \le d_{i,1}) = F_i(d_{i,1}) = \mathbb{P}(Y_i = 1).$$

The number of defaults and the portfolio loss are

$$M = \sum_{i=1}^{m} Y_i, \qquad L = \sum_{i=1}^{m} \delta_i e_i Y_i,$$

where $e_i$ is the exposure for firm $i$ and $\delta_i$ is the fraction lost in default. Risk measures such as VaR and CTE are then computed from the distributions of $M$ and $L$.

**Definition 12.6** For two default indicators $Y_i$ and $Y_j$ with marginal default probabilities $\bar{p}_i$ and $\bar{p}_j$, the **default correlation** is

$$\rho(Y_i, Y_j) = \frac{\mathbb{E}[Y_i Y_j] - \bar{p}_i \bar{p}_j}{\sqrt{(\bar{p}_i - \bar{p}_i^2)(\bar{p}_j - \bar{p}_j^2)}}.$$

**Example 12.7** Suppose there are two obligors, $\mathbb{P}(Y_1 = 1) = \mathbb{P}(Y_2 = 1) = 0.1$, and $\delta_1 e_1 = \delta_2 e_2 = 10^6$. If $Y_1$ and $Y_2$ are independent, then

$$\mathbb{P}(L = 2 \cdot 10^6) = 0.01, \qquad \mathbb{P}(L = 10^6) = 2(0.1)(0.9) = 0.18, \qquad \mathbb{P}(L = 0) = 0.81.$$

If instead

$$\mathbb{P}(Y_1 = 1, Y_2 = 1) = 0.09, \qquad \mathbb{P}(Y_1 = 1, Y_2 = 0) = \mathbb{P}(Y_1 = 0, Y_2 = 1) = 0.01,$$

and

$$\mathbb{P}(Y_1 = 0, Y_2 = 0) = 0.89,$$

then the marginal default probabilities remain 0.1, but

$$\mathbb{P}(L = 2 \cdot 10^6) = 0.09.$$

Thus the same marginal default probabilities can imply very different portfolio tail risk once dependence changes.

Copulas are natural in threshold models because marginal default probabilities are often easier to estimate than the joint dependence structure. By Sklar's theorem (Theorem 8.4), the marginal cdfs and the copula can be chosen separately. In credit modelling, this means the default or migration thresholds are calibrated to match marginal probabilities, while the copula specifies the dependence among creditworthiness variables.

In **Li's Gaussian copula model**, $X_i$ is the default time of firm $i$ and the default threshold is $d_i = T$ for all $i$. It assumes that the default times are exponentially distributed with parameters $\lambda_i$:

$$X_i \sim \text{Exp}(\lambda_i), \qquad F_i(t) = 1 - e^{-\lambda_i t}.$$

A Gaussian copula $C_{\boldsymbol{\Sigma}}$ is used for the joint distribution:

$$\mathbb{P}(X_1 \le t_1, \ldots, X_m \le t_m) = C_{\boldsymbol{\Sigma}}\left(1 - e^{-\lambda_1 t_1}, \ldots, 1 - e^{-\lambda_m t_m}\right).$$

If $\bar{p}_i = \mathbb{P}(X_i \le T)$, then

$$\bar{p}_i = 1 - e^{-\lambda_i T} \qquad \Longleftrightarrow \qquad \lambda_i = -\frac{\log(1 - \bar{p}_i)}{T}.$$

For a more general threshold model with marginal cdfs $F_1, \ldots, F_m$ and copula $C$, default probabilities involving the indicators require only the copula and the marginal default probabilities. For example, if $m = 4$, then

$$\mathbb{P}(Y_1 = 1, Y_2 = 1) = \mathbb{P}(X_1 \le d_1, X_2 \le d_2, X_3 < \infty, X_4 < \infty) = C(\bar{p}_1, \bar{p}_2, 1, 1).$$

Exact enumeration becomes impossible for large portfolios: with $m = 1000$, the vector $\boldsymbol{Y} =$

$(Y_1, \dots, Y_m)$ has $2^{1000}$ possible default configurations.

**Example 12.8** Consider Li's model for two firms with one-year default probabilities 0.1 and 0.15.

1. Determine $\lambda_1$ and $\lambda_2$ such that $X_1 \sim \mathrm{Exp}(\lambda_1)$ and $X_2 \sim \mathrm{Exp}(\lambda_2)$.
2. If $X_1$ and $X_2$ are instead modelled as standard normal critical variables, determine the thresholds $d_1$ and $d_2$.
3. Suppose $\boldsymbol{X} = (X_1, X_2)$ is modelled using a Gaussian copula with correlation $\rho = 0.5$. Give expressions for $\mathbb{P}(Y_1 = 1, Y_2 = 1)$ and $\mathbb{P}(Y_1 = 0, Y_2 = 0)$ in terms of the bivariate normal CDF $\Phi_\rho$.
4. Repeat the previous part using the Frank copula
$$C(u_1, u_2) = -\frac{1}{\theta} \log\left(1 + \frac{(e^{-\theta u_1} - 1)(e^{-\theta u_2} - 1)}{e^{-\theta} - 1}\right),$$
with $\theta = 2$, and compute the probabilities.
5. Let $L = 500{,}000 Y_1 + 750{,}000 Y_2$. Compute $q_{0.95}(L)$, the smallest $q$ such that $\mathbb{P}(L > q) \le 0.05$, and repeat for $q_{0.99}(L)$.

**Solution.** For (1), with a one-year horizon $T = 1$, Li's model gives

$$\bar{p}_i = 1 - e^{-\lambda_i} \qquad \Longrightarrow \qquad \lambda_i = -\log(1 - \bar{p}_i).$$

Therefore

$$\lambda_1 = -\log(0.9) \approx 0.1054, \qquad \lambda_2 = -\log(0.85) \approx 0.1625.$$

For (2), if the critical variables are standard normal, the thresholds satisfy

$$\Phi(d_1) = 0.1, \qquad \Phi(d_2) = 0.15,$$

so

$$d_1 = \Phi^{-1}(0.1) \approx -1.2816, \qquad d_2 = \Phi^{-1}(0.15) \approx -1.0364.$$

For (3), under the Gaussian copula with correlation $\rho = 0.5$,

$$\mathbb{P}(Y_1 = 1, Y_2 = 1) = \Phi_{0.5}(d_1, d_2),$$

and by inclusion-exclusion,

$$\mathbb{P}(Y_1 = 0, Y_2 = 0) = 1 - 0.1 - 0.15 + \Phi_{0.5}(d_1, d_2).$$

Numerically,

$$\Phi_{0.5}(d_1, d_2) \approx 0.0428, \qquad \mathbb{P}(Y_1 = 0, Y_2 = 0) \approx 0.7928.$$

For (4), with the Frank copula and $\theta = 2$,

$$\mathbb{P}(Y_1 = 1, Y_2 = 1) = C(0.1, 0.15) = -\frac{1}{2}\log\left(1 + \frac{(e^{-0.2} - 1)(e^{-0.3} - 1)}{e^{-2} - 1}\right) \approx 0.0279.$$

Again by inclusion-exclusion,

$$\mathbb{P}(Y_1 = 0, Y_2 = 0) = 1 - 0.1 - 0.15 + C(0.1, 0.15) \approx 0.7779.$$

For (5), the possible loss values are

$$0, \qquad 500000, \qquad 750000, \qquad 1250000.$$

Under the Gaussian copula,

$$\begin{aligned}
\mathbb{P}(L = 1250000) &= \mathbb{P}(Y_1 = 1, Y_2 = 1) = \Phi_{0.5}(d_1, d_2) \approx 0.0428,\\
\mathbb{P}(L = 750000) &= \mathbb{P}(Y_1 = 0, Y_2 = 1) = \mathbb{P}(Y_2 = 1) - \mathbb{P}(Y_1 = 1, Y_2 = 1) = 0.15 - 0.0428 = 0.1072,\\
\mathbb{P}(L = 500000) &= \mathbb{P}(Y_1 = 1, Y_2 = 0) = \mathbb{P}(Y_1 = 1) - \mathbb{P}(Y_1 = 1, Y_2 = 1) = 0.1 - 0.0428 = 0.0572.
\end{aligned}$$

Thus

$$\mathbb{P}(L > 500000) = 0.1500 > 0.05, \qquad \mathbb{P}(L > 750000) = 0.0428 < 0.05,$$

so

$$q_{0.95}(L) = 750000.$$

Also,

$$\mathbb{P}(L > 750000) = 0.0428 > 0.01, \qquad \mathbb{P}(L > 1250000) = 0,$$

so

$$q_{0.99}(L) = 1250000.$$

Under the Frank copula, the joint default probability is smaller,

$$\mathbb{P}(L = 1250000) = 0.0279,$$

but the same quantiles result:

$$q_{0.95}(L) = 750000, \qquad q_{0.99}(L) = 1250000.$$

$\square$

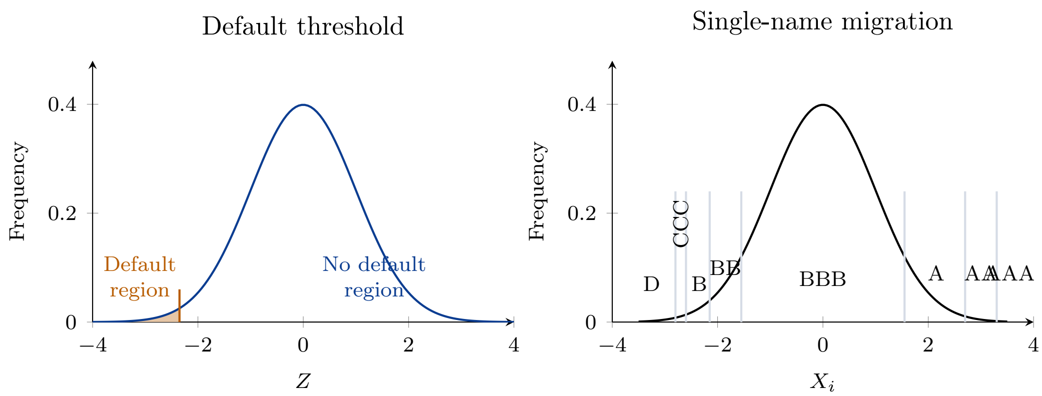


Figure 13: Thresholds convert a continuous creditworthiness variable into default and rating states.

## Vasicek's Single-Factor Credit Model

For a name $n$, define a latent creditworthiness index

$$Y_n = \beta_n Z + \sqrt{1-\beta_n^2}\,\varepsilon_n,$$

where $Z$ and $\varepsilon_1, \ldots, \varepsilon_N$ are iid standard normal random variables. The factor $Z$ is the **systematic component**, $\varepsilon_n$ is the **idiosyncratic component**, and $\beta_n$ is the **factor correlation** of name $n$. Default occurs when

$$Y_n \le H_n, \qquad H_n = \Phi^{-1}(\mathrm{PD}_n).$$

This model is based on Merton's structural threshold idea. It is called the **Vasicek model** (for credit risk, not interest rates) and the **single-factor Gaussian copula model**, and it forms the basis for the Bank for International Settlements credit-risk capital calculations. The construction makes $Y_n$ standard normal:

$$\mathrm{Var}(Y_n) = \beta_n^2 + (1-\beta_n^2) = 1.$$

The default indicator

$$\mathbb{1}_{\{Y_n \le H_n\}}$$

is Bernoulli with mean $\mathrm{PD}_n$ and variance $\mathrm{PD}_n(1-\mathrm{PD}_n)$.

The analogy with CAPM is direct. CAPM writes excess return as a systematic component plus idiosyncratic noise:

$$R_i = r_f + \beta_i(R_M - r_f) + \varepsilon_i.$$

Taking expectations gives the security market line

$$\mathbb{E}[R_i] = r_f + \beta_i(\mathbb{E}[R_M] - r_f),$$

where

$$\beta_i = \frac{\mathrm{cov}(R_i, R_M)}{\sigma_M^2}.$$

The regression form is

$$R_i = \alpha + \beta R_M + \varepsilon.$$

Investors can diversify away idiosyncratic risk, so they demand compensation for systematic risk. In the credit model, a large diversified portfolio behaves similarly: idiosyncratic credit risk diversifies away, while systematic credit risk remains.

The correlation between the creditworthiness indices of names $n$ and $m$ is $\beta_n\beta_m$. The correlation between their default indicators is

$$\rho_{nm} = \frac{\mathbb{E}[\mathbb{1}_{\{Y_n \le H_n\}}\mathbb{1}_{\{Y_m \le H_m\}}] - \mathrm{PD}_n\,\mathrm{PD}_m}{\sqrt{\mathrm{PD}_n(1-\mathrm{PD}_n)\,\mathrm{PD}_m(1-\mathrm{PD}_m)}} = \frac{\Phi_2(H_n, H_m; \beta_n\beta_m) - \mathrm{PD}_n\,\mathrm{PD}_m}{\sqrt{\mathrm{PD}_n(1-\mathrm{PD}_n)\,\mathrm{PD}_m(1-\mathrm{PD}_m)}},$$

where $\Phi_2(x, y; \rho)$ is the usual the cumulative bivariate normal distribution function with correlation $\rho$:

$$\Phi_2(x, y; \rho) = \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{x}\int_{-\infty}^{y} \exp\left(-\frac{u^2 - 2\rho uv + v^2}{2(1-\rho^2)}\right) \mathrm{d}v\ \mathrm{d}u.$$

If exposures at default are $\mathrm{EAD}_n$ and recovery rates are $\mathrm{RR}_n$, the total portfolio loss is

$$L_T = \sum_{n=1}^{N} \mathrm{EAD}_n(1-\mathrm{RR}_n)\mathbb{1}_{\{Y_n \le H_n\}}.$$

**Proposition 12.9** For a sequence of increasingly granular portfolios with

$$c_n = \mathrm{EAD}_n(1-\mathrm{RR}_n) \qquad \text{and} \qquad \sum_{n=1}^{N} c_n^2 \to 0 \quad \text{as } N \to \infty,$$

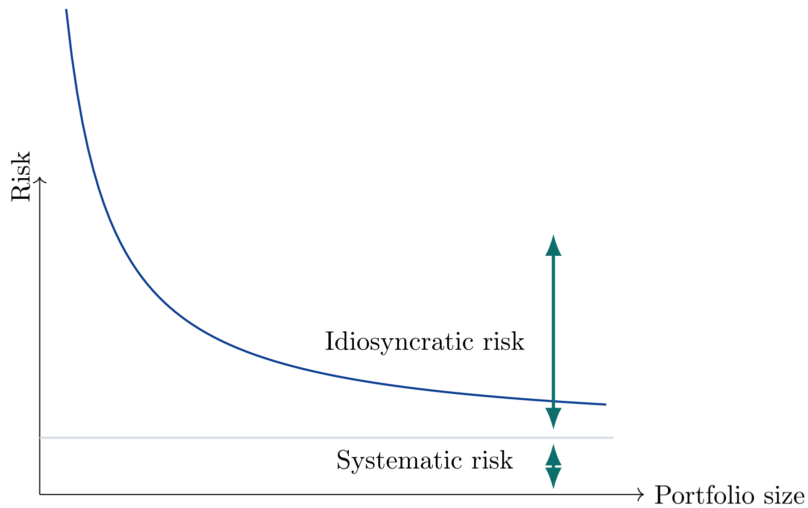


Figure 14: As the portfolio becomes large and granular, idiosyncratic risk is diversified away but systematic risk remains.

the total loss $L_T$ tends to its conditional expectation given the systematic factor $Z$, which we call the **systematic loss**:

$$L_S = \mathbb{E}[L_T \mid Z] = \sum_{n=1}^{N} \mathrm{EAD}_n (1 - \mathrm{RR}_n) \Phi\left( \frac{\Phi^{-1}(\mathrm{PD}_n) - \beta_n Z}{\sqrt{1 - \beta_n^2}} \right).$$

The conditional expectation for each name is

$$\mathbb{E}[\mathbb{1}_{\{Y_n \le H_n\}} \mid Z] = \mathbb{P}\left( \varepsilon_n \le \frac{H_n - \beta_n Z}{\sqrt{1 - \beta_n^2}} \,\middle|\, Z \right) = \Phi\left( \frac{\Phi^{-1}(\mathrm{PD}_n) - \beta_n Z}{\sqrt{1 - \beta_n^2}} \right).$$

**Proof.** Conditional on the systematic factor $Z$, the default indicators

$$\mathbb{1}_{\{Y_n \le H_n\}}, \qquad n = 1, \dots, N,$$

are independent, because each one depends on $Z$ and on its own idiosyncratic shock $\varepsilon_n$, and the shocks $\varepsilon_n$ are independent across names. Therefore

$$\mathbb{E}[L_T \mid Z] = \sum_{n=1}^{N} \mathrm{EAD}_n (1 - \mathrm{RR}_n) \mathbb{E}[\mathbb{1}_{\{Y_n \le H_n\}} \mid Z].$$

Since

$$Y_n = \beta_n Z + \sqrt{1-\beta_n^2}\,\varepsilon_n \qquad \text{and} \qquad H_n = \Phi^{-1}(\mathrm{PD}_n),$$

we obtain

$$\mathbb{E}[\mathbb{1}_{\{Y_n \le H_n\}} \mid Z] = \mathbb{P}(Y_n \le H_n \mid Z) = \mathbb{P}\left(\varepsilon_n \le \frac{H_n - \beta_n Z}{\sqrt{1-\beta_n^2}} \,\middle|\, Z\right) = \Phi\left(\frac{\Phi^{-1}(\mathrm{PD}_n) - \beta_n Z}{\sqrt{1-\beta_n^2}}\right),$$

which gives the stated expression for $L_S = \mathbb{E}[L_T \mid Z]$.

Now work along the given sequence of portfolios and write

$$p_n(Z) = \Phi\left(\frac{\Phi^{-1}(\mathrm{PD}_n) - \beta_n Z}{\sqrt{1-\beta_n^2}}\right).$$

Conditional on $Z$, the variables $\mathbb{1}_{\{Y_n \le H_n\}}$ are independent Bernoulli random variables with parameter $p_n(Z)$, so

$$\mathrm{Var}(L_T \mid Z) = \sum_{n=1}^{N} c_n^2 \,\mathrm{Var}(\mathbb{1}_{\{Y_n \le H_n\}} \mid Z) = \sum_{n=1}^{N} c_n^2 p_n(Z)(1-p_n(Z)) \le \frac{1}{4}\sum_{n=1}^{N} c_n^2.$$

By the assumption $\sum_{n=1}^{N} c_n^2 \to 0$ as $N \to \infty$, it follows that $\mathrm{Var}(L_T \mid Z) \to 0$. Therefore

$$\mathbb{E}\big[(L_T - \mathbb{E}[L_T \mid Z])^2 \mid Z\big] \to 0,$$

so $L_T - \mathbb{E}[L_T \mid Z] \to 0$ in conditional $L^2$, and hence in probability, given $Z$. This is the precise sense in which idiosyncratic fluctuations diversify away. Therefore $L_T$ tends to its conditional expectation $L_S$, as claimed. $\square$

The systematic loss is a decreasing function of the global factor $Z$: smaller values of $Z$ produce larger losses. Therefore its quantiles can be computed explicitly:

$$L_{S,\alpha} = \sum_{n=1}^{N} \mathrm{EAD}_n (1-\mathrm{RR}_n)\Phi\left(\frac{\Phi^{-1}(\mathrm{PD}_n) - \beta_n z_{1-\alpha}}{\sqrt{1-\beta_n^2}}\right).$$

Thus **portfolio VaR** is the sum of the individual instrument contributions in the asymptotic single-factor model.

Gordy (2003) proved that in the asymptotic single-factor model, the capital charge of a portfolio is the sum of the capital charges of the individual loans. The capital charge of a loan does not depend on the portfolio in which it is held. This is the **portfolio invariance** property, which is convenient for regulation, although not always realistic. It implies that the capital charge of a loan depends only on its own characteristics and not on the other loans in the portfolio, so it does not reflect diversification benefits.

## Basel II and Extensions

The single-factor model is the basis for the Bank for International Settlements (BIS) credit-risk capital calculations. The BIS is an international organization that promotes cooperation among central banks and sets global standards for banking regulation. The BIS's **Basel II** framework, which was implemented in 2007, uses the single-factor credit model to determine the minimum capital requirements for banks' credit risk. The regulatory capital requirement is based on VaR at the 99.9% confidence level over a one-year horizon. The core quantities are probability of default PD, exposure at default EAD, loss given default LGD, and maturity $M$.

The **worst-case default rate** (**WCDR**) at confidence level 99.9% is

$$\mathrm{WCDR} = \Phi\left(\frac{\Phi^{-1}(\mathrm{PD}) - \beta z_{0.001}}{\sqrt{1-\beta^2}}\right),$$

so the instrument **capital charge** is

$$\mathrm{EAD} \cdot \mathrm{LGD} \cdot (\mathrm{WCDR} - \mathrm{PD}) \cdot \mathrm{MA}\,.$$

Basel often expresses capital requirements in terms of **risk-weighted assets (RWA)** rather than directly in dollars of capital. RWA are not the actual book value of the assets; instead, they are a regulatory rescaling of exposure that puts safer and riskier positions on a common basis. The idea is that a bank must hold capital equal to a fixed fraction of its RWA, so a larger capital charge corresponds to a larger amount of risk-weighted assets. Under Basel II, the minimum total capital requirement is 8% of RWA, so the corresponding RWA for the exposure are

$$\mathrm{RWA} = \frac{\text{capital charge}}{0.08} = 12.5 \times \text{capital charge}.$$

Thus, with the capital-charge formula above,

$$\mathrm{RWA} = 12.5 \cdot \mathrm{EAD} \cdot \mathrm{LGD} \cdot (\mathrm{WCDR} - \mathrm{PD}) \cdot \mathrm{MA}\,.$$

Equivalently, if one defines

$$K = \mathrm{LGD} \cdot (\mathrm{WCDR} - \mathrm{PD}) \cdot \mathrm{MA},$$

then $K$ is the capital requirement per dollar of exposure and

$$\text{RWA} = 12.5\,K\ \text{EAD}\,.$$

Basel specifies the factor correlation through an empirical function of PD:

$$\beta^2 = 0.12\frac{1-e^{-50\,\text{PD}}}{1-e^{-50}} + 0.24\left(1 - \frac{1-e^{-50\,\text{PD}}}{1-e^{-50}}\right),$$

and the **maturity adjustment** is

$$\text{MA} = \frac{1+(M-2.5)b}{1-1.5b}, \qquad b = \big(0.11852 - 0.05478\log(\text{PD})\big)^2.$$

Here $M$ is the actual maturity of the exposure. As the company's default probability increases, $\beta$ decreases. The interpretation is that less creditworthy firms tend to be driven more by idiosyncratic risk and less by the market factor.

The **foundation internal ratings based (F-IRB) approach** lets the bank supply PD while Basel specifies LGD, EAD, and $M$. The **advanced internal ratings based (A-IRB) approach** lets the bank supply all four quantities. The corporate, sovereign, and bank exposure formulas are modified for other exposure classes; for example, retail exposures have no maturity adjustment. The trading book is more complicated because the exposures themselves are random.

Credit risk and liquidity risk are closely connected. Firms with lower credit risk are viewed as more homogeneous and can usually be bought and sold more easily and in larger quantities without substantial price impact. This liquidity component is reflected in bond spreads: the difference between BBB and AA bond spreads is too large to be explained entirely by default risk.

**Example 12.10** Consider two banks with loan portfolios of equal size and quality. Bank A lends only to Texas oil companies, while Bank B lends nationwide to businesses in many sectors. A purely portfolio-invariant one-factor capital rule does not reflect the diversification benefit in Bank B's portfolio.

A multi-factor model generalizes the single-factor credit model by writing

$$Y_n = \sum_{k=1}^{K}\beta_{nk}Z_k + \sqrt{1-\sum_{k=1}^{K}\beta_{nk}^2}\,\varepsilon_n,$$

where $Z_1,\dots,Z_K,\varepsilon_1,\dots,\varepsilon_N$ are iid standard normal random variables. The threshold rule is unchanged:

$$\mathbb{1}_{\{Y_n \le H_n\}}, \qquad H_n = \Phi^{-1}(\mathrm{PD}_n).$$

Let $\boldsymbol{Z} = (Z_1, \dots, Z_K)$. In the large-portfolio limit,

$$\mathbb{E}\left[L_T \mid \boldsymbol{Z}\right] = \sum_{n=1}^{N} \mathrm{EAD}_n(1-\mathrm{RR}_n)\Phi\left(\frac{\Phi^{-1}(\mathrm{PD}_n) - \sum_{k=1}^{K}\beta_{nk}Z_k}{\sqrt{1-\sum_{k=1}^{K}\beta_{nk}^2}}\right).$$

Unlike the one-factor case, VaR is generally not available explicitly unless the portfolio is homogeneous, in which case the model effectively reduces to a one-factor model.

The Gaussian single-factor and multi-factor models use Gaussian copulas: the creditworthiness indicators have a multivariate Gaussian distribution. More general threshold models allow arbitrary marginal distributions for the creditworthiness indicators and use a copula to determine their joint distribution. The thresholds are calibrated to match default and migration probabilities, while the copula describes codependence.

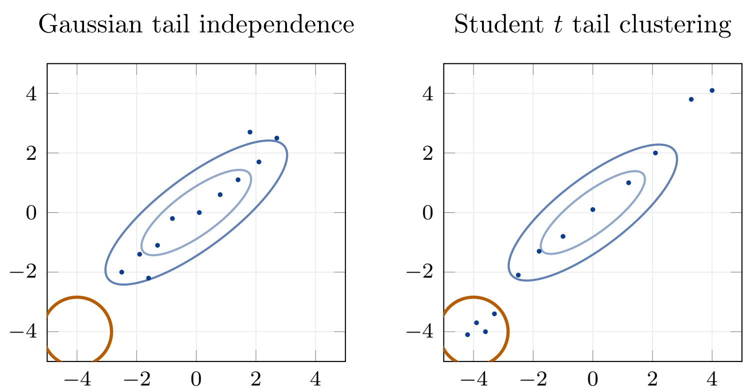


Figure 15: A Gaussian copula has tail independence, while a Student $t$ copula can generate joint lower-tail events more frequently.

The QRM textbook provides a simulation with 10,000 names that illustrates the impact of the copula choice. Each portfolio has a common default probability for all names and a common correlation between creditworthiness indicators:

| Portfolio | PD | Creditworthiness-indicator correlation |
|---|---|---|
| A | 0.06% | 2.58% |
| B | 0.50% | 3.80% |
| C | 7.50% | 9.21% |

The following table gives the 99% VaR of $M$, the number of defaults. The case $\nu = \infty$ is the Gaussian copula:

| Portfolio | $\nu = \infty$ | $\nu = 50$ | $\nu = 10$ |
|---|---|---|---|
| A | 21 | 49 | 118 |
| B | 157 | 261 | 589 |
| C | 2206 | 2400 | 3067 |

For Portfolio A, the 99th percentile of $M$ under the Student $t$ copula with $\nu = 10$ is more than five times the corresponding Gaussian value. Dependence choice therefore has a major effect on credit tail risk.

**Proposition 12.11** Consider the one-factor threshold model

$$X_i = \sqrt{\rho}\, F + \sqrt{1-\rho}\, \varepsilon_i, \qquad i = 1, \ldots, m,$$

where $F, \varepsilon_1, \ldots, \varepsilon_m$ are iid standard normal. If $\bar{p}_i$ is the default probability of firm $i$, then for large $m$,

$$q_\alpha(L) \approx \sum_{i=1}^{m} \delta_i e_i \Phi\left( \frac{\Phi^{-1}(\bar{p}_i) + \sqrt{\rho}\, \Phi^{-1}(\alpha)}{\sqrt{1-\rho}} \right),$$

where $q_\alpha(L)$ is the $100\alpha$ percentile of

$$L = \sum_{i=1}^{m} \delta_i e_i Y_i.$$

For $\alpha = 0.999$, each term in the sum represents the risk capital associated with firm $i$, up to any adjustment factor.

**Example 12.12** Capital requirements for a portfolio with credit loss

$$L = \sum_{i=1}^{m} \delta_i e_i Y_i$$

are set by

$$q_{0.999}(L) = \sum_{i=1}^{m} \delta_i e_i \Phi\left( \frac{\Phi^{-1}(\bar{p}_i) + \sqrt{\rho}\, \Phi^{-1}(0.999)}{\sqrt{1-\rho}} \right),$$

where $\rho$ is the correlation between $X_i$ and $X_j$ for all $i \neq j$. A portfolio contains investments in 100 firms. For 50% of the firms, the exposure is 5,000,000; for 30% it is 7,500,000; and for 20% it is 15,000,000. Suppose the probability of default is 0.05 for all firms. Compute $q_{0.999}(L)$ for $\rho = 0.2$ and for $\rho = 0.7$.

**Solution.** The total exposure is

$$\sum_{i=1}^{m} \delta_i e_i = 50(5 \cdot 10^6) + 30(7.5 \cdot 10^6) + 20(15 \cdot 10^6) = 775 \cdot 10^6.$$

Since all firms have the same default probability $\bar{p}_i = 0.05$, the quantile formula simplifies to

$$q_{0.999}(L) = 775 \cdot 10^6 \Phi\left(\frac{\Phi^{-1}(0.05) + \sqrt{\rho}\,\Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right).$$

Using

$$\Phi^{-1}(0.05) \approx -1.6449, \qquad \Phi^{-1}(0.999) \approx 3.0902,$$

we obtain the following.

If $\rho = 0.2$, then

$$\Phi\left(\frac{-1.6449 + \sqrt{0.2}\,(3.0902)}{\sqrt{0.8}}\right) \approx 0.3844,$$

so

$$q_{0.999}(L) \approx 775 \cdot 10^6 \times 0.3844 \approx 2.979 \times 10^8.$$

Thus

$$q_{0.999}(L) \approx 297.9 \text{ million}.$$

If $\rho = 0.7$, then

$$\Phi\left(\frac{-1.6449 + \sqrt{0.7}\,(3.0902)}{\sqrt{0.3}}\right) \approx 0.9570,$$

so

$$q_{0.999}(L) \approx 775 \cdot 10^6 \times 0.9570 \approx 7.417 \times 10^8.$$

Thus

$$q_{0.999}(L) \approx 741.7 \text{ million}.$$

The much larger value for $\rho = 0.7$ reflects the stronger default clustering when dependence is high. □

**Example 12.13** Consider again

$$q_{0.999}(L) = \sum_{i=1}^{m} \delta_i e_i \Phi\left(\frac{\Phi^{-1}(\bar{p}_i) + \sqrt{\rho}\,\Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right),$$

which comes from the model

$$X_i = \sqrt{\rho}\, F + \sqrt{1-\rho}\, \varepsilon_i,$$

where $F, \varepsilon_1, \ldots, \varepsilon_m$ are iid standard normal. Suppose $m = 100$, $\delta_i e_i = 10^6$, $\bar{p}_i = 0.08$ for all $i$, and $\rho = 0.5$. Determine $q_{0.999}(L)$.

**Solution.** Here every term in the sum is identical, so

$$q_{0.999}(L) = 100 \cdot 10^6 \Phi\left(\frac{\Phi^{-1}(0.08) + \sqrt{0.5}\, \Phi^{-1}(0.999)}{\sqrt{0.5}}\right).$$

Using

$$\Phi^{-1}(0.08) \approx -1.4051, \qquad \Phi^{-1}(0.999) \approx 3.0902,$$

we get

$$\Phi\left(\frac{-1.4051 + \sqrt{0.5}\,(3.0902)}{\sqrt{0.5}}\right) \approx 0.8650.$$

Therefore

$$q_{0.999}(L) \approx 100 \cdot 10^6 \times 0.8650 = 8.650 \times 10^7.$$

So the portfolio capital is approximately

$$q_{0.999}(L) \approx 86.5 \text{ million.}$$

□